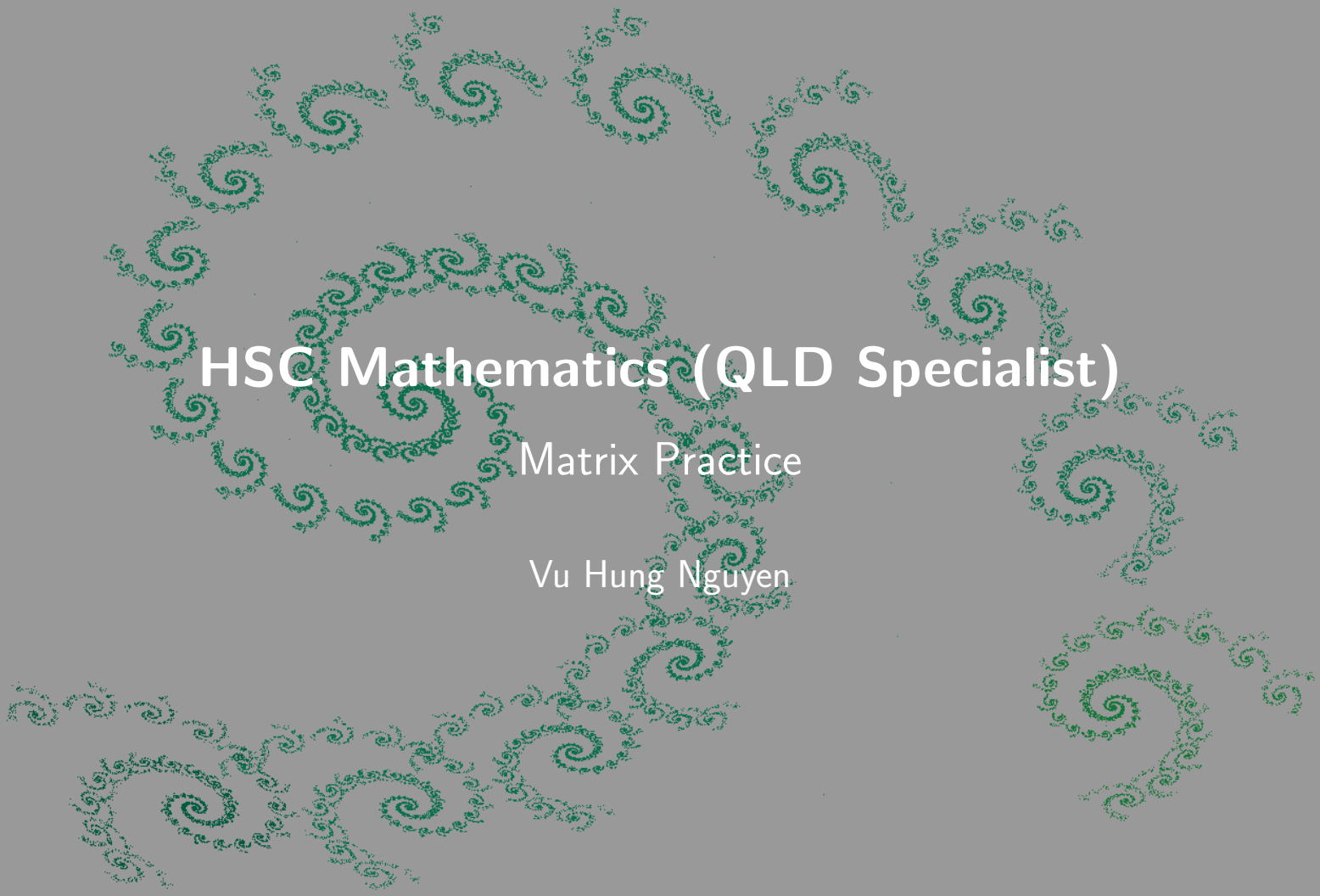


HSC Mathematics (QLD Specialist)

Matrix Practice

Vu Hung Nguyen



Download PDF and resources:

`https://github.com/vuhung16au/math-olympiad-ml`



Read online at Vu's Maths Hub:

`https://vumaths.com/`

This work is licensed under CC BY-NC 4.0 (Attribution-Non-Commercial), see the LICENSE file on the github page for more info.

"As for everything else, so for a mathematical theory:
beauty can be perceived but not explained."

Arthur Cayley (Founder of matrix algebra)

Contents

1 Introduction **8**
1.1 The Lost Heritage: When NSW Extension 2 Had Matrices (1966–1980) 8
1.2 About This Booklet 9
1.3 Target Audience 10
1.4 How to Use This Booklet 10
1.5 Knowledge Needed 11
1.6 What You Will Learn 11

2 Matrix Fundamentals **11**
2.1 Introduction to Matrix Algebra 12
P 2.1: Matrix Algebra — Basic Operations 13
S 2.1 14
P 2.2: Matrix Algebra — Transpose 15
S 2.2 15
2.2 Inverses and Determinants for 2×2 Matrices 16
P 2.3: Calculating Inverses and Determinants 16
S 2.3 17
2.3 Simultaneous Linear Equations with Two Variables 17
P 2.4: Algebraic and Geometric Interpretations 18
S 2.4 19
2.4 Using Matrix Algebra for Systems of Linear Equations in Two Variables 19
P 2.5: Matrix Solution to a 2×2 System 20
S 2.5 21
2.5 Inverses and Determinants for $n \times n$ Matrices 21
P 2.6: Inverting a 3×3 Matrix 23
S 2.6 24
2.6 Simultaneous Linear Equations with More Than Two Variables 25
P 2.7: Translating a Word Problem into a System 26
S 2.7 26
2.7 Using Matrix Algebra for Systems of Linear Equations in More Than Two Variables 27
P 2.8: Matrix Solution to a 3×3 System 27
S 2.8 28
2.8 Using Augmented Matrices for Systems of Equations 29
P 2.9: Gaussian Elimination 30
S 2.9 30
2.9 Matrices as Linear Transformations 31
P 2.10: Transformation Composition 32
S 2.10 33
2.10 Matrix Powers, Iteration and Dynamical Systems 33
P 2.11: Iterating a System 34
S 2.11 35
2.11 Dominance Matrices 35
P 2.12: Tournament Rankings 36
S 2.12 37
2.12 Leslie Matrices 38
P 2.13: Projecting a Rodent Population 39
S 2.13 39

3 Part 1: Worked Problems **40**
3.1 Basic . . . 40
P 3.1: School Demographics Matrix . . . 40
S 3.1 . . . 41
P 3.2: Matrix Properties and Transformations . . . 42
S 3.2 . . . 43
3.2 Medium . . . 44
P 3.3: Matrix Algebra and Commutativity . . . 44
S 3.3 . . . 45
P 3.4: Matrix Inverses and Matrix Equations . . . 46
S 3.4 . . . 47
3.3 Advanced . . . 48
P 3.5: Matrix Inverses and the Zero Matrix . . . 48
S 3.5 . . . 49
P 3.6: Matrix Methods for Simultaneous Equations . . . 50
S 3.6 . . . 51
P 3.7: Determinant Properties and Matrix Algebra . . . 52
S 3.7 . . . 53
P 3.8: Dominance Arrays and Tournament Rankings . . . 54
S 3.8 . . . 55
P 3.9: Adjacency Matrices and Network Paths . . . 56
S 3.9 . . . 57
P 3.10: Parametric Conics: The Matrix Shortcut . . . 59
S 3.10 . . . 60
P 3.11: NSW HSC Heritage (1975 & 1976) . . . 61
S 3.11 . . . 62

4 Part 2: Practice and Challenge Bank **63**
4.1 Warm-Up Drills . . . 63
P 4.1: Matrix Systems and Polynomial Curve Fitting . . . 63
S 4.1 . . . 64
P 4.2: Matrix Systems and Intersecting Planes . . . 65
S 4.2 . . . 66
4.2 Stretch Problems . . . 67
P 4.3: Leslie Matrices and Parametric Population Modeling . . . 67
S 4.3 . . . 69
P 4.4: Leslie Matrices and Stable Populations . . . 70
S 4.4 . . . 72
P 4.5: Leslie Matrices and Periodic Populations . . . 73
S 4.5 . . . 74
4.3 Challenge Corner . . . 75
P 4.6: Parametric Simultaneous Equations and Matrices . . . 75
S 4.6 . . . 76
P 4.7: Matrix Systems and Geometric Transformations . . . 77
S 4.7 . . . 78
P 4.8: Matrix Determinants and System Solutions . . . 79
S 4.8 . . . 80
P 4.9: Determinants, Area and Orientation . . . 81
S 4.9 . . . 83
P 4.10: Matrix Algebra and Transformations . . . 84

S 4.10 . . . 85
P 4.11: Systems of Linear Equations and 3D Geometry . . . 86
S 4.11 . . . 87
P 4.12: Augmented Matrices and Parametric Solutions . . . 88
S 4.12 . . . 89
P 4.13: Advanced Matrix Products and Inverses . . . 90
S 4.13 . . . 91
P 4.14: Matrix Systems and Planes in 3D Space . . . 92
S 4.14 . . . 94
P 4.15: Matrix Polynomials and System Solutions . . . 95
S 4.15 . . . 97
P 4.16: The Geometry of Mirrors . . . 98
S 4.16 . . . 100
P 4.17: The Hidden Bridge . . . 101
S 4.17 . . . 103
P 4.18: Linearising Trigonometry . . . 104
S 4.18 . . . 106
P 4.19: The Matrix Code of Fibonacci . . . 107
S 4.19 . . . 109
P 4.20: The Nilpotent Shortcut . . . 110
S 4.20 . . . 112
P 4.21: Matrices and Probability: Predicting Changing Systems . . . 114
S 4.21 . . . 116
P 4.22: The Algebra of Transformations . . . 117
S 4.22 . . . 119
P 4.23: The Calculus Matrix . . . 121
S 4.23 . . . 122
P 4.24: The Commutator Conundrum . . . 123
S 4.24 . . . 125
P 4.25: The Net Strength Theorem . . . 126
S 4.25 . . . 128
P 4.26: A Hidden Pattern in Matrix Powers . . . 130
S 4.26 . . . 131
P 4.27: Esports Dominance Networks . . . 133
S 4.27 . . . 135
P 4.28: Matrix Powers and Convergence . . . 137
S 4.28 . . . 138
P 4.29: A Coupled Recurrence and Rational Approximations to $\sqrt{2}$ . . . 140
S 4.29: Brief . . . 141

5 Matrix Toolkit (Quick Reference) 142

6 Summary and Where Next 144

A Appendix A: Bridge to University Linear Algebra 145
A.1 Linear Combinations and Span . . . 145
A.2 Linear Independence . . . 145
A.3 Basis and Dimension . . . 146
A.4 Rank and Null Space . . . 147
A.5 Eigenvalues and Eigenvectors . . . 148
A.6 The Characteristic Polynomial . . . 149

A.7 Where Linear Algebra Goes Next: Least Squares, SVD, etc. 151

B Appendix B: Matrices in Computing and Applied Mathematics 152
B.1 NumPy and SymPy 152
B.2 Matrices and Graph Theory 153
B.3 Computer Graphics 155
B.4 Data Science and Machine Learning 155

1 Introduction

Matrices are one of the central mathematical tools used in modern science, engineering, computing, economics, data science, computer graphics, and artificial intelligence. At first, a matrix may look like little more than a rectangular table of numbers. However, matrix algebra provides a powerful language for representing and solving many problems involving systems of equations, geometric transformations, networks, population models, and multidimensional data.

For senior secondary students, matrices also provide an important bridge between school mathematics and university mathematics. Ideas that are usually studied separately at school — such as simultaneous equations, vectors, transformations, functions, and geometry — can often be expressed within a single mathematical framework using matrices.

This booklet introduces these ideas through short explanations, worked examples, structured exercises, and more challenging problems. The aim is not simply to learn how to perform matrix calculations, but to develop an understanding of **why matrix methods work, what they represent, and where they are useful**.

Remark 1.1 (Syllabus Note and Historical Context)

Matrices are not part of the current NSW NESA Mathematics Extension 1 or Extension 2 syllabuses (where they are currently taught in Queensland and Victorian Specialist Mathematics). Therefore, NSW students should regard this booklet as **enrichment material rather than examinable HSC content**.
However, matrices are **not foreign to NSW**. From 1966 to 1980, the highest level of senior mathematics in NSW (originally called **First Level** under the Wyndham Scheme, and later renamed **4 Unit**) was a **two-year senior course** studied across both Year 11 and Year 12. Matrices and linear algebra formed the theoretical foundation of that course: Section 4 of the 1965 syllabus was explicitly titled *"Plane Geometry, Geometric Algebra, Matrices, Geometry in Three Dimensions"*.
In **1981**, the NSW Board restructured 4 Unit into a **one-year course taken only in Year 12**. With instructional time halved, matrices, linear transformations, group theory, and 3D geometry were excised from the curriculum. Studying this booklet is therefore not merely looking across state borders—it is a **restoration of NSW's own mathematical heritage**.

1.1 The Lost Heritage: When NSW Extension 2 Had Matrices (1966–1980)

A perennial debate among mathematics educators across Australia is whether matrices belong in the senior secondary school syllabus:

- **The argument against:** When rushed or taught in isolation, matrices often devolve into students blindly memorising mechanical algorithms (such as the "row-by-column" recipe) without appreciating the underlying linear transformations or geometric meaning.
- **The argument for:** When connected to geometry and algebra, matrices offer an extraordinarily powerful, unifying language. They provide structural shortcuts that turn pages of exhausting algebra into clean, conceptual deductions. Furthermore, they are the foundational language of 3D computer graphics, data science, quantum physics, and artificial intelligence.

The historical trajectory of the NSW curriculum provides the ultimate case study in this debate. When matrices were excised in 1981 due to the compression of the course into a

single year, one major topic was left behind: **Conic Sections** (ellipses and hyperbolas), which remained in the syllabus until 2019.

This led to a 40-year historical paradox. In the pre-2020 NSW HSC Extension 2 examinations, classic questions regularly required students to find the area of a triangle formed by points on an ellipse or hyperbola. Because these points were defined parametrically ($P(a\cos\theta, b\sin\theta)$ and $Q(a\cos\phi, b\sin\phi)$), students were forced to use the only tools remaining in the syllabus:

1. Calculate the chord length PQ using the distance formula: $\sqrt{(x_2-x_1)^2+(y_2-y_1)^2}$, producing massive radical expressions.
2. Derive the Cartesian equation of the secant line passing through P and Q.
3. Apply the perpendicular distance formula $h = \dfrac{|Ax_0+By_0+C|}{\sqrt{A^2+B^2}}$ to find the altitude from the third vertex.
4. Calculate Area $= \frac{1}{2} \times$ base $\times$ height, filling pages of algebra waiting for nested square roots to cancel out.

Students and teachers endured decades of algebraic torture, unaware that **these problems were originally written when matrix determinants were on the syllabus to solve them!**

If one uses the matrix tools from the 1966–1980 syllabus, the area of any triangle with vertices (x_1, y_1), (x_2, y_2), and (x_3, y_3) is given directly by the 3×3 coordinate determinant:

$$\text{Area} = \frac{1}{2}\left|\det\begin{pmatrix} x_1 & y_1 & 1 \\ x_2 & y_2 & 1 \\ x_3 & y_3 & 1 \end{pmatrix}\right|.$$

When parametric coordinates are substituted into this matrix, the determinant evaluates in **two short lines of elementary algebra** ($\frac{1}{2}ab\sin(\phi-\theta)$).

Furthermore, this matrix determinant is intimately connected to 3D vectors: it is the algebraic counterpart of the vector cross-product magnitude $\frac{1}{2}\|\mathbf{u}\times\mathbf{v}\|$ flattened into two dimensions.

The goal of this booklet is to teach matrices not as rote drills, but as a **conceptual superpower**—reconnecting modern students with the depth, elegance, and coherence that once defined senior mathematics in NSW.

1.2 About This Booklet

This booklet is designed as a self-contained introduction to matrix mathematics for senior secondary students.

It begins with the fundamental language and operations of matrices: matrix notation, dimensions, addition, scalar multiplication, matrix multiplication, transpose, determinants, and inverses. These ideas are then connected to familiar school mathematics, particularly simultaneous linear equations and geometric transformations.

As the booklet progresses, the emphasis shifts from learning individual techniques to **using matrices as a problem-solving tool**. Applications include systems of equations, dominance matrices, population modelling with Leslie matrices, polynomial curve fitting, and problems involving geometry in two and three dimensions.

The worked-problem and challenge sections are intended to encourage deeper mathematical reasoning rather than mechanical calculation. Some problems combine several ideas and require students to decide which matrix method is appropriate.

The appendices then provide a glimpse beyond the secondary-school syllabus. Topics such as Gaussian elimination, linear independence, eigenvalues and eigenvectors, computer-based matrix calculations, and adjacency matrices introduce students to the broader subject of **linear algebra**.

The overall progression is therefore:

school algebra → matrix algebra → applications → university linear algebra.

The intention is not to replace a formal linear algebra course, but to make the transition into one considerably more familiar.

1.3 Target Audience

This booklet is primarily written for senior secondary mathematics students who would like to explore mathematics beyond their immediate syllabus.

For **NSW Mathematics Extension 1 and Extension 2 students**, matrices provide useful enrichment and an opportunity to see familiar topics such as vectors, simultaneous equations, transformations, and three-dimensional geometry from a different perspective.

For **Queensland Specialist Mathematics students**, the booklet can be used for revision, additional worked examples, problem-solving practice, and extension beyond the examinable course.

It is also suitable for students elsewhere in Australia who are preparing for university study in mathematics-intensive fields such as engineering, computer science, physics, data science, economics, and artificial intelligence.

Students beginning university mathematics may also find parts of the booklet useful as revision. The later appendices are particularly intended to provide a gentle introduction to ideas that appear in first-year linear algebra.

No previous formal study of matrices is assumed.

1.4 How to Use This Booklet

There are several ways to use this booklet depending on your background.

If matrices are completely new to you, begin with **Chapter 2: Matrix Fundamentals** and work through the topics in order. Do not try to memorise all the formulas immediately. Instead, pay attention to the meaning of each operation and the dimensions of the matrices involved.

After reading an explanation, attempt the associated problem **before reading the solution**. Matrix algebra is best learned by doing calculations yourself and checking where mistakes occur.

Once you are comfortable with the fundamentals, move to **Part 1: Worked Problems**. These problems demonstrate how several matrix ideas can be combined in a single question.

You can then use **Part 2: Practice and Challenge Bank** for independent practice. The later problems are deliberately less routine and may require you to connect matrices with algebra, geometry, modelling, or other areas of mathematics.

Finally, the appendices can be read selectively. They are not required for understanding the earlier chapters. Instead, they are intended for students who are curious about where matrix mathematics leads at university and in computing.

A useful study routine is therefore:

Learn → Attempt → Check → Reflect → Extend.

When checking a solution, do not only ask whether your numerical answer is correct. Ask why the method works and whether another method could also have been used.

1.5 Knowledge Needed

This booklet assumes only the mathematical background normally developed during senior secondary mathematics.

Students should be comfortable with basic algebraic manipulation, substitution and elimination, solving simultaneous linear equations, working with fractions and indices, and interpreting equations geometrically.

Some familiarity with coordinate geometry and vectors is useful for the sections involving transformations and three-dimensional geometry, but the necessary matrix ideas are introduced within the booklet.

Importantly, **no prior knowledge of matrices or linear algebra is required**. Chapter 2 begins from the definition of a matrix and develops the required notation and operations step by step.

Some topics in the appendices require greater mathematical maturity, but these sections are explicitly intended as enrichment and may be skipped without affecting the main sequence of the booklet.

1.6 What You Will Learn

By working through this booklet, you should develop both computational skills and a broader understanding of matrix mathematics.

You will learn how to represent information using matrices; perform matrix arithmetic and multiplication; work with transposes, determinants, and inverses; translate systems of linear equations into matrix form; solve systems using inverse matrices and augmented matrices; interpret matrices geometrically; and apply matrices to modelling problems.

More importantly, you should begin to recognise several important ideas that continue into university mathematics:

$$AX = B$$

is not merely a compact way of writing simultaneous equations. It is one of the fundamental mathematical forms used throughout linear algebra, numerical computing, optimisation, statistics, engineering, and machine learning.

Similarly,

$$AB$$

is more than an arithmetic operation. Matrix multiplication can represent the **composition of transformations**, the propagation of a population model, or movement through a network.

The later enrichment sections introduce this wider perspective. They show how elementary matrix calculations develop naturally into concepts such as Gaussian elimination, linear independence, eigenvalues and eigenvectors, numerical computation, and graph theory.

The goal of the booklet is therefore not simply to teach students to **calculate with matrices**, but to help them begin to **think in matrices**.

2 Matrix Fundamentals

This section covers the essential matrix theory that underpins the worked problems and practice exercises in the rest of this booklet. Since matrices are not part of the NSW NESA syllabus, this section serves as a self-contained primer adapted from the Queensland (QCAA) Specialist Mathematics curriculum (Chapters 8 and 9 of *Cambridge Maths Unit 3 & 4 Specialist*).

Each topic presents the key ideas, followed by one or more worked examples.

2.1 Introduction to Matrix Algebra

A **matrix** is a rectangular array of numbers arranged in rows and columns. Matrices are used extensively in mathematics, science, economics, and computer science to organise data, represent systems of equations, and describe geometric transformations.

Notation and Terminology

A matrix with m rows and n columns is called an $m \times n$ matrix (read "m by n"). We typically denote matrices by uppercase letters (A, B, C, ...) and their individual entries by lowercase letters with subscripts. The entry in row i and column j of matrix A is written a_{ij}.

$$A = \begin{bmatrix} a_{11} & a_{12} & \cdots & a_{1n} \\ a_{21} & a_{22} & \cdots & a_{2n} \\ \vdots & \vdots & \ddots & \vdots \\ a_{m1} & a_{m2} & \cdots & a_{mn} \end{bmatrix}$$

Special Types of Matrices

- **Row matrix:** A matrix with exactly one row, e.g. $\begin{bmatrix} 1 & 2 & 3 \end{bmatrix}$ is a 1×3 row matrix.
- **Column matrix:** A matrix with exactly one column, e.g. $\begin{bmatrix} 4 \\ 5 \end{bmatrix}$ is a 2×1 column matrix.
- **Square matrix:** A matrix where $m = n$ (same number of rows and columns).
- **Zero matrix** (O): A matrix in which every entry is 0. For example, $O_{2\times 2} = \begin{bmatrix} 0 & 0 \\ 0 & 0 \end{bmatrix}$.
- **Identity matrix** (I): A square matrix with 1s on the main diagonal and 0s everywhere else. For any square matrix A of the same size, $AI = IA = A$.

$$I_2 = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}, \qquad I_3 = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

- **Diagonal matrix:** A square matrix where all off-diagonal entries are 0, e.g. $\begin{bmatrix} 3 & 0 \\ 0 & -7 \end{bmatrix}$.
- **Symmetric matrix:** A square matrix A where $A = A^T$ (equal to its transpose), e.g. $\begin{bmatrix} 1 & 4 \\ 4 & 2 \end{bmatrix}$.

Matrix Operations

1. Matrix Addition and Subtraction

Two matrices can be added (or subtracted) only if they have the *same dimensions*. The operation is performed entry by entry:

$$A + B = \begin{bmatrix} a_{11} + b_{11} & a_{12} + b_{12} \\ a_{21} + b_{21} & a_{22} + b_{22} \end{bmatrix}$$

2. Scalar Multiplication

To multiply a matrix by a scalar k, multiply every entry by k:

$$kA = \begin{bmatrix} ka_{11} & ka_{12} \\ ka_{21} & ka_{22} \end{bmatrix}$$

3. Matrix Multiplication

The product AB is defined only when the number of *columns* of A equals the number of *rows* of B. If A is $m \times n$ and B is $n \times p$, then AB is $m \times p$.

The entry in row i, column j of AB is the dot product of row i of A with column j of B:

$$(AB)_{ij} = \sum_{k=1}^{n} a_{ik}\, b_{kj}$$

Remark 2.1 (Key Properties of Matrix Multiplication)

- **Not commutative:** In general, $AB \neq BA$ (even when both products are defined).
- **Associative:** $(AB)C = A(BC)$.
- **Distributive:** $A(B + C) = AB + AC$ and $(A + B)C = AC + BC$.
- **Identity:** $AI = IA = A$.
- **Zero:** $AO = OA = O$ (where O is the zero matrix of appropriate size).

4. Transpose

The **transpose** of an $m \times n$ matrix A, written A^T, is the $n \times m$ matrix obtained by swapping its rows and columns: the entry in row i, column j of A^T is a_{ji}.

Key properties of the transpose:

- $(A^T)^T = A$
- $(A + B)^T = A^T + B^T$
- $(kA)^T = kA^T$
- $(AB)^T = B^T A^T$ *(note the reversal of order)*

Problem 2.1: Matrix Algebra — Basic Operations

Let $A = \begin{bmatrix} 2 & -1 \\ 3 & 4 \end{bmatrix}$, $B = \begin{bmatrix} 0 & 5 \\ -2 & 1 \end{bmatrix}$, and $k = 3$.

(a) Calculate $A + B$.

(b) Calculate kA.

(c) Calculate AB.

(d) Calculate BA.

(e) Is $AB = BA$? What does this tell you about matrix multiplication?

Solution 2.1

(a)

$$A+B=\begin{bmatrix}2+0 & -1+5\\ 3+(-2) & 4+1\end{bmatrix}=\begin{bmatrix}2 & 4\\ 1 & 5\end{bmatrix}$$

(b)

$$kA=3\begin{bmatrix}2 & -1\\ 3 & 4\end{bmatrix}=\begin{bmatrix}6 & -3\\ 9 & 12\end{bmatrix}$$

(c)

$$AB=\begin{bmatrix}2 & -1\\ 3 & 4\end{bmatrix}\begin{bmatrix}0 & 5\\ -2 & 1\end{bmatrix}$$

Row 1 of A · columns of B:
$(2)(0)+(-1)(-2)=2, \quad (2)(5)+(-1)(1)=9$
Row 2 of A · columns of B:
$(3)(0)+(4)(-2)=-8, \quad (3)(5)+(4)(1)=19$

$$AB=\begin{bmatrix}2 & 9\\ -8 & 19\end{bmatrix}$$

(d)

$$BA=\begin{bmatrix}0 & 5\\ -2 & 1\end{bmatrix}\begin{bmatrix}2 & -1\\ 3 & 4\end{bmatrix}$$

$(0)(2)+(5)(3)=15, \quad (0)(-1)+(5)(4)=20$
$(-2)(2)+(1)(3)=-1, \quad (-2)(-1)+(1)(4)=6$

$$BA=\begin{bmatrix}15 & 20\\ -1 & 6\end{bmatrix}$$

(e)

$$AB=\begin{bmatrix}2 & 9\\ -8 & 19\end{bmatrix}\neq\begin{bmatrix}15 & 20\\ -1 & 6\end{bmatrix}=BA$$

Matrix multiplication is **not commutative**. The order in which you multiply matrices matters. This is one of the most important differences between matrix algebra and ordinary number algebra.

Takeaways 2.1

- **Dimension Compatibility:** Always check dimensions before multiplying — an $m\times n$ matrix can only multiply an $n\times p$ matrix.
- **Non-Commutativity:** $AB\neq BA$ in general. Never assume you can swap the order of matrix multiplication.
- **Entry-by-Entry vs. Row-by-Column:** Addition is entry-by-entry (same dimensions required). Multiplication uses the row-by-column dot product rule (inner dimensions must match).

Problem 2.2: Matrix Algebra — Transpose

Let $P = \begin{bmatrix} 1 & 2 & 3 \\ 4 & 5 & 6 \end{bmatrix}$ and $Q = \begin{bmatrix} 7 & 8 \\ 9 & 10 \\ 11 & 12 \end{bmatrix}$.

(a) Find P^T and state its dimensions.

(b) Calculate PQ and then find $(PQ)^T$.

(c) Calculate $Q^T P^T$ and verify that $(PQ)^T = Q^T P^T$.

Solution 2.2

(a)

$$P^T = \begin{bmatrix} 1 & 4 \\ 2 & 5 \\ 3 & 6 \end{bmatrix}$$

P is 2×3, so P^T is 3×2.

(b)

P is 2×3 and Q is 3×2, so PQ is 2×2.

$$PQ = \begin{bmatrix} 1 & 2 & 3 \\ 4 & 5 & 6 \end{bmatrix} \begin{bmatrix} 7 & 8 \\ 9 & 10 \\ 11 & 12 \end{bmatrix}$$

$(1)(7) + (2)(9) + (3)(11) = 58, \quad (1)(8) + (2)(10) + (3)(12) = 64$

$(4)(7) + (5)(9) + (6)(11) = 139, \quad (4)(8) + (5)(10) + (6)(12) = 154$

$$PQ = \begin{bmatrix} 58 & 64 \\ 139 & 154 \end{bmatrix}, \quad (PQ)^T = \begin{bmatrix} 58 & 139 \\ 64 & 154 \end{bmatrix}$$

(c)

$$Q^T = \begin{bmatrix} 7 & 9 & 11 \\ 8 & 10 & 12 \end{bmatrix} \quad (2 \times 3), \quad P^T = \begin{bmatrix} 1 & 4 \\ 2 & 5 \\ 3 & 6 \end{bmatrix} \quad (3 \times 2)$$

$$Q^T P^T = \begin{bmatrix} 7 & 9 & 11 \\ 8 & 10 & 12 \end{bmatrix} \begin{bmatrix} 1 & 4 \\ 2 & 5 \\ 3 & 6 \end{bmatrix}$$

$(7)(1) + (9)(2) + (11)(3) = 58, \quad (7)(4) + (9)(5) + (11)(6) = 139$

$(8)(1) + (10)(2) + (12)(3) = 64, \quad (8)(4) + (10)(5) + (12)(6) = 154$

$$Q^T P^T = \begin{bmatrix} 58 & 139 \\ 64 & 154 \end{bmatrix}$$

Confirmed: $(PQ)^T = Q^T P^T = \begin{bmatrix} 58 & 139 \\ 64 & 154 \end{bmatrix}$. ✓

Takeaways 2.2

- **Transpose Reverses Dimensions:** If A is $m \times n$, then A^T is $n \times m$.
- **Shoe–Sock Rule:** The transpose of a product reverses the order: $(AB)^T = B^T A^T$. Think of it like putting on socks then shoes — to reverse, you remove shoes first, then socks.

2.2 Inverses and Determinants for 2×2 Matrices

In standard algebra, the inverse of a number x (where $x \neq 0$) is $\frac{1}{x}$, such that $x \times \frac{1}{x} = 1$. Similarly, in matrix algebra, the **inverse** of a square matrix A, denoted A^{-1}, is the matrix that multiplies with A to yield the identity matrix I.

$$AA^{-1} = A^{-1}A = I$$

The Determinant

The **determinant** of a matrix is a special scalar value that tells us whether the matrix has an inverse. For a 2×2 matrix:

$$A = \begin{bmatrix} a & b \\ c & d \end{bmatrix}$$

The determinant is denoted by $\det(A)$ or $|A|$ and is calculated as:

$$\det(A) = ad - bc$$

The Inverse Formula

If $\det(A) \neq 0$, the inverse of the 2×2 matrix A exists and is given by:

$$A^{-1} = \frac{1}{\det(A)} \begin{bmatrix} d & -b \\ -c & a \end{bmatrix}$$

Notice the pattern for finding the inverse of a 2×2 matrix:

- **Swap** the elements on the main diagonal (a and d).
- **Negate** the elements on the other diagonal (b and c).
- **Multiply** the resulting matrix by the scalar $\frac{1}{\det(A)}$.

Remark 2.2 (Singular Matrices)

If $\det(A) = 0$, then $\frac{1}{\det(A)}$ is undefined. In this case, the matrix A **does not have an inverse** and is called a **singular matrix**. If $\det(A) \neq 0$, the matrix is called **non-singular**.

Problem 2.3: Calculating Inverses and Determinants

Consider the following two matrices:

$$M = \begin{bmatrix} 4 & 1 \\ 2 & 3 \end{bmatrix}, \qquad N = \begin{bmatrix} 6 & -2 \\ -9 & 3 \end{bmatrix}$$

(a) Calculate $\det(M)$. Hence, find M^{-1}.

(b) Verify that $MM^{-1} = I$.

(c) Determine whether matrix N has an inverse.

Solution 2.3

(a)
First, find the determinant of M:
$\det(M) = (4)(3) - (1)(2) = 12 - 2 = 10$.
Since $\det(M) = 10 \neq 0$, the inverse exists.
Apply the 2×2 inverse formula (swap main diagonal, negate the other):
$M^{-1} = \frac{1}{10}\begin{bmatrix} 3 & -1 \\ -2 & 4 \end{bmatrix}$.
(Note: Leaving the scalar $1/10$ at the front is usually preferred unless decimals/fractions are specifically required for further calculation.)

(b)
To verify, calculate the product MM^{-1}:
$MM^{-1} = \begin{bmatrix} 4 & 1 \\ 2 & 3 \end{bmatrix}\left(\frac{1}{10}\begin{bmatrix} 3 & -1 \\ -2 & 4 \end{bmatrix}\right)$
Because scalar multiplication can be pulled to the front, we calculate the matrix product first:
$MM^{-1} = \frac{1}{10}\left(\begin{bmatrix} 4 & 1 \\ 2 & 3 \end{bmatrix}\begin{bmatrix} 3 & -1 \\ -2 & 4 \end{bmatrix}\right)$

$= \frac{1}{10}\begin{bmatrix} (4)(3)+(1)(-2) & (4)(-1)+(1)(4) \\ (2)(3)+(3)(-2) & (2)(-1)+(3)(4) \end{bmatrix}$

$= \frac{1}{10}\begin{bmatrix} 12-2 & -4+4 \\ 6-6 & -2+12 \end{bmatrix} = \frac{1}{10}\begin{bmatrix} 10 & 0 \\ 0 & 10 \end{bmatrix} = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} = I$. ✓

(c)
First, find the determinant of N:
$\det(N) = (6)(3) - (-2)(-9) = 18 - 18 = 0$.
Because $\det(N) = 0$, matrix N is a **singular matrix**. It does not have an inverse.

Takeaways 2.3

- **Determinant First:** Always calculate the determinant before attempting to find the inverse. If it is zero, you can state that the matrix is singular and stop immediately.
- **Scalar Extraction:** When multiplying a matrix by its inverse to check your answer, it is much easier to leave the scalar fraction (like $\frac{1}{10}$) outside the matrix multiplication until the very end.
- **Algebraic vs. Geometric:** A singular matrix ($\det = 0$) cannot be inverted algebraically (as this would require division by zero). Geometrically, this means the transformation squashes 2D space down into a 1D line or a point, losing information that cannot be uniquely reconstructed.

2.3 Simultaneous Linear Equations with Two Variables

Before using matrices to solve equations, it is important to review what a system of simultaneous linear equations actually represents. A linear equation in two variables, such as $ax + by = c$, geometrically represents a straight line on a Cartesian plane.

When we solve two simultaneous linear equations, we are looking for the (x, y) coordinates that satisfy both equations at the same time. Geometrically, this means finding the point(s) where the two lines intersect.

Three Possible Outcomes

For any system of two linear equations:

$$\begin{cases} a_1x + b_1y = c_1 \\ a_2x + b_2y = c_2 \end{cases}$$

There are three possible geometric configurations and algebraic outcomes:

- **Unique Solution:** The lines have different gradients and intersect at exactly **one point**.
- **No Solution:** The lines have the same gradient but different y-intercepts. They are **parallel** and never intersect. Algebraically, solving them yields a contradiction (e.g., $0 = 5$).
- **Infinitely Many Solutions:** The lines have the same gradient and the same y-intercept. They are **coincident** (the exact same line). Algebraically, solving them yields a tautology (e.g., $0 = 0$).

Problem 2.4: Algebraic and Geometric Interpretations

For each of the following systems of equations, solve algebraically using either substitution or elimination, and describe the geometric relationship between the two lines.

(a)

$2x + 3y = 8$
$5x - y = 3$

(b)

$x + 2y = 4$
$2x + 4y = 10$

Solution 2.4

(a)
Let the equations be:

$$2x + 3y = 8 \quad \text{— (1)}$$
$$5x - y = 3 \quad \text{— (2)}$$

Using substitution, rearrange equation (2) to make y the subject:
$y = 5x - 3$
Substitute this into equation (1):

$$2x + 3(5x - 3) = 8$$
$$2x + 15x - 9 = 8$$
$$17x = 17$$
$$x = 1$$

Substitute $x = 1$ back into the rearranged equation (2):
$y = 5(1) - 3 = 2$
Algebraic Solution: $x = 1, y = 2$.
Geometric Relationship: The two lines have a unique solution. They intersect at the single point $(1, 2)$.

(b)
Let the equations be:

$$x + 2y = 4 \quad \text{— (1)}$$
$$2x + 4y = 10 \quad \text{— (2)}$$

Using elimination, multiply equation (1) by 2:

$$2x + 4y = 8 \quad \text{— (3)}$$

Subtract equation (3) from equation (2):

$$(2x + 4y) - (2x + 4y) = 10 - 8$$
$$0 = 2$$

This is a mathematical contradiction (0 can never equal 2).
Algebraic Solution: No solution.
Geometric Relationship: The two equations represent distinct parallel lines. Because they are parallel, they never intersect.

Takeaways 2.4

- **Algebra meets Geometry:** Solving simultaneous equations is identical to finding the intersection points of lines on a graph.
- **Contradictions reveal Parallel Lines:** If your algebraic steps lead to an impossible statement like $0 = 2$, it is not a mistake. It is the mathematical way of stating the lines are parallel.

2.4 Using Matrix Algebra for Systems of Linear Equations in Two Variables

Any system of linear equations can be translated into a single matrix equation. This allows us to use the power of matrix inverses to solve for the variables directly, bypassing manual substitution or elimination.

Forming the Matrix Equation

Consider the general 2×2 system:

$$\begin{cases} ax + by = e \\ cx + dy = f \end{cases}$$

By using the rules of matrix multiplication, we can separate the coefficients, the variables, and the constants into three matrices:

$$\begin{bmatrix} a & b \\ c & d \end{bmatrix} \begin{bmatrix} x \\ y \end{bmatrix} = \begin{bmatrix} e \\ f \end{bmatrix}$$

This is often written compactly as $AX = B$, where:

- A is the **coefficient matrix**.
- X is the **variable matrix** (a column vector).
- B is the **constant matrix** (a column vector).

Solving the Matrix Equation

To isolate the variable matrix X, we cannot "divide" by matrix A. Instead, we multiply both sides of the equation by the inverse matrix, A^{-1}.

Because matrix multiplication is not commutative, we must multiply from the **left** on both sides:

$$\begin{aligned} AX &= B \\ A^{-1}(AX) &= A^{-1}B \\ (A^{-1}A)X &= A^{-1}B \\ IX &= A^{-1}B \\ X &= A^{-1}B \end{aligned}$$

If $\det(A) = 0$, then A^{-1} does not exist. This algebraically confirms the situations discussed in the previous topic: the system has either no unique solution (parallel lines) or infinitely many solutions (coincident lines).

Problem 2.5: Matrix Solution to a 2×2 System

Solve the following system of equations using matrix algebra:

$$\begin{aligned} 2x + 3y &= 8 \\ 5x - y &= 3 \end{aligned}$$

(Note: This is the same system solved algebraically in Topic 3.)

Solution 2.5

Step 1: Write the system as a matrix equation.
Ensure the equations are in standard form (variables on the left, constants on the right). They are.

$$AX = B \implies \begin{bmatrix} 2 & 3 \\ 5 & -1 \end{bmatrix} \begin{bmatrix} x \\ y \end{bmatrix} = \begin{bmatrix} 8 \\ 3 \end{bmatrix}$$

Step 2: Find the determinant of A.
$\det(A) = (2)(-1) - (3)(5) = -2 - 15 = -17$.
Since $\det(A) \neq 0$, a unique solution exists.

Step 3: Find the inverse matrix, A^{-1}.

$$A^{-1} = \frac{1}{-17} \begin{bmatrix} -1 & -3 \\ -5 & 2 \end{bmatrix} = -\frac{1}{17} \begin{bmatrix} -1 & -3 \\ -5 & 2 \end{bmatrix}$$

Step 4: Solve for X using $X = A^{-1}B$.

$$X = -\frac{1}{17} \begin{bmatrix} -1 & -3 \\ -5 & 2 \end{bmatrix} \begin{bmatrix} 8 \\ 3 \end{bmatrix}$$

Calculate the matrix product first:

$$X = -\frac{1}{17} \begin{bmatrix} (-1)(8) + (-3)(3) \\ (-5)(8) + (2)(3) \end{bmatrix}$$

$$X = -\frac{1}{17} \begin{bmatrix} -8 - 9 \\ -40 + 6 \end{bmatrix} = -\frac{1}{17} \begin{bmatrix} -17 \\ -34 \end{bmatrix}$$

Finally, multiply the scalar into the matrix:

$$X = \begin{bmatrix} 1 \\ 2 \end{bmatrix}$$

Conclusion:
Since $X = \begin{bmatrix} x \\ y \end{bmatrix} = \begin{bmatrix} 1 \\ 2 \end{bmatrix}$, the solution is $x = 1, y = 2$.
This precisely matches the geometric point of intersection $(1, 2)$ found previously!

Takeaways 2.5

- **Left-Multiplication is Critical:** You must write $X = A^{-1}B$. Writing $X = BA^{-1}$ is mathematically incorrect because the inner dimensions for multiplication would not match (a 2×1 matrix cannot be multiplied by a 2×2 matrix in that order).
- **Scalar Timing:** As seen in the example, delay multiplying the scalar fraction (like $-\frac{1}{17}$) until the very last step. This avoids messy fraction arithmetic during the matrix multiplication.
- **Determinant Check:** The determinant provides an immediate check. If $\det(A) = 0$, you can stop and state that no unique geometric intersection exists.

2.5 Inverses and Determinants for $n \times n$ Matrices

While the inverse of a 2×2 matrix has a simple formula, extending this to 3×3 matrices (and larger) requires a more systematic process using **minors** and **cofactors**.

Minors and Cofactors

For any element a_{ij} in a square matrix A:

- The **minor**, M_{ij}, is the determinant of the smaller matrix that remains after deleting row i and column j.

- The **cofactor**, C_{ij}, is the minor multiplied by a positive or negative sign, based on its position:

$$C_{ij} = (-1)^{i+j} M_{ij}$$

This creates a "checkerboard" pattern of signs. For a 3×3 matrix:

$$\begin{bmatrix} + & - & + \\ - & + & - \\ + & - & + \end{bmatrix}$$

Evaluating a 3×3 Determinant

The determinant of a 3×3 matrix can be evaluated by **cofactor expansion** along any row or column. You multiply each element in that row (or column) by its corresponding cofactor and add them up.

Expanding along the first row:

$$\det(A) = a_{11}C_{11} + a_{12}C_{12} + a_{13}C_{13}$$

Using the 2×2 determinant formula for the minors, this expands to:

$$\det(A) = a_{11}(a_{22}a_{33} - a_{23}a_{32}) - a_{12}(a_{21}a_{33} - a_{23}a_{31}) + a_{13}(a_{21}a_{32} - a_{22}a_{31})$$

The Inverse of an $n \times n$ Matrix

The formula for the inverse of any non-singular $n \times n$ matrix is:

$$A^{-1} = \frac{1}{\det(A)} \text{adj}(A)$$

Where $\text{adj}(A)$, called the **adjugate matrix**, is the *transpose* of the matrix of cofactors.

Geometric Meaning: Determinants as Area Engines

Determinants are not merely computational recipes for finding inverses; they possess a direct geometric interpretation. In two dimensions, the determinant of a 2×2 matrix gives the signed area of the parallelogram formed by its column vectors.

For any triangle with three vertices $P_1(x_1, y_1)$, $P_2(x_2, y_2)$, and $P_3(x_3, y_3)$ in the Cartesian plane, its area is given by the 3×3 determinant:

$$\text{Area}(\triangle P_1P_2P_3) = \frac{1}{2} \left| \det \begin{bmatrix} x_1 & y_1 & 1 \\ x_2 & y_2 & 1 \\ x_3 & y_3 & 1 \end{bmatrix} \right|.$$

We can understand why this formula works using elementary row operations. Subtracting the first row from the second and third rows (which geometrically translates the triangle so that vertex P_1 moves to the origin $(0, 0)$):

$$\det \begin{bmatrix} x_1 & y_1 & 1 \\ x_2 & y_2 & 1 \\ x_3 & y_3 & 1 \end{bmatrix} = \det \begin{bmatrix} x_1 & y_1 & 1 \\ x_2 - x_1 & y_2 - y_1 & 0 \\ x_3 - x_1 & y_3 - y_1 & 0 \end{bmatrix}.$$

Expanding along the third column, the 3×3 determinant collapses immediately into a 2×2 determinant:

$$\text{Area} = \frac{1}{2} \left| (x_2 - x_1)(y_3 - y_1) - (x_3 - x_1)(y_2 - y_1) \right|.$$

If one vertex is already at the origin $O(0,0)$, the formula simplifies directly to:

$$\text{Area}(\triangle OP_2P_3) = \frac{1}{2}|x_2y_3 - x_3y_2| = \frac{1}{2}\left|\det\begin{bmatrix} x_2 & y_2 \\ x_3 & y_3 \end{bmatrix}\right|.$$

Remark 2.3 (Collinearity Criterion)

Three points P_1, P_2, P_3 are **collinear** (lie on the same straight line) if and only if the area of the triangle they form is zero:

$$\det\begin{bmatrix} x_1 & y_1 & 1 \\ x_2 & y_2 & 1 \\ x_3 & y_3 & 1 \end{bmatrix} = 0.$$

Remark 2.4 (Concurrence Criterion for Three Lines)

Historically, the 1965 NSW First Level syllabus Section 4(a) explicitly paired the triangle area formula with the determinant condition for the concurrence of three straight lines. Three distinct lines in the Cartesian plane:

$$L_1 : a_1x + b_1y + c_1 = 0, \qquad L_2 : a_2x + b_2y + c_2 = 0, \qquad L_3 : a_3x + b_3y + c_3 = 0$$

are **concurrent** (pass through a common single point) if and only if the determinant of their coefficients vanishes:

$$\det\begin{bmatrix} a_1 & b_1 & c_1 \\ a_2 & b_2 & c_2 \\ a_3 & b_3 & c_3 \end{bmatrix} = 0.$$

This highlights a remarkable projective duality in coordinate geometry:

- **Three points are collinear** $\iff$ the coordinate matrix has determinant zero.
- **Three lines are concurrent** $\iff$ the line coefficient matrix has determinant zero.

Problem 2.6: Inverting a 3×3 Matrix

Let $A = \begin{bmatrix} 1 & 2 & -1 \\ 0 & 3 & 1 \\ 2 & 0 & -2 \end{bmatrix}$.

(a) Evaluate $\det(A)$ by expanding along the first row.

(b) Find the matrix of cofactors, C.

(c) Find the adjugate matrix, $\text{adj}(A)$.

(d) State the inverse matrix, A^{-1}.

Solution 2.6

(a) Determinant

Expanding along the first row:

$$\begin{aligned}\det(A) &= 1\begin{vmatrix}3 & 1\\ 0 & -2\end{vmatrix} - 2\begin{vmatrix}0 & 1\\ 2 & -2\end{vmatrix} + (-1)\begin{vmatrix}0 & 3\\ 2 & 0\end{vmatrix}\\ &= 1((3)(-2)-(1)(0)) - 2((0)(-2)-(1)(2)) - 1((0)(0)-(3)(2))\\ &= 1(-6) - 2(-2) - 1(-6)\\ &= -6+4+6 = 4\end{aligned}$$

Since $\det(A) = 4 \neq 0$, the inverse exists.

(b) Matrix of Cofactors

We calculate the cofactor for all 9 elements. Remember to apply the checkerboard signs!

$$\begin{aligned}C_{11} &= +((3)(-2)-(1)(0)) = -6\\ C_{12} &= -((0)(-2)-(1)(2)) = -(-2) = 2\\ C_{13} &= +((0)(0)-(3)(2)) = -6\end{aligned}$$

$$\begin{aligned}C_{21} &= -((2)(-2)-(-1)(0)) = -(-4) = 4\\ C_{22} &= +((1)(-2)-(-1)(2)) = -2+2 = 0\\ C_{23} &= -((1)(0)-(2)(2)) = -(-4) = 4\end{aligned}$$

$$\begin{aligned}C_{31} &= +((2)(1)-(-1)(3)) = 2+3 = 5\\ C_{32} &= -((1)(1)-(-1)(0)) = -(1) = -1\\ C_{33} &= +((1)(3)-(2)(0)) = 3\end{aligned}$$

The cofactor matrix is:

$$C = \begin{bmatrix}-6 & 2 & -6\\ 4 & 0 & 4\\ 5 & -1 & 3\end{bmatrix}$$

(c) Adjugate Matrix

The adjugate is the transpose of the cofactor matrix (C^T):

$$\text{adj}(A) = \begin{bmatrix}-6 & 4 & 5\\ 2 & 0 & -1\\ -6 & 4 & 3\end{bmatrix}$$

(d) Inverse Matrix

Using the formula $A^{-1} = \frac{1}{\det(A)}\text{adj}(A)$:

$$A^{-1} = \frac{1}{4}\begin{bmatrix}-6 & 4 & 5\\ 2 & 0 & -1\\ -6 & 4 & 3\end{bmatrix}$$

Takeaways 2.6

- **Row Choice for Determinants:** You can expand along *any* row or column to find the determinant. For speed, always choose the row or column containing the most zeros.
- **Don't Forget the Transpose:** A common error is writing the cofactor matrix directly into the inverse formula. You must take the transpose of the cofactor matrix to get the adjugate.
- **Sign Errors:** Calculating a 3×3 inverse manually involves many small multiplications and subtractions. Write out your working explicitly to avoid dropping minus signs.
- **Geometric Area Engine:** The determinant of a 3×3 matrix with a column of 1s computes the area of any triangle in the plane, bypassing the need for distance and perpendicular height formulas.

2.6 Simultaneous Linear Equations with More Than Two Variables

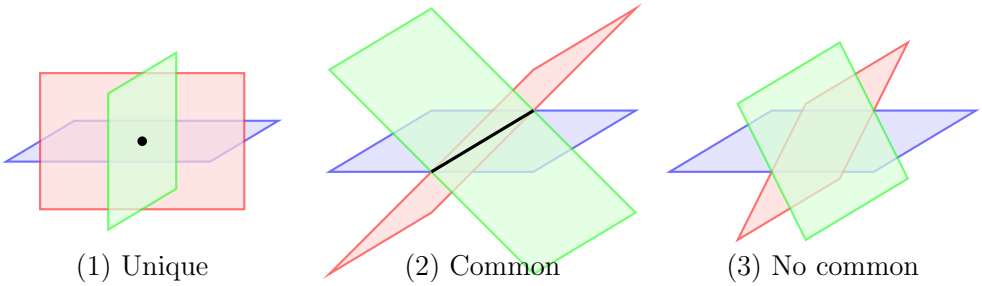
When we move from two variables (x, y) to three variables (x, y, z), the geometric interpretation expands from two-dimensional space to three-dimensional space.

A linear equation in two variables $(ax + by = c)$ represents a **line**.
A linear equation in three variables $(ax + by + cz = d)$ represents a **flat plane** in 3D space.

Geometric Interpretations in 3D

When solving a system of three linear equations in three variables, we are looking for the point(s) in 3D space where all three planes intersect. There are three main categories of outcomes:

- **Unique Solution:** The three planes meet at a single, distinct point.
- **Infinitely Many Solutions:** The planes intersect along a common line (like the pages of a book meeting at the spine), or all three equations represent the exact same plane.
- **No Solution:** There is no single point that belongs to all three planes. For example, two planes might be parallel, or they might form a triangular prism (like a tent) where they intersect in pairs but never all three together.



(1) Unique common point (2) Common intersection line (3) No common intersection

Setting Up Systems

To find a unique solution for n unknown variables, you generally need n independent linear equations. A common application is translating real-world scenarios into a 3×3 system of equations.

Problem 2.7: Translating a Word Problem into a System

A local bakery sells three types of cakes: chocolate, vanilla, and strawberry. The bakery records its sales and revenue for three consecutive days:

- **Monday:** 2 chocolate, 3 vanilla, and 1 strawberry cake were sold for a total of \$115.
- **Tuesday:** 1 chocolate, 2 vanilla, and 2 strawberry cakes were sold for a total of \$110.
- **Wednesday:** 3 chocolate, 1 vanilla, and 3 strawberry cakes were sold for a total of \$165.

Assuming the price of each cake type remains constant, define appropriate variables and set up a system of three simultaneous linear equations that models this situation.

Solution 2.7

Step 1: Define the variables.
Let x = the price of a chocolate cake (in \$).
Let y = the price of a vanilla cake (in \$).
Let z = the price of a strawberry cake (in \$).

Step 2: Translate each day's sales into an equation.
Monday's sales (2 choc, 3 van, 1 straw yield \$115):

$$2x + 3y + z = 115$$

Tuesday's sales (1 choc, 2 van, 2 straw yield \$110):

$$x + 2y + 2z = 110$$

Wednesday's sales (3 choc, 1 van, 3 straw yield \$165):

$$3x + y + 3z = 165$$

Step 3: State the final system.
The system of simultaneous linear equations is:

$$\begin{aligned} 2x + 3y + z &= 115 \\ x + 2y + 2z &= 110 \\ 3x + y + 3z &= 165 \end{aligned}$$

(Note: We will solve this exact system using matrices in the next topic!)

Takeaways 2.7

- **Geometry of 3 Variables:** Always visualize 3×3 systems as three flat planes in 3D space. When a unique solution exists, it is the common intersection point.
- **Information Requirement:** Solving for 3 unknowns uniquely requires 3 independent pieces of information (equations). If you only have two equations, two independent consistent planes normally intersect in a line; parallel planes have no common points and coincident planes give a plane of solutions.
- **Careful Translation:** When reading word problems, ensure the coefficients (quantities) align correctly with their corresponding variables (prices) in each equation.

2.7 Using Matrix Algebra for Systems of Linear Equations in More Than Two Variables

The elegance of matrix algebra is that for a square system with an invertible coefficient matrix, the fundamental equation used to solve it, $X = A^{-1}B$, remains exactly the same regardless of whether you are solving a 2×2 system, a 3×3 system, or a 100×100 system.

The only difference is the computational effort required to find the inverse matrix, A^{-1}.

Problem 2.8: Matrix Solution to a 3×3 System

Recall the bakery word problem from the previous topic, which gave us the following system:

$$\begin{aligned} 2x + 3y + z &= 115 \\ x + 2y + 2z &= 110 \\ 3x + y + 3z &= 165 \end{aligned}$$

Use matrix algebra to solve for x, y, and z, finding the exact price of each cake.

Solution 2.8

Step 1: Write the system in the form $AX = B$.

$$\begin{bmatrix} 2 & 3 & 1 \\ 1 & 2 & 2 \\ 3 & 1 & 3 \end{bmatrix} \begin{bmatrix} x \\ y \\ z \end{bmatrix} = \begin{bmatrix} 115 \\ 110 \\ 165 \end{bmatrix}$$

Where A is the 3×3 coefficient matrix, and B is the constants matrix.

Step 2: Find the determinant of A.
Expanding along the first row:

$$\begin{aligned} \det(A) &= 2\begin{vmatrix} 2 & 2 \\ 1 & 3 \end{vmatrix} - 3\begin{vmatrix} 1 & 2 \\ 3 & 3 \end{vmatrix} + 1\begin{vmatrix} 1 & 2 \\ 3 & 1 \end{vmatrix} \\ &= 2(6-2) - 3(3-6) + 1(1-6) \\ &= 2(4) - 3(-3) + 1(-5) \\ &= 8 + 9 - 5 = 12 \end{aligned}$$

Since $\det(A) = 12 \neq 0$, a unique solution exists.

Step 3: Find the matrix of cofactors, C.
Calculating the 9 cofactors (with the checkerboard sign pattern):

$$\begin{aligned} C_{11} &= +(6-2) = 4, & C_{12} &= -(3-6) = 3, & C_{13} &= +(1-6) = -5 \\ C_{21} &= -(9-1) = -8, & C_{22} &= +(6-3) = 3, & C_{23} &= -(2-9) = 7 \\ C_{31} &= +(6-2) = 4, & C_{32} &= -(4-1) = -3, & C_{33} &= +(4-3) = 1 \end{aligned}$$

So, the cofactor matrix is $C = \begin{bmatrix} 4 & 3 & -5 \\ -8 & 3 & 7 \\ 4 & -3 & 1 \end{bmatrix}$.

Step 4: Find the adjugate and inverse of A.
Transpose C to get the adjugate:

$$\text{adj}(A) = C^T = \begin{bmatrix} 4 & -8 & 4 \\ 3 & 3 & -3 \\ -5 & 7 & 1 \end{bmatrix}$$

The inverse is $A^{-1} = \frac{1}{\det(A)}\text{adj}(A) = \frac{1}{12}\begin{bmatrix} 4 & -8 & 4 \\ 3 & 3 & -3 \\ -5 & 7 & 1 \end{bmatrix}$.

Step 5: Solve for X using $X = A^{-1}B$.

$$X = \frac{1}{12}\begin{bmatrix} 4 & -8 & 4 \\ 3 & 3 & -3 \\ -5 & 7 & 1 \end{bmatrix}\begin{bmatrix} 115 \\ 110 \\ 165 \end{bmatrix}$$

Calculate the matrix product first:

$$X = \frac{1}{12}\begin{bmatrix} 4(115) + (-8)(110) + 4(165) \\ 3(115) + 3(110) + (-3)(165) \\ -5(115) + 7(110) + 1(165) \end{bmatrix}$$

$$X = \frac{1}{12}\begin{bmatrix} 460 - 880 + 660 \\ 345 + 330 - 495 \\ -575 + 770 + 165 \end{bmatrix} = \frac{1}{12}\begin{bmatrix} 240 \\ 180 \\ 360 \end{bmatrix} = \begin{bmatrix} 20 \\ 15 \\ 30 \end{bmatrix}$$

Conclusion:
The variables are $x = 20, y = 15, z = 30$.
The prices are \$20 for a chocolate cake, \$15 for a vanilla cake, and \$30 for a strawberry cake.

Takeaways 2.8

- **Universal Logic:** For a square system with an invertible coefficient matrix, the framework $X = A^{-1}B$ scales to any number of dimensions. Only the mechanics of finding A^{-1} change.
- **Computational Load:** As systems get larger (e.g., 4×4), finding inverses by hand becomes highly prone to arithmetic errors. This is why computers and graphics cards are used to solve these systems in real life!
- **Checking your work:** You can easily verify the final solution by substituting the values back into any of the original equations (e.g., $2(20)+3(15)+30 = 40+45+30 = 115$, which matches Monday's total).

2.8 Using Augmented Matrices for Systems of Equations

Finding the inverse of a 3×3 or larger matrix is computationally heavy. Furthermore, if a system has no solutions or infinitely many solutions, the inverse method fails because $\det(A) = 0$.

A more robust and efficient method is **Gaussian elimination**, which uses **augmented matrices** and **elementary row operations**.

The Augmented Matrix

Instead of writing three separate matrices (A, X, and B), we combine the coefficient matrix A and the constant matrix B into a single matrix, separated by a vertical line. For a 3×3 system, it looks like this:

$$[A \,|\, B] = \left[\begin{array}{ccc|c} a_{11} & a_{12} & a_{13} & b_1 \\ a_{21} & a_{22} & a_{23} & b_2 \\ a_{31} & a_{32} & a_{33} & b_3 \end{array}\right]$$

Elementary Row Operations

Because each row represents an algebraic equation, we can perform operations on the rows without changing the solution to the system. There are three valid **elementary row operations**:

1. **Swap:** Interchange any two rows (e.g., $R_1 \leftrightarrow R_2$).
2. **Scale:** Multiply an entire row by a non-zero scalar (e.g., $R_2 \to \frac{1}{2}R_2$).
3. **Add/Subtract:** Add a multiple of one row to another row (e.g., $R_3 \to R_3 - 2R_1$).

Row Echelon Form (REF)

The goal of Gaussian elimination is to use these operations to transform the augmented matrix into **Row Echelon Form**. In REF, the bottom-left corner of the matrix is filled with zeros, creating a "staircase" pattern:

$$\left[\begin{array}{ccc|c} 1 & * & * & * \\ 0 & 1 & * & * \\ 0 & 0 & 1 & * \end{array}\right]$$

Once in this form, the system can be easily solved from the bottom up using **back-substitution**.

Problem 2.9: Gaussian Elimination

Solve the following system of linear equations using an augmented matrix and row reduction:

$$\begin{aligned} x + y + z &= 6 \\ 2x - y + 2z &= 6 \\ x + 2y - z &= 2 \end{aligned}$$

Solution 2.9

Step 1: Write the augmented matrix.

$$\left[\begin{array}{ccc|c} 1 & 1 & 1 & 6 \\ 2 & -1 & 2 & 6 \\ 1 & 2 & -1 & 2 \end{array}\right]$$

Step 2: Eliminate the x-coefficients in Rows 2 and 3.
To get a zero in R_2, column 1, we subtract $2 \times R_1$ from R_2 ($R_2 \to R_2 - 2R_1$):

$$\left[\begin{array}{ccc|c} 1 & 1 & 1 & 6 \\ 0 & -3 & 0 & -6 \\ 1 & 2 & -1 & 2 \end{array}\right]$$

To get a zero in R_3, column 1, we subtract R_1 from R_3 ($R_3 \to R_3 - R_1$):

$$\left[\begin{array}{ccc|c} 1 & 1 & 1 & 6 \\ 0 & -3 & 0 & -6 \\ 0 & 1 & -2 & -4 \end{array}\right]$$

Step 3: Simplify and eliminate the y-coefficient in Row 3.
Let's simplify R_2 by dividing by -3 ($R_2 \to -\frac{1}{3}R_2$):

$$\left[\begin{array}{ccc|c} 1 & 1 & 1 & 6 \\ 0 & 1 & 0 & 2 \\ 0 & 1 & -2 & -4 \end{array}\right]$$

Now, eliminate the y-coefficient in R_3 by subtracting R_2 ($R_3 \to R_3 - R_2$):

$$\left[\begin{array}{ccc|c} 1 & 1 & 1 & 6 \\ 0 & 1 & 0 & 2 \\ 0 & 0 & -2 & -6 \end{array}\right]$$

Step 4: Finalize Row Echelon Form and Back-Substitute.
Divide R_3 by -2 ($R_3 \to -\frac{1}{2}R_3$):

$$\left[\begin{array}{ccc|c} 1 & 1 & 1 & 6 \\ 0 & 1 & 0 & 2 \\ 0 & 0 & 1 & 3 \end{array}\right]$$

Translate the matrix back into equations:

$$\begin{aligned} x + y + z &= 6 \quad \text{— (1)} \\ y &= 2 \quad \text{— (2)} \\ z &= 3 \quad \text{— (3)} \end{aligned}$$

We already have $y = 2$ and $z = 3$. Substitute these back into equation (1):

$$\begin{aligned} x + (2) + (3) &= 6 \\ x + 5 &= 6 \\ x &= 1 \end{aligned}$$

The unique solution is $x = 1, y = 2, z = 3$.

Takeaways 2.9

- **Computational Efficiency:** Gaussian elimination requires far fewer calculations than finding a 3×3 inverse, making it the preferred method for solving systems by hand.
- **Detecting Anomalies:** If a system has no solution, row reduction will eventually produce an impossible row like $\left[\begin{array}{ccc|c} 0 & 0 & 0 & 5 \end{array}\right]$, which algebraically means $0 = 5$.
- **Infinite Solutions:** If a system has infinite solutions, row reduction will produce a redundant row like $\left[\begin{array}{ccc|c} 0 & 0 & 0 & 0 \end{array}\right]$, indicating one equation was just a multiple of another.

2.9 Matrices as Linear Transformations

One of the most important applications of matrices is in representing geometric transformations. While you may be familiar with column vectors as representing points or translations, multiplying a vector by a matrix can scale, reflect, rotate, or shear that vector.

A linear transformation can be thought of as a function that maps an input vector $\mathbf{v} = \begin{bmatrix} x \\ y \end{bmatrix}$ to an output vector $\mathbf{v}' = \begin{bmatrix} x' \\ y' \end{bmatrix}$ via matrix multiplication:

$$\mathbf{v}' = A\mathbf{v} \quad \Longrightarrow \quad \begin{bmatrix} x' \\ y' \end{bmatrix} = \begin{bmatrix} a & b \\ c & d \end{bmatrix} \begin{bmatrix} x \\ y \end{bmatrix}$$

Common 2D Transformations

Here are standard matrices for common geometric transformations in the 2D Cartesian plane:

Transform	Matrix	Geometric Effect
Scaling	$\begin{bmatrix} k_x & 0 \\ 0 & k_y \end{bmatrix}$	Enlarges or shrinks shapes by factor k_x in x-dir, k_y in y-dir.
Reflection	$\begin{bmatrix} \cos 2\alpha & \sin 2\alpha \\ \sin 2\alpha & -\cos 2\alpha \end{bmatrix}$	Reflects across a line through the origin making angle α with the positive x-axis ($y = x \tan \alpha$).
Rotation	$\begin{bmatrix} \cos\theta & -\sin\theta \\ \sin\theta & \cos\theta \end{bmatrix}$	Rotates counter-clockwise by angle θ around the origin.
Shear	$\begin{bmatrix} 1 & k \\ 0 & 1 \end{bmatrix}$ (horizontal) $\begin{bmatrix} 1 & 0 \\ k & 1 \end{bmatrix}$ (vertical)	Slants the shape along the axis by a factor k.

Composition of Transformations and Inverses

If we apply a transformation A and then a transformation B, this is equivalent to a single transformation given by the matrix product BA. Notice the order: $B(A\mathbf{v}) = (BA)\mathbf{v}$. Because

matrix multiplication is generally **not commutative** ($AB \neq BA$), the order of transformations matters. For example, rotating an object and then reflecting it is usually different from reflecting it and then rotating it.

Furthermore, if a matrix A represents a transformation, its inverse A^{-1} (if it exists) represents **undoing** that transformation. The absolute value $|\det(A)|$ gives the area scale factor; the sign indicates orientation. If $\det(A) = 0$, the transformation collapses the 2D plane into a line or a point, making it non-invertible.

Remark 2.5 (NSW HSC Heritage: Reflections and Rotations (1975–1980))

A celebrated theorem examined in the NSW HSC 4 Unit examinations (notably 1975 Question 6 and 1976 Question 9) states that **the product of any two reflection matrices is a rotation matrix**.
Let R_α and R_β be reflection matrices across lines inclined at angles α and β respectively:

$$R_\beta R_\alpha = \begin{bmatrix} \cos 2\beta & \sin 2\beta \\ \sin 2\beta & -\cos 2\beta \end{bmatrix} \begin{bmatrix} \cos 2\alpha & \sin 2\alpha \\ \sin 2\alpha & -\cos 2\alpha \end{bmatrix} = \begin{bmatrix} \cos 2(\beta-\alpha) & -\sin 2(\beta-\alpha) \\ \sin 2(\beta-\alpha) & \cos 2(\beta-\alpha) \end{bmatrix}.$$

The composite transformation is a rotation through an angle $\theta = 2(\beta-\alpha)$, twice the angle between the two lines of reflection! Notice also that $\det(R_\alpha) = -\cos^2 2\alpha - \sin^2 2\alpha = -1$ (reversing orientation), while $\det(R_\beta R_\alpha) = (-1)(-1) = +1$ (preserving orientation), confirming geometrically why two reflections yield a rotation.

Problem 2.10: Transformation Composition

Let point P have coordinates $(2, 3)$.

(a) Find the coordinates of P' after a counter-clockwise rotation of $90°$ around the origin.

(b) Find the matrix that represents a reflection across the x-axis, followed by a counter-clockwise rotation of $90°$.

(c) Use this combined matrix to transform point P.

Solution 2.10

(a) Single Transformation

The rotation matrix for $\theta = 90°$ is $R_{90°} = \begin{bmatrix} \cos 90° & -\sin 90° \\ \sin 90° & \cos 90° \end{bmatrix} = \begin{bmatrix} 0 & -1 \\ 1 & 0 \end{bmatrix}$. Applying this to $P = \begin{bmatrix} 2 \\ 3 \end{bmatrix}$:

$$P' = \begin{bmatrix} 0 & -1 \\ 1 & 0 \end{bmatrix} \begin{bmatrix} 2 \\ 3 \end{bmatrix} = \begin{bmatrix} -3 \\ 2 \end{bmatrix}$$

So P' is at $(-3, 2)$.

(b) Composition Matrix

Let M_1 be the reflection across the x-axis: $M_1 = \begin{bmatrix} 1 & 0 \\ 0 & -1 \end{bmatrix}$.

Let M_2 be the 90° rotation: $M_2 = \begin{bmatrix} 0 & -1 \\ 1 & 0 \end{bmatrix}$.

The combined transformation is $M = M_2 M_1$ (since M_1 acts first):

$$M = \begin{bmatrix} 0 & -1 \\ 1 & 0 \end{bmatrix} \begin{bmatrix} 1 & 0 \\ 0 & -1 \end{bmatrix} = \begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix}$$

Notice that this resulting matrix corresponds to a reflection across the line $y = x$.

(c) Applying the Combined Matrix

Applying M to P:

$$P'' = \begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix} \begin{bmatrix} 2 \\ 3 \end{bmatrix} = \begin{bmatrix} 3 \\ 2 \end{bmatrix}$$

So the final coordinate is $(3, 2)$.

Takeaways 2.10

- **Order Matters:** When chaining geometric transformations, you multiply the matrices from right to left.
- **Geometric Meaning of Determinant:** If you apply the transformation matrix M from part (b), $\det(M) = (0)(0) - (1)(1) = -1$. The absolute value of 1 means the area is preserved, and the negative sign indicates that the orientation (e.g., clockwise vs counter-clockwise ordering of vertices) has been reversed, typical of a reflection.

2.10 Matrix Powers, Iteration and Dynamical Systems

We have seen how matrices can transform vectors. A surprising number of mathematical and real-world models rely on applying a transformation repeatedly to see how a system evolves over time.

If we have an initial state represented by a vector x_0, applying a matrix transformation A produces the next state:

$$x_1 = Ax_0$$

If we apply the same transformation again to get the subsequent state:

$$x_2 = Ax_1 = A(Ax_0) = A^2 x_0$$

This pattern continues. The state after n steps is given by explicitly calculating the n-th power of the matrix A:

$$x_n = A^n x_0$$

This relationship $x_{n+1} = Ax_n \implies x_n = A^n x_0$ is the foundational idea behind **dynamical systems**. It forms a conceptual bridge between basic matrix multiplication and advanced applications.

Applications of Matrix Iteration

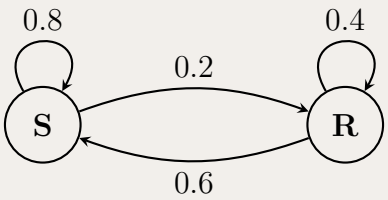
By studying the long-term behaviour of A^n as n becomes very large ($n \to \infty$), we can answer questions about the stability and equilibrium of systems. For example:

- **Probability and Markov Chains:** Tracking the probabilities of states transitioning over time.
- **Graph Theory:** Finding the number of pathways of length n between nodes in a network or computing total dominance scores (e.g., $A + A^2$).
- **Population Models:** Projecting future age demographics (like the Leslie matrix model).
- **Fibonacci and Recurrence Relations:** Expressing sequential terms compactly to compute the n-th term quickly.

Problem 2.11: Iterating a System

Suppose the weather in a city is either Sunny (S) or Rainy (R) on a given day. The probability of tomorrow's weather depends only on today's weather, given by the transition rules:

- If today is Sunny, tomorrow has a 80% chance of being Sunny and 20% chance of being Rainy.
- If today is Rainy, tomorrow has a 60% chance of being Sunny and 40% chance of being Rainy.



Let $x_n = \begin{bmatrix} P(\text{Sunny on day } n) \\ P(\text{Rainy on day } n) \end{bmatrix}$.

(a) Construct the transition matrix A such that $x_{n+1} = Ax_n$.

(b) If Monday (day 0) is definitely Sunny, so $x_0 = \begin{bmatrix} 1 \\ 0 \end{bmatrix}$, find the probabilities for Wednesday (day 2) by calculating x_2.

(c) Calculate A^2 directly, and verify that $x_2 = A^2 x_0$.

Solution 2.11

(a) Transition Matrix
We place the probabilities of transitioning *to* a state in the rows and *from* a state in the columns:

$$A = \begin{matrix} \\ \text{From } S \\ \text{From } R \end{matrix} \begin{matrix} \begin{matrix} \text{To } S & \text{To } R \end{matrix} \\ \begin{bmatrix} 0.8 & 0.6 \\ 0.2 & 0.4 \end{bmatrix} \end{matrix}$$

So $x_{n+1} = \begin{bmatrix} 0.8 & 0.6 \\ 0.2 & 0.4 \end{bmatrix} x_n$.

(b) Finding x_2 iteratively
First, find x_1 (Tuesday):

$$x_1 = Ax_0 = \begin{bmatrix} 0.8 & 0.6 \\ 0.2 & 0.4 \end{bmatrix} \begin{bmatrix} 1 \\ 0 \end{bmatrix} = \begin{bmatrix} 0.8 \\ 0.2 \end{bmatrix}$$

Next, find x_2 (Wednesday):

$$x_2 = Ax_1 = \begin{bmatrix} 0.8 & 0.6 \\ 0.2 & 0.4 \end{bmatrix} \begin{bmatrix} 0.8 \\ 0.2 \end{bmatrix} = \begin{bmatrix} 0.64 + 0.12 \\ 0.16 + 0.08 \end{bmatrix} = \begin{bmatrix} 0.76 \\ 0.24 \end{bmatrix}$$

There is a 76% chance of sun and 24% chance of rain on Wednesday.

(c) Direct calculation via A^2

$$A^2 = \begin{bmatrix} 0.8 & 0.6 \\ 0.2 & 0.4 \end{bmatrix} \begin{bmatrix} 0.8 & 0.6 \\ 0.2 & 0.4 \end{bmatrix} = \begin{bmatrix} 0.64 + 0.12 & 0.48 + 0.24 \\ 0.16 + 0.08 & 0.12 + 0.16 \end{bmatrix} = \begin{bmatrix} 0.76 & 0.72 \\ 0.24 & 0.28 \end{bmatrix}$$

Now calculate A^2x_0:

$$A^2x_0 = \begin{bmatrix} 0.76 & 0.72 \\ 0.24 & 0.28 \end{bmatrix} \begin{bmatrix} 1 \\ 0 \end{bmatrix} = \begin{bmatrix} 0.76 \\ 0.24 \end{bmatrix}$$

This perfectly matches our iterative result.

Takeaways 2.11

- **Efficiency:** For small n, iterating step-by-step (x_1, x_2) is easy. But for large n, calculating A^n (often using computer software or advanced techniques like diagonalisation) and directly applying $x_n = A^n x_0$ is vastly more efficient.
- **Long-Term Behaviour:** Repeatedly squaring a matrix (e.g., $A \to A^2 \to A^4 \to A^8$) is a quick way to see where a system eventually settles, known as the steady-state or equilibrium.

2.11 Dominance Matrices

Matrices are not only used for solving algebraic equations; they are incredibly powerful tools in discrete mathematics and network theory. A **dominance matrix** is used to represent and analyze directed relationships, such as the outcomes of a round-robin sports tournament or a "pecking order" in a group of animals.

Constructing the Dominance Matrix

If a group of n competitors play each other exactly once (no draws allowed), the results can be stored in an $n \times n$ matrix D:

- $d_{ij} = 1$ if competitor i defeated competitor j.
- $d_{ij} = 0$ if competitor i lost to competitor j.

The main diagonal is always entirely zeros, as a competitor cannot play themselves.

Direct and Indirect Wins

- **One-step dominance (Direct wins):** The sum of the elements in row i of matrix D gives the total number of matches won by competitor i. However, this often results in ties.
- **Two-step dominance (Indirect wins):** If Team A beats Team B, and Team B beats Team C, Team A has a two-step dominance over Team C. Mathematically, this is found by squaring the matrix to get D^2. This rewards teams for beating *strong* opponents.
- **Total Dominance Score:** To break ties, we calculate a power score matrix $T = D + D^2$. The row sums of T provide a much more robust ranking system.

Problem 2.12: Tournament Rankings

Four teams (A, B, C, and D) compete in a round-robin tournament. The results are as follows:

- Team A defeated Teams B and C.
- Team B defeated Teams C and D.
- Team C defeated Team D.
- Team D defeated Team A.

(a) Construct the 4×4 dominance matrix D. Calculate the row sums to find the direct rankings and identify any ties.

(b) Calculate D^2, the two-step dominance matrix.

(c) Calculate the total dominance matrix $T = D + D^2$, and use its row sums to determine the final, tie-broken ranking of the four teams.

Solution 2.12

(a) Matrix D and Direct Rankings
Following the rules, we place a 1 in row i, column j if the row team beat the column team.

$$D = \begin{bmatrix} 0 & 1 & 1 & 0 \\ 0 & 0 & 1 & 1 \\ 0 & 0 & 0 & 1 \\ 1 & 0 & 0 & 0 \end{bmatrix}$$

Row Sums (Direct Wins):

- Team A sum $= 2$
- Team B sum $= 2$
- Team C sum $= 1$
- Team D sum $= 1$

We have a tie for 1st place (A and B) and a tie for 3rd place (C and D).

(b) Matrix D^2 (Indirect Wins)
Multiply D by itself. For example, to find Row 1, Col 4 of D^2, we calculate $(0)(0)+(1)(1)+(1)(1)+(0)(0) = 2$. This means Team A has two indirect pathways to beat Team D (via B and via C).

$$D^2 = D \times D = \begin{bmatrix} 0 & 1 & 1 & 0 \\ 0 & 0 & 1 & 1 \\ 0 & 0 & 0 & 1 \\ 1 & 0 & 0 & 0 \end{bmatrix} \begin{bmatrix} 0 & 1 & 1 & 0 \\ 0 & 0 & 1 & 1 \\ 0 & 0 & 0 & 1 \\ 1 & 0 & 0 & 0 \end{bmatrix} = \begin{bmatrix} 0 & 0 & 1 & 2 \\ 1 & 0 & 0 & 1 \\ 1 & 0 & 0 & 0 \\ 0 & 1 & 1 & 0 \end{bmatrix}$$

(c) Total Dominance and Final Ranking
Calculate $T = D + D^2$ by adding the corresponding elements:

$$T = \begin{bmatrix} 0 & 1 & 1 & 0 \\ 0 & 0 & 1 & 1 \\ 0 & 0 & 0 & 1 \\ 1 & 0 & 0 & 0 \end{bmatrix} + \begin{bmatrix} 0 & 0 & 1 & 2 \\ 1 & 0 & 0 & 1 \\ 1 & 0 & 0 & 0 \\ 0 & 1 & 1 & 0 \end{bmatrix} = \begin{bmatrix} 0 & 1 & 2 & 2 \\ 1 & 0 & 1 & 2 \\ 1 & 0 & 0 & 1 \\ 1 & 1 & 1 & 0 \end{bmatrix}$$

Row Sums of T:

- Team A $= 0+1+2+2 = \mathbf{5}$
- Team B $= 1+0+1+2 = \mathbf{4}$
- Team D $= 1+1+1+0 = \mathbf{3}$
- Team C $= 1+0+0+1 = \mathbf{2}$

Final Ranking: 1st: Team A, 2nd: Team B, 3rd: Team D, 4th: Team C.
The ties have been successfully broken!

Takeaways 2.12

- **Quality of Wins:** Team A and B both had 2 direct wins. However, Team A beat Team B (a strong team), giving A more indirect pathways to dominance. The $D + D^2$ method algebraically measures the "quality" of an opponent.
- **Networks and Graphs:** Dominance matrices are adjacency matrices for directed graphs. Squaring the matrix algebraically counts the number of 2-step paths between nodes in a network.
- **Real-World Algorithms:** This logic is the foundational mathematics behind modern search engines (like Google's PageRank algorithm), which rank web pages based on the quantity and quality of incoming links.

2.12 Leslie Matrices

Matrices can model dynamic, real-world systems that change over time. In biology and ecology, the **Leslie matrix** is a standard model used to predict the population growth of organisms by breaking the population down into distinct age groups (classes). Because females typically drive population reproduction rates, Leslie matrices generally only track the female portion of a population.

The Structure of a Leslie Matrix

Suppose a population is divided into n age classes of equal time duration.

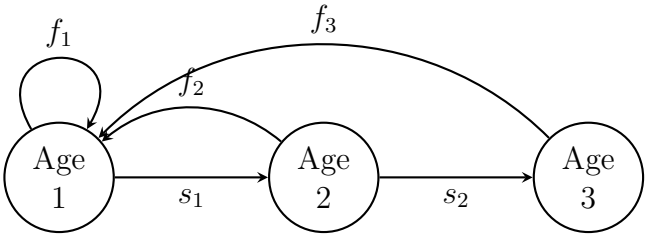
- Let f_i be the **fecundity (fertility) rate**: the average number of female offspring born to a female in age class i.
- Let s_i be the **survival rate**: the probability that a female in age class i survives to enter age class $i+1$.

The Leslie matrix L is a square $n \times n$ matrix built with a very specific layout:

- The **first row** contains the fecundity rates $(f_1, f_2, \ldots, f_n)$.
- The **sub-diagonal** (the diagonal immediately below the main diagonal) contains the survival rates $(s_1, s_2, \ldots, s_{n-1})$.
- All other entries are exactly zero.

For a 3-age-class population, the matrix looks like this:

$$L = \begin{bmatrix} f_1 & f_2 & f_3 \\ s_1 & 0 & 0 \\ 0 & s_2 & 0 \end{bmatrix}$$



Population Projection

If we represent the number of individuals in each age class at time t as a column matrix (a state vector) $\mathbf{n}_t$, we can find the population at the next time interval by multiplying by L:

$$\mathbf{n}_{t+1} = L\mathbf{n}_t$$

To project k time periods into the future from an initial population $\mathbf{n}_0$, we use matrix exponentiation:

$$\mathbf{n}_k = L^k\mathbf{n}_0$$

Problem 2.13: Projecting a Rodent Population

A species of rodent is divided into three age classes, each spanning one month: 0–1 month, 1–2 months, and 2–3 months. The maximum lifespan is 3 months.

- **Fertility rates:** Class 1 females produce 0 offspring. Class 2 females produce 2.0 offspring. Class 3 females produce 1.5 offspring.
- **Survival rates:** 40% (0.4) of Class 1 survive to Class 2. 30% (0.3) of Class 2 survive to Class 3.
- **Initial Population ($\mathbf{n}_0$):** There are currently 100 females in Class 1, 50 in Class 2, and 20 in Class 3.

(a) Construct the Leslie matrix, L, for this population.

(b) Calculate the population state vector after 1 month ($\mathbf{n}_1$).

(c) Calculate the population state vector after 2 months ($\mathbf{n}_2$).

Solution 2.13

(a) Construct the Leslie matrix

Row 1 contains the fertility rates: $0, 2.0, 1.5$.
The sub-diagonal contains the survival rates: 0.4 and 0.3.

$$L = \begin{bmatrix} 0 & 2.0 & 1.5 \\ 0.4 & 0 & 0 \\ 0 & 0.3 & 0 \end{bmatrix}$$

(b) Population after 1 month ($\mathbf{n}_1$)

Multiply the Leslie matrix by the initial population vector $\mathbf{n}_0$:

$$\mathbf{n}_1 = L\mathbf{n}_0 = \begin{bmatrix} 0 & 2.0 & 1.5 \\ 0.4 & 0 & 0 \\ 0 & 0.3 & 0 \end{bmatrix} \begin{bmatrix} 100 \\ 50 \\ 20 \end{bmatrix} = \begin{bmatrix} 0(100) + 2.0(50) + 1.5(20) \\ 0.4(100) + 0(50) + 0(20) \\ 0(100) + 0.3(50) + 0(20) \end{bmatrix}$$

$$\mathbf{n}_1 = \begin{bmatrix} 0 + 100 + 30 \\ 40 \\ 15 \end{bmatrix} = \begin{bmatrix} 130 \\ 40 \\ 15 \end{bmatrix}$$

After 1 month, there are 130 newborns, 40 adults (1-2 mo), and 15 elders (2-3 mo).

(c) Population after 2 months ($\mathbf{n}_2$)

Multiply the Leslie matrix by the new population vector $\mathbf{n}_1$:

$$\mathbf{n}_2 = L\mathbf{n}_1 = \begin{bmatrix} 0 & 2.0 & 1.5 \\ 0.4 & 0 & 0 \\ 0 & 0.3 & 0 \end{bmatrix} \begin{bmatrix} 130 \\ 40 \\ 15 \end{bmatrix} = \begin{bmatrix} 0(130) + 2.0(40) + 1.5(15) \\ 0.4(130) + 0(40) + 0(15) \\ 0(130) + 0.3(40) + 0(15) \end{bmatrix}$$

$$\mathbf{n}_2 = \begin{bmatrix} 0 + 80 + 22.5 \\ 52 \\ 12 \end{bmatrix} = \begin{bmatrix} 102.5 \\ 52 \\ 12 \end{bmatrix}$$

(Note: In mathematical biology, it is standard practice to keep decimals during intermediate calculation steps to avoid compounding rounding errors, and only round to whole animals when stating the final interpretive answer. So we have roughly 103 newborns, 52 adults, and 12 elders).

Takeaways 2.13

- **Structured Flow:** The rigid structure of the Leslie matrix ensures mathematical logic: the top row calculates all new births and drops them into age class 1, while the sub-diagonal pushes the survivors from class i down into class $i+1$.
- **Iteration:** A system that updates based on its previous state ($\mathbf{n}_{t+1} = L\mathbf{n}_t$) is called a dynamical system. Matrices are perfect for iterating large blocks of data simultaneously.
- **Long-term Behavior:** While outside the scope of this topic, the long-term survival or extinction of a population depends entirely on the largest *eigenvalue* of its Leslie matrix!

3 Part 1: Worked Problems

3.1 Basic

Problem 3.1: School Demographics Matrix

Based on the demographic data provided in the school census: At a certain school, there are 110 girls and 110 boys in Year 7. The numbers of girls and boys in the other year levels are 120 and 105 in Year 8, 140 and 130 in Year 9, 100 and 95 in Year 10, 90 and 85 in Year 11, and 60 and 65 in Year 12.

(a) Summarise this information in a 6×2 matrix, S, where the rows represent the year levels (7 to 12) and the columns represent the gender (girls, boys).

(b) Define a column matrix C such that the matrix product SC results in a 6×1 matrix representing the total number of students in each year level. Calculate SC.

(c) Define a row matrix R such that the matrix product RS results in a 1×2 matrix representing the total number of girls and boys in the school overall. Calculate RS.

(d) The school receives an extracurricular grant of \$50 per girl and \$45 per boy. Write down a column matrix G representing these amounts, and calculate the product SG to find the total grant amount allocated to each year level.

Hint:

- **Part (a):** Organise the data sequentially. Ensure your rows correspond to Years 7 through 12, and the first and second columns correspond to girls and boys, respectively.
- **Part (b):** To sum the elements across the rows of a matrix using matrix multiplication, you need to post-multiply by a column matrix of 1s. Check your dimensions: a (6×2) matrix multiplied by a (2×1) matrix yields a (6×1) matrix.
- **Part (c):** To sum the elements down the columns, you must pre-multiply by a row matrix of 1s. Verify the dimensions: (1×6) multiplied by (6×2).
- **Part (d):** Matrix multiplication pairs the row elements of the first matrix with the column elements of the second matrix. Make sure the \$50 aligns with the girls' column and the \$45 aligns with the boys' column.

Solution 3.1

(a)

$$S = \begin{bmatrix} 110 & 110 \\ 120 & 105 \\ 140 & 130 \\ 100 & 95 \\ 90 & 85 \\ 60 & 65 \end{bmatrix}$$

(b)

$$C = \begin{bmatrix} 1 \\ 1 \end{bmatrix}, \quad SC = \begin{bmatrix} 110 & 110 \\ 120 & 105 \\ 140 & 130 \\ 100 & 95 \\ 90 & 85 \\ 60 & 65 \end{bmatrix} \begin{bmatrix} 1 \\ 1 \end{bmatrix} = \begin{bmatrix} 220 \\ 225 \\ 270 \\ 195 \\ 175 \\ 125 \end{bmatrix}$$

(c)

$$R = \begin{bmatrix} 1 & 1 & 1 & 1 & 1 & 1 \end{bmatrix}$$

$$RS = \begin{bmatrix} 1 & 1 & 1 & 1 & 1 & 1 \end{bmatrix} \begin{bmatrix} 110 & 110 \\ 120 & 105 \\ 140 & 130 \\ 100 & 95 \\ 90 & 85 \\ 60 & 65 \end{bmatrix} = \begin{bmatrix} 620 & 590 \end{bmatrix}$$

(d)

$$G = \begin{bmatrix} 50 \\ 45 \end{bmatrix}, \quad SG = \begin{bmatrix} 110 & 110 \\ 120 & 105 \\ 140 & 130 \\ 100 & 95 \\ 90 & 85 \\ 60 & 65 \end{bmatrix} \begin{bmatrix} 50 \\ 45 \end{bmatrix} = \begin{bmatrix} 10450 \\ 10725 \\ 12850 \\ 9275 \\ 8325 \\ 5925 \end{bmatrix}$$

Takeaways 3.1

- **Weighted Totals:** Post-multiplying a data matrix by a column of weights forms a weighted sum for each row. Here, SG calculates each year level's grant; the same technique applies to costs, marks, prices and weighted averages.
- **Summation Vectors:** Matrix multiplication using vectors of 1s is an efficient algebraic method for summing rows or columns of a data matrix without writing out lengthy addition equations.
- **Dimensionality Checks:** Always verify conformability before multiplying matrices. The number of columns in the leading matrix must equal the number of rows in the trailing matrix (e.g., $m \times n$ multiplied by $n \times p$ gives an $m \times p$ matrix).

Problem 3.2: Matrix Properties and Transformations

Consider a general 2×2 matrix $A = \begin{bmatrix} p & q \\ r & s \end{bmatrix}$ and its associated adjugate matrix $A^* = \begin{bmatrix} s & -q \\ -r & p \end{bmatrix}$.

(a) Evaluate the matrix product AA^* and express the result in the form kI, where I is the 2×2 identity matrix and k is a scalar expressed in terms of p, q, r, and s.

(b) Given that $k \neq 0$, use your result from part (a) to algebraically deduce the standard expression for the inverse matrix, A^{-1}.

(c) Evaluate the reverse product A^*A and hence show that for this specific pair of matrices, multiplication is commutative.

(d) A linear transformation in the plane is represented by the matrix $M = \begin{bmatrix} 2x & 3 \\ 8 & x \end{bmatrix}$. Using the concept of k from part (a), determine the exact possible values of x for which this transformation cannot be reversed (i.e., the matrix is singular).

Hint:

- **Part (a):** Perform standard row-by-column matrix multiplication. Expand the terms, simplify the diagonals, and factor out any common algebraic expressions to reveal the identity matrix.
- **Part (b):** Recall that the definition of an inverse matrix is $AA^{-1} = I$. Take your equation from part (a), $AA^* = kI$, and manipulate it algebraically to isolate I on one side of the equation.
- **Part (c):** Multiply the matrices again, but this time with A^* on the left and A on the right. Compare the resulting matrix to the one you found in part (a).
- **Part (d):** A matrix transformation does not have an inverse when its associated scalar k (the determinant) is equal to zero. Set up an equation for $k = 0$ using the elements of matrix M and solve for x.

Solution 3.2

(a)

$$AA^* = \begin{bmatrix} p & q \\ r & s \end{bmatrix} \begin{bmatrix} s & -q \\ -r & p \end{bmatrix} = \begin{bmatrix} ps - qr & -pq + qp \\ rs - sr & -rq + sp \end{bmatrix} = \begin{bmatrix} ps - qr & 0 \\ 0 & ps - qr \end{bmatrix}$$

Factoring out the common term: $AA^* = (ps - qr) \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} = (ps - qr)I$.

Therefore, $k = ps - qr$.

(b)

From part (a), we have $AA^* = (ps - qr)I$.
Assuming $ps - qr \neq 0$, we can divide both sides by this scalar:
$A\left(\frac{1}{ps-qr}A^*\right) = I$

Since $AA^{-1} = I$, it follows that $A^{-1} = \frac{1}{ps-qr}\begin{bmatrix} s & -q \\ -r & p \end{bmatrix}$.

(c)

$$A^*A = \begin{bmatrix} s & -q \\ -r & p \end{bmatrix} \begin{bmatrix} p & q \\ r & s \end{bmatrix} = \begin{bmatrix} sp - qr & sq - qs \\ -rp + pr & -rq + ps \end{bmatrix} = \begin{bmatrix} ps - qr & 0 \\ 0 & ps - qr \end{bmatrix} = (ps - qr)I$$

Since $AA^* = (ps - qr)I$ and $A^*A = (ps - qr)I$, we have shown that $AA^* = A^*A$. Thus, multiplication of a matrix by its adjugate is commutative.

(d)

For the transformation to have no inverse, $k = ps - qr = 0$.
For matrix M, $p = 2x$, $q = 3$, $r = 8$, and $s = x$.
$k = (2x)(x) - (3)(8) = 2x^2 - 24$
Setting $k = 0$: $2x^2 - 24 = 0 \implies x^2 = 12 \implies x = \pm\sqrt{12} = \pm 2\sqrt{3}$.

Takeaways 3.2

- **Determinant:** The scalar value $k = ps - qr$ derived in part (a) is the determinant of a 2×2 matrix. It is the fundamental factor that dictates whether a matrix has an inverse.
- **Adjugate and Inverse Formula:** For $A = \begin{bmatrix} p & q \\ r & s \end{bmatrix}$, $AA^* = A^*A = \det(A)I$. Hence, if $ps - qr \neq 0$,
$$A^{-1} = \frac{1}{ps - qr}\begin{bmatrix} s & -q \\ -r & p \end{bmatrix}.$$
- **Singular Matrices:** Matrices with a determinant of zero are called singular. Geometrically, these matrices represent transformations that collapse 2D space into a 1D line or a point, which is why the mapping cannot be algebraically reversed.

3.2 Medium

Problem 3.3: Matrix Algebra and Commutativity

Consider general 2×2 matrices A and B.

(a) By expanding the product, show that $(A+B)^2 = A^2 + AB + BA + B^2$.

(b) Hence, state the necessary mathematical condition for matrices A and B such that the standard binomial expansion $(A+B)^2 = A^2 + 2AB + B^2$ is valid.

(c) Let $A = \begin{bmatrix} 1 & 0 \\ 0 & -1 \end{bmatrix}$ representing a reflection in the x-axis, and $B = \begin{bmatrix} k & 0 \\ 0 & k \end{bmatrix}$ representing a dilation by scale factor k. Verify algebraically that for these specific matrices, $(A+B)^2 = A^2 + 2AB + B^2$.

(d) Let $C = \begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix}$ and $D = \begin{bmatrix} 0 & -1 \\ 1 & 0 \end{bmatrix}$. Evaluate $(C+D)^2$ and $C^2 + 2CD + D^2$, demonstrating they are not equal, and explain why this algebraic difference occurs in terms of matrix multiplication properties.

Hint:

- **Part (a):** Write $(A+B)^2$ as $(A+B)(A+B)$ and apply the distributive law carefully, ensuring you maintain the exact left-to-right order of matrix multiplication.
- **Part (b):** Compare your expanded result from part (a) to the target expression $A^2 + 2AB + B^2$. What must the middle terms $AB + BA$ simplify to?
- **Part (c):** Calculate the individual products AB and BA first. Check if they satisfy the condition you identified in part (b).
- **Part (d):** For the first expression, perform the matrix addition $C+D$ before squaring the result. For the second expression, calculate C^2, D^2, and CD separately before summing them.

Solution 3.3

(a)
$(A+B)^2 = (A+B)(A+B) = A(A+B) + B(A+B) = A^2 + AB + BA + B^2$.

(b)
For $A^2 + AB + BA + B^2$ to equal $A^2 + 2AB + B^2$, we must have:
$AB + BA = 2AB \implies BA = AB$.
The necessary condition is that matrices A and B must commute.

(c)
$$AB = \begin{bmatrix} 1 & 0 \\ 0 & -1 \end{bmatrix} \begin{bmatrix} k & 0 \\ 0 & k \end{bmatrix} = \begin{bmatrix} k & 0 \\ 0 & -k \end{bmatrix}$$
$$BA = \begin{bmatrix} k & 0 \\ 0 & k \end{bmatrix} \begin{bmatrix} 1 & 0 \\ 0 & -1 \end{bmatrix} = \begin{bmatrix} k & 0 \\ 0 & -k \end{bmatrix}$$
Since $AB = BA$, the matrices commute, and therefore the binomial expansion $(A+B)^2 = A^2+2AB+B^2$ holds valid for this specific pair.

(d)
$$C + D = \begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix} + \begin{bmatrix} 0 & -1 \\ 1 & 0 \end{bmatrix} = \begin{bmatrix} 0 & 0 \\ 2 & 0 \end{bmatrix}.$$
$$(C+D)^2 = \begin{bmatrix} 0 & 0 \\ 2 & 0 \end{bmatrix} \begin{bmatrix} 0 & 0 \\ 2 & 0 \end{bmatrix} = \begin{bmatrix} 0 & 0 \\ 0 & 0 \end{bmatrix}.$$
Next, evaluating the components of the expanded form:
$$C^2 = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} = I \text{ and } D^2 = \begin{bmatrix} -1 & 0 \\ 0 & -1 \end{bmatrix} = -I.$$
$$CD = \begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix} \begin{bmatrix} 0 & -1 \\ 1 & 0 \end{bmatrix} = \begin{bmatrix} 1 & 0 \\ 0 & -1 \end{bmatrix}.$$
$$C^2 + 2CD + D^2 = I + 2\begin{bmatrix} 1 & 0 \\ 0 & -1 \end{bmatrix} - I = \begin{bmatrix} 2 & 0 \\ 0 & -2 \end{bmatrix}.$$
Since $\begin{bmatrix} 0 & 0 \\ 0 & 0 \end{bmatrix} \neq \begin{bmatrix} 2 & 0 \\ 0 & -2 \end{bmatrix}$, the expressions are unequal. This occurs because general matrix multiplication is non-commutative ($CD \neq DC$), preventing the cross-terms from simplifying to $2CD$.

Takeaways 3.3

- **Matrix Non-Commutativity:** The most critical difference between standard real number algebra and matrix algebra is that matrix multiplication is generally non-commutative ($AB \neq BA$).
- **Binomial Expansion:** Standard polynomial identities like $(x+y)^2 = x^2+2xy+y^2$ cannot be blindly applied to matrices unless the specific matrices are proven to commute.
- **Geometric Connection:** The algebraic failure of commutativity mirrors the geometric reality of transformations. For instance, applying a reflection then a rotation often results in a different final orientation compared to applying the rotation then the reflection.

Problem 3.4: Matrix Inverses and Matrix Equations

Let A and B be 2×2 matrices defined as $A = \begin{bmatrix} k & 2 \\ -1 & 1 \end{bmatrix}$ and $B = \begin{bmatrix} 1 & -1 \\ 2 & 3 \end{bmatrix}$, where k is a real constant.

(a) Determine the determinant of A in terms of k, and hence state the value of k for which matrix A is singular.

(b) Let $k = 3$. Determine the inverse matrices A^{-1} and B^{-1}.

(c) For $k = 3$, calculate the matrix product AB and hence determine $(AB)^{-1}$.

(d) Evaluate the product $B^{-1}A^{-1}$ to verify the property $(AB)^{-1} = B^{-1}A^{-1}$. Hence, use this property to make the matrix X the subject of the matrix equation $ABX = C$, where C is a 2×2 matrix.

Hint:

- **Part (a):** The determinant of a 2×2 matrix $\begin{bmatrix} a & b \\ c & d \end{bmatrix}$ is $ad - bc$. A matrix is singular (has no inverse) when its determinant equals zero.
- **Part (b):** Use the formula for the inverse of a 2×2 matrix: $\frac{1}{\det(M)} \begin{bmatrix} d & -b \\ -c & a \end{bmatrix}$.
- **Part (c):** Multiply matrix A by matrix B using the standard row-by-column method. Then find the inverse of the resulting 2×2 matrix.
- **Part (d):** Multiply B^{-1} (on the left) by A^{-1} (on the right) and factor out the scalar multipliers. To solve $ABX = C$, remember that you must multiply both sides of the equation by the inverse of AB on the left.

Solution 3.4

(a)
$\det(A) = (k)(1) - (2)(-1) = k + 2$.
For matrix A to be singular, $\det(A) = 0$. Therefore, $k = -2$.

(b)
Given $k = 3$, $A = \begin{bmatrix} 3 & 2 \\ -1 & 1 \end{bmatrix}$.
$\det(A) = 3 + 2 = 5$. Thus, $A^{-1} = \frac{1}{5}\begin{bmatrix} 1 & -2 \\ 1 & 3 \end{bmatrix}$.
$\det(B) = (1)(3) - (-1)(2) = 3 + 2 = 5$. Thus, $B^{-1} = \frac{1}{5}\begin{bmatrix} 3 & 1 \\ -2 & 1 \end{bmatrix}$.

(c)
$AB = \begin{bmatrix} 3 & 2 \\ -1 & 1 \end{bmatrix}\begin{bmatrix} 1 & -1 \\ 2 & 3 \end{bmatrix} = \begin{bmatrix} 3(1)+2(2) & 3(-1)+2(3) \\ -1(1)+1(2) & -1(-1)+1(3) \end{bmatrix} = \begin{bmatrix} 7 & 3 \\ 1 & 4 \end{bmatrix}$.
$\det(AB) = (7)(4) - (3)(1) = 28 - 3 = 25$.
$(AB)^{-1} = \frac{1}{25}\begin{bmatrix} 4 & -3 \\ -1 & 7 \end{bmatrix}$.

(d)
$B^{-1}A^{-1} = \left(\frac{1}{5}\begin{bmatrix} 3 & 1 \\ -2 & 1 \end{bmatrix}\right)\left(\frac{1}{5}\begin{bmatrix} 1 & -2 \\ 1 & 3 \end{bmatrix}\right) = \frac{1}{25}\begin{bmatrix} 3(1)+1(1) & 3(-2)+1(3) \\ -2(1)+1(1) & -2(-2)+1(3) \end{bmatrix} = \frac{1}{25}\begin{bmatrix} 4 & -3 \\ -1 & 7 \end{bmatrix}$.
This verifies that $(AB)^{-1} = B^{-1}A^{-1}$.
To solve $ABX = C$:
$(AB)^{-1}(AB)X = (AB)^{-1}C$
$IX = (AB)^{-1}C$
$X = B^{-1}A^{-1}C$.

Takeaways 3.4

- **The "Socks and Shoes" Property:** The inverse of a matrix product reverses the order of the original matrices: $(AB)^{-1} = B^{-1}A^{-1}$.
- **Solving Matrix Equations:** For $ABX = C$,

$$X = (AB)^{-1}C = B^{-1}A^{-1}C.$$

  Matrix "division" is performed using inverses on the correct side, and the order of factors must not be changed.
- **Determinant of a Product:** Notice that $\det(AB) = 25$, while $\det(A) = 5$ and $\det(B) = 5$. This verifies another key property: $\det(AB) = \det(A) \times \det(B)$.

3.3 Advanced

Problem 3.5: Matrix Inverses and the Zero Matrix

Let A, B, C, and D be 2×2 matrices, where O is the zero matrix and I is the identity matrix.

(a) Assume A is an invertible matrix and that $AB = O$. Prove algebraically that B must be the zero matrix.

(b) Suppose instead that C and D are both strictly non-zero matrices, yet their product is the zero matrix ($CD = O$). By referencing your proof in part (a), deduce whether C can be an invertible matrix. Support your answer with logical reasoning.

(c) A matrix M is defined as being self-inverse if $M^{-1} = M$. By showing this implies $M^2 = I$, determine the algebraic condition on real numbers a, b, and c such that $M = \begin{bmatrix} a & b \\ c & -a \end{bmatrix}$ is a self-inverse matrix.

(d) Using your condition from part (c), construct a specific matrix M where $a = 3$, and b and c are integers. Verify that your chosen matrix is self-inverse by explicitly evaluating M^2.

Hint:

- **Part (a):** Start with the given equation $AB = O$. Since A is invertible, you are permitted to pre-multiply both sides of the equation by A^{-1}. What does $A^{-1}A$ simplify to?
- **Part (b):** Use a proof by contradiction. What would happen if C *was* invertible? Apply the exact same logic established in part (a) and compare the result to the condition that D is a non-zero matrix.
- **Part (c):** Multiply both sides of $M^{-1} = M$ by M. Then, square the given matrix $M = \begin{bmatrix} a & b \\ c & -a \end{bmatrix}$ using standard row-by-column multiplication and equate the resulting elements to the corresponding elements of the identity matrix, I.
- **Part (d):** Substitute $a = 3$ into your condition from part (c). Find two integers b and c that multiply together to satisfy the equation, construct the matrix, and manually multiply $M \times M$ to check it yields I.

Solution 3.5

(a)
Given $AB = O$ and A is invertible, A^{-1} exists. Pre-multiplying both sides by A^{-1}:
$A^{-1}(AB) = A^{-1}O$
$(A^{-1}A)B = O$
$IB = O \implies B = O$.

(b)
Assume for the sake of contradiction that C is invertible.
Then, given $CD = O$, we could pre-multiply by C^{-1} to get $C^{-1}(CD) = C^{-1}O \implies D = O$.
However, the problem states that D is a non-zero matrix. This is a contradiction. Therefore, the assumption that C is invertible must be false; C must be a singular matrix.

(c)
If $M^{-1} = M$, then multiplying both sides by M gives $MM^{-1} = MM \implies I = M^2$.
Evaluating M^2:
$$M^2 = \begin{bmatrix} a & b \\ c & -a \end{bmatrix}\begin{bmatrix} a & b \\ c & -a \end{bmatrix} = \begin{bmatrix} a^2+bc & ab-ab \\ ac-ac & bc+(-a)^2 \end{bmatrix} = \begin{bmatrix} a^2+bc & 0 \\ 0 & a^2+bc \end{bmatrix}.$$
For $M^2 = I = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}$, we must have the condition: $a^2 + bc = 1$.

(d)
Let $a = 3$. From part (c), $3^2 + bc = 1 \implies 9 + bc = 1 \implies bc = -8$.
Choose integers $b = 2$ and $c = -4$ (other integer pairs like $4, -2$ are also valid).
$$M = \begin{bmatrix} 3 & 2 \\ -4 & -3 \end{bmatrix}.$$
Verification:
$$M^2 = \begin{bmatrix} 3 & 2 \\ -4 & -3 \end{bmatrix}\begin{bmatrix} 3 & 2 \\ -4 & -3 \end{bmatrix} = \begin{bmatrix} 9-8 & 6-6 \\ -12+12 & -8+9 \end{bmatrix} = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} = I.$$

Takeaways 3.5

- **Zero Product Property Failure:** In real number algebra, $xy = 0$ implies $x = 0$ or $y = 0$. In matrix algebra, $CD = O$ does *not* necessarily mean $C = O$ or $D = O$. Non-zero matrices can multiply to give the zero matrix.
- **Singularity is Key:** For two non-zero matrices to multiply to the zero matrix, they must both be singular (determinant of zero). If either were invertible, the other would be forced to be the zero matrix.
- **Self-Inverse Matrices:** $M^{-1} = M$ if and only if $M^2 = I$. For
$$M = \begin{bmatrix} a & b \\ c & -a \end{bmatrix},$$
this reduces to the single condition $a^2+bc = 1$. Such matrices are called involutions.

Problem 3.6: Matrix Methods for Simultaneous Equations

Consider the following system of simultaneous linear equations with real parameter k:

$$\begin{aligned}(k-1)x + 3y &= k+2\\ 2x + (k-2)y &= 4\end{aligned}$$

(a) Express this system as a matrix equation of the form $AX = B$, and find a simplified quadratic expression for the determinant of the coefficient matrix A.

(b) Determine the values of k for which the matrix A is singular (i.e., does not have an inverse).

(c) For each value of k found in part (b), determine whether the system has no solutions or infinitely many solutions. Provide a brief geometric justification for your answers.

(d) Assuming the matrix A is invertible, use the matrix inverse method to find the unique solution for x and y in terms of k, expressing your final answers in their simplest rational form.

Hint:

- **Part (a):** The coefficient matrix A contains the multipliers for x and y. The determinant of a 2×2 matrix $\begin{bmatrix} a & b \\ c & d \end{bmatrix}$ is calculated as $ad - bc$.
- **Part (b):** A matrix is singular when its determinant is exactly zero. Set your expression from part (a) to zero and solve the resulting quadratic equation for k.
- **Part (c):** Substitute your k values back into the original system of equations. If the two equations represent distinct parallel lines (e.g., yielding a false statement like $0 = 5$), there is no solution. If they represent the exact same line, there are infinitely many solutions.
- **Part (d):** Use the inverse matrix formula $X = A^{-1}B = \frac{1}{\det(A)} \begin{bmatrix} d & -b \\ -c & a \end{bmatrix} B$. After multiplying the matrices, factor the resulting numerators to see if they cancel with the determinant in the denominator.

Solution 3.6

(a)

Matrix equation: $\begin{bmatrix} k-1 & 3 \\ 2 & k-2 \end{bmatrix} \begin{bmatrix} x \\ y \end{bmatrix} = \begin{bmatrix} k+2 \\ 4 \end{bmatrix}$

$\det(A) = (k-1)(k-2) - (3)(2) = k^2 - 3k + 2 - 6 = k^2 - 3k - 4.$

Factored form: $\det(A) = (k-4)(k+1)$.

(b)

Matrix A is singular when $\det(A) = 0$.

$k^2 - 3k - 4 = 0 \implies (k-4)(k+1) = 0.$

Therefore, A is singular when $k = 4$ or $k = -1$.

(c)

Case $k = 4$:

Equations become $3x + 3y = 6$ (which simplifies to $x + y = 2$) and $2x + 2y = 4$ (which simplifies to $x + y = 2$).

Because the equations represent coincident (identical) lines, there are **infinitely many solutions**.

Case $k = -1$:

Equations become $-2x + 3y = 1$ and $2x - 3y = 4$.

Adding the two equations yields $0 = 5$, which is mathematically inconsistent. Because the equations represent distinct parallel lines, there are **no solutions**.

(d)

For $k \neq 4$ and $k \neq -1$, A is invertible.

$A^{-1} = \frac{1}{(k-4)(k+1)} \begin{bmatrix} k-2 & -3 \\ -2 & k-1 \end{bmatrix}$

$X = \frac{1}{(k-4)(k+1)} \begin{bmatrix} k-2 & -3 \\ -2 & k-1 \end{bmatrix} \begin{bmatrix} k+2 \\ 4 \end{bmatrix}$

$x = \frac{(k-2)(k+2)-12}{(k-4)(k+1)} = \frac{k^2-4-12}{(k-4)(k+1)} = \frac{k^2-16}{(k-4)(k+1)} = \frac{(k-4)(k+4)}{(k-4)(k+1)} = \frac{k+4}{k+1}$

$y = \frac{-2(k+2)+4(k-1)}{(k-4)(k+1)} = \frac{-2k-4+4k-4}{(k-4)(k+1)} = \frac{2k-8}{(k-4)(k+1)} = \frac{2(k-4)}{(k-4)(k+1)} = \frac{2}{k+1}$

Unique solution: $x = \frac{k+4}{k+1}$ and $y = \frac{2}{k+1}$.

Takeaways 3.6

- **Determinant as a Geometric Indicator:** Just as the discriminant reveals the nature of roots in a quadratic, the determinant of a coefficient matrix dictates the geometry of linear systems. A non-zero determinant guarantees intersecting lines (one unique solution).
- **Interpreting Singularity:** A singular matrix ($\det = 0$) means the matrix transformation collapses 2D space. For a system of equations, this means the lines share the exact same gradient, resulting in either parallel lines (no intersection) or coincident lines (infinite intersections).
- **Do Not Lose Excluded Values:** Cancelling a factor while simplifying $A^{-1}B$ does not remove the original restriction $\det A \neq 0$. Here, even though factors cancel, $k = 4$ and $k = -1$ must still be excluded from the unique-solution formula.

Problem 3.7: Determinant Properties and Matrix Algebra

Let $A = \begin{bmatrix} a & b \\ c & d \end{bmatrix}$ and $B = \begin{bmatrix} p & q \\ r & s \end{bmatrix}$ be arbitrary 2×2 matrices.

(a) Prove algebraically that $\det(AB) = \det(A)\det(B)$.

(b) Using the result from part (a), prove that if a matrix M is invertible, then $\det(M^{-1}) = \frac{1}{\det(M)}$.

(c) Consider the matrices $C = \begin{bmatrix} k & 2 \\ 1 & -1 \end{bmatrix}$ and $D = \begin{bmatrix} 2 & 0 \\ -1 & k \end{bmatrix}$, where k is a non-zero real constant. Using the determinant properties established above, determine the exact values of k for which $\det(C^2 D^{-1}) = \frac{9}{2}$.

(d) Using the larger value of k found in part (c), calculate $\det(C + D)$ and explicitly show that $\det(C + D) \neq \det(C) + \det(D)$.

Hint:

- **Part (a):** Multiply the matrices A and B first using standard row-by-column multiplication. Then apply the $ad - bc$ determinant rule to the resulting matrix. Expand all terms and factor them to show the expression matches $(ad - bc)(ps - qr)$.
- **Part (b):** Start with the fundamental definition of an inverse matrix: $MM^{-1} = I$. Apply the determinant operator to both sides of this equation.
- **Part (c):** Apply the rules $\det(C^2) = (\det(C))^2$ and $\det(D^{-1}) = 1/\det(D)$. Form an algebraic equation purely in terms of k and solve the resulting quadratic.
- **Part (d):** Add the matrices C and D element-by-element first to find the matrix $(C + D)$, then find its determinant. Compare this final value to the sum of their individual determinants.

Solution 3.7

(a)

$AB = \begin{bmatrix} a & b \\ c & d \end{bmatrix} \begin{bmatrix} p & q \\ r & s \end{bmatrix} = \begin{bmatrix} ap+br & aq+bs \\ cp+dr & cq+ds \end{bmatrix}$.

$\det(AB) = (ap+br)(cq+ds) - (aq+bs)(cp+dr)$

$\det(AB) = (apcq + apds + brcq + brds) - (aqcp + aqdr + bscp + bsdr)$

The terms $apcq$ and $bsdr$ cancel out:

$\det(AB) = apds + brcq - aqdr - bscp$

Factoring by grouping:

$\det(AB) = ad(ps-qr) - bc(ps-qr) = (ad-bc)(ps-qr)$

Since $\det(A) = ad-bc$ and $\det(B) = ps-qr$, we have proven $\det(AB) = \det(A)\det(B)$.

(b)

By definition, if M is invertible, $MM^{-1} = I$.

Taking the determinant of both sides: $\det(MM^{-1}) = \det(I)$.

Using the property from part (a): $\det(M)\det(M^{-1}) = 1$.

Because M is invertible, $\det(M) \neq 0$. Thus, we can divide by $\det(M)$ to yield $\det(M^{-1}) = \frac{1}{\det(M)}$.

(c)

$\det(C) = (k)(-1) - (2)(1) = -k - 2$.

$\det(D) = (2)(k) - (0)(-1) = 2k$.

Using the properties: $\det(C^2D^{-1}) = \det(C)^2 \times \det(D^{-1}) = \frac{\det(C)^2}{\det(D)}$.

$\frac{(-k-2)^2}{2k} = \frac{9}{2} \implies \frac{k^2+4k+4}{2k} = \frac{9}{2}$

Cross-multiplying: $2(k^2+4k+4) = 18k \implies 2k^2+8k+8 = 18k$

$2k^2 - 10k + 8 = 0 \implies k^2 - 5k + 4 = 0$.

Factoring gives $(k-4)(k-1) = 0$, so $k = 4$ or $k = 1$.

(d)

Using the larger value $k = 4$:

$C = \begin{bmatrix} 4 & 2 \\ 1 & -1 \end{bmatrix} \implies \det(C) = -4 - 2 = -6$.

$D = \begin{bmatrix} 2 & 0 \\ -1 & 4 \end{bmatrix} \implies \det(D) = 8 - 0 = 8$.

$\det(C) + \det(D) = -6 + 8 = 2$.

Now, finding $C + D$:

$C + D = \begin{bmatrix} 4+2 & 2+0 \\ 1-1 & -1+4 \end{bmatrix} = \begin{bmatrix} 6 & 2 \\ 0 & 3 \end{bmatrix}$.

$\det(C+D) = (6)(3) - (2)(0) = 18$.

Since $18 \neq 2$, we have shown that $\det(C+D) \neq \det(C) + \det(D)$.

Takeaways 3.7

- **Multiplicative Nature of Determinants:** The determinant is a strictly multiplicative function. Matrix exponents and inverses can be evaluated directly through their determinants without calculating the resulting complex matrices first.
- **Non-Linearity of Determinants:** The determinant is *not* a linear operator. $\det(A+B)$ is rarely equal to $\det(A)+\det(B)$.
- **Composite Determinant Shortcut:** Use

$$\det(M^n) = \det(M)^n, \qquad \det(M^{-1}) = \frac{1}{\det(M)}$$

to evaluate complicated expressions without calculating the matrices themselves; for example,

$$\det(C^2D^{-1}) = \frac{\det(C)^2}{\det(D)}.$$

Problem 3.8: Dominance Arrays and Tournament Rankings

Five schools—A, B, C, D, and E—compete in a round-robin hockey tournament where no matches end in a draw. The results are as follows:

- School A defeated B, C, and D.
- School B defeated C and D.
- School C defeated D and E.
- School D defeated E.
- School E defeated A and B.

(a) Construct the 5×5 dominance array, M, for this tournament, where a 1 in row i and column j indicates that team i defeated team j.

(b) Evaluate the squared array M^2. What does the numerical value located in row 5, column 3 of M^2 represent contextually?

(c) The tournament committee decides to rank the schools using a total power score vector P, calculated by the formula $P = (M + M^2)\mathbf{v}$, where $\mathbf{v}$ is a 5×1 column vector of ones. Calculate P and state the final ranking of the five schools.

(d) For any standard $n \times n$ round-robin dominance array (where there are no draws), the sum of all elements in the array is a fixed constant depending only on n. Determine this algebraic expression in terms of n, and provide a brief logical justification based on the properties of a round-robin tournament.

Hint:

- **Part (a):** Set up a 5×5 grid. If A beats B, place a 1 in the row for A and column for B, and a 0 in the row for B and column for A. The main diagonal will contain only zeros.
- **Part (b):** Multiply array M by itself using standard row-by-column multiplication. The value in row 5, column 3 corresponds to the pathways from E to C.
- **Part (c):** Adding M and M^2 combines the 1-step and 2-step dominance scores. Multiplying by a column vector of ones simply sums the rows of the resulting array.
- **Part (d):** In a round-robin tournament, how many total games are played between n teams? Remember that each match results in exactly one '1' being placed somewhere in the dominance array.

Solution 3.8

(a)

$$M = \begin{bmatrix} 0 & 1 & 1 & 1 & 0 \\ 0 & 0 & 1 & 1 & 0 \\ 0 & 0 & 0 & 1 & 1 \\ 0 & 0 & 0 & 0 & 1 \\ 1 & 1 & 0 & 0 & 0 \end{bmatrix}$$

(b)

$$M^2 = \begin{bmatrix} 0 & 0 & 1 & 2 & 2 \\ 0 & 0 & 0 & 1 & 2 \\ 1 & 1 & 0 & 0 & 1 \\ 1 & 1 & 0 & 0 & 0 \\ 0 & 1 & 2 & 2 & 0 \end{bmatrix}$$

The value in row 5, column 3 is 2. This represents the number of 2-step dominances School E has over School C (i.e., School E defeated two teams—A and B—who both subsequently defeated School C).

(c)

$$M + M^2 = \begin{bmatrix} 0 & 1 & 2 & 3 & 2 \\ 0 & 0 & 1 & 2 & 2 \\ 1 & 1 & 0 & 1 & 2 \\ 1 & 1 & 0 & 0 & 1 \\ 1 & 2 & 2 & 2 & 0 \end{bmatrix}$$

Summing the rows yields the total power score vector P:

$$P = \begin{bmatrix} 8 \\ 5 \\ 5 \\ 3 \\ 7 \end{bmatrix}$$

The rankings are: 1st: A, 2nd: E, Tied 3rd: B and C, 5th: D.

(d)

In an n-team round-robin tournament with no draws, every pair of teams plays exactly one match. The number of unique pairs is given by the combination formula $\frac{n(n-1)}{2}$. Since each match contributes exactly one '1' to the dominance array (and '0' to the reverse), the total sum of all elements in the array is exactly $\frac{n(n-1)}{2}$.

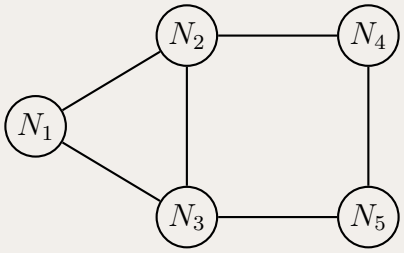
Takeaways 3.8

- **2-Step Dominance:** Squaring a dominance array reveals indirect victories, which are crucial for breaking ties or uncovering hidden strengths in a round-robin format.
- **Power Scores:** The expression $(M+M^2)\mathbf{v}$ is a standard algebraic method for calculating a comprehensive ranking score, weighing direct and indirect wins equally.
- **Array Properties:** Recognizing general algebraic properties of tournament structures (like the total sum of elements being $\frac{n(n-1)}{2}$) connects discrete combinatorics with array operations.

Problem 3.9: Adjacency Matrices and Network Paths

Consider a small communications network of five computer nodes, labeled N_1 to N_5. The network has bidirectional links connecting the following pairs of nodes:

- N_1 is connected to N_2 and N_3.
- N_2 is connected to N_1, N_3, and N_4.
- N_3 is connected to N_1, N_2, and N_5.
- N_4 is connected to N_2 and N_5.
- N_5 is connected to N_3 and N_4.



(a) Construct the 5×5 adjacency matrix, A, for this network, where $a_{ij} = 1$ if there is a direct link between N_i and N_j, and 0 otherwise.

(b) Calculate the matrix A^2. What does the entry in row 2, column 3 of A^2 represent in the context of this network?

(c) What is the significance of the entries on the main diagonal of A^2? Use these entries to determine the total number of edges in the network.

(d) Calculate the trace of A^3 (the sum of its main diagonal entries). What does this number tell you about the physical structure of the network?

Hint:

- **Part (a):** Since the links are bidirectional, the matrix A will be symmetric across its main diagonal. A node is not directly connected to itself, so the main diagonal consists of zeros.
- **Part (b):** The entries of A^k represent the number of walks of exactly length k between two vertices.
- **Part (c):** A walk of length 2 from a node to itself must go to an adjacent node and immediately return. How does this relate to the degree (number of connections) of that node?
- **Part (d):** A walk of length 3 from a node to itself is a closed loop of three edges. Think about what geometric shape this forms in the network graph.

Solution 3.9

(a)

$$A = \begin{bmatrix} 0 & 1 & 1 & 0 & 0 \\ 1 & 0 & 1 & 1 & 0 \\ 1 & 1 & 0 & 0 & 1 \\ 0 & 1 & 0 & 0 & 1 \\ 0 & 0 & 1 & 1 & 0 \end{bmatrix}$$

(b)

$$A^2 = \begin{bmatrix} 2 & 1 & 1 & 1 & 1 \\ 1 & 3 & 1 & 0 & 2 \\ 1 & 1 & 3 & 2 & 0 \\ 1 & 0 & 2 & 2 & 0 \\ 1 & 2 & 0 & 0 & 2 \end{bmatrix}$$

The entry in row 2, column 3 of A^2 is 1. This represents the number of 2-step walks from N_2 to N_3. The single path is $N_2 \to N_1 \to N_3$.

(c)

The entries on the main diagonal of A^2 represent the number of 2-step walks from a node to itself. In a simple network, this means traveling to a neighbor and returning immediately. Therefore, each diagonal entry represents the degree (number of direct connections) of that node.

The sum of these entries is $2+3+3+2+2 = 12$. Since each edge in the network connects two nodes, it contributes 2 to the sum of the degrees. Thus, the total number of edges is $12/2 = 6$.

(d)

The main diagonal of A^3 represents the number of 3-step walks from a node to itself. $(A^3)_{11} = 2, (A^3)_{22} = 2, (A^3)_{33} = 2, (A^3)_{44} = 0, (A^3)_{55} = 0$.

The trace (sum of the main diagonal) is 6. In a simple graph, a 3-step walk from a node to itself is a triangle (a closed loop of 3 edges). Each triangle N_i, N_j, N_k can be traversed in two directions starting from any of its 3 nodes, so a single triangle is counted $3 \times 2 = 6$ times in the trace of A^3.

Therefore, there is exactly $6/6 = 1$ triangle in the network structure (which is formed by nodes N_1, N_2, N_3).

Takeaways 3.9

- **Adjacency Matrix Powers:** While A records direct connections, its powers A^k count the number of walks of exactly length k between any two nodes. This connects matrix multiplication directly to combinatorial counting.
- **Graph Invariants:** Simple graph properties can be extracted from matrix properties. For example, the sum of the main diagonal of A^2 gives twice the number of edges, and the trace of A^3 gives six times the number of triangles.
- **Symmetry:** Because the network is undirected (bidirectional links), the adjacency matrix A is symmetric ($A^T = A$), which in turn guarantees that all powers of A are also symmetric.

Problem 3.10: Parametric Conics: The Matrix Shortcut

In the Cartesian plane, consider the ellipse $\dfrac{x^2}{a^2}+\dfrac{y^2}{b^2}=1$ with semi-major axis $a>0$ and semi-minor axis $b>0$. Let $O(0,0)$ denote the origin, and let $P(a\cos\theta, b\sin\theta)$ and $Q(a\cos\phi, b\sin\phi)$ be two points on the ellipse $(0\le\theta<\phi<2\pi)$.

Historical Context (The Pre-2020 4U Coordinate Geometry Ordeal). In the old NSW HSC Extension 2 (4 Unit) syllabus, finding the area of $\triangle OPQ$ was a classic examination question. Under the standard syllabus coordinate geometry method, students had to:

(i) Apply the distance formula to find the base chord length: $PQ = 2\sin\left(\frac{\phi-\theta}{2}\right)\sqrt{a^2\sin^2\left(\frac{\theta+\phi}{2}\right)+b^2\cos^2\left(\frac{\theta+\phi}{2}\right)}$.

(ii) Derive the Cartesian equation of chord PQ: $\left(b\cos\frac{\theta+\phi}{2}\right)x+\left(a\sin\frac{\theta+\phi}{2}\right)y = ab\cos\frac{\phi-\theta}{2}$.

(iii) Apply the perpendicular distance formula from $O(0,0)$ to chord PQ: $h=\dfrac{ab\cos\left(\frac{\phi-\theta}{2}\right)}{\sqrt{a^2\sin^2\left(\frac{\theta+\phi}{2}\right)+b^2\cos^2\left(\frac{\theta+\phi}{2}\right)}}$.

(iv) Multiply Area $=\frac{1}{2}\times PQ\times h$, filling pages of algebra waiting for the radicals to cancel and yield $\frac{1}{2}ab\sin(\phi-\theta)$.

In this problem, we explore how matrix determinants and vector methods bypass this entire ordeal.

(a) The Matrix Determinant Shortcut: Construct the 3×3 coordinate matrix for vertices $O(0,0)$, $P(a\cos\theta, b\sin\theta)$, and $Q(a\cos\phi, b\sin\phi)$. By evaluating the determinant, derive the exact triangle area $\text{Area}(\triangle OPQ)=\frac{1}{2}ab\sin(\phi-\theta)$ in two lines of algebra.

(b) The Vector Cross-Product Bridge: Express the position vectors of P and Q in 3D as $\mathbf{p}=(a\cos\theta, b\sin\theta, 0)$ and $\mathbf{q}=(a\cos\phi, b\sin\phi, 0)$. Use the cross product $\mathbf{p}\times\mathbf{q}$ to calculate the area, and explain why the result is mathematically identical to the matrix determinant.

(c) Geometric Deduction and Optimization: Determine the maximum possible area of $\triangle OPQ$ as P and Q vary on the ellipse. What is the value of $\phi-\theta$ when this maximum occurs, and what does this mean geometrically for the radii OP and OQ?

Hint:

- **Part (a):** Expand the 3×3 determinant along the first row containing $(0, 0, 1)$, reducing it immediately to a 2×2 determinant. Factor ab out and apply the sine subtraction formula $\sin(\phi - \theta) = \sin\phi\cos\theta - \cos\phi\sin\theta$.
- **Part (b):** Compute the cross product $\mathbf{p} \times \mathbf{q}$. Since both vectors lie in the xy-plane ($z = 0$), the cross product points purely in the $\mathbf{k}$ direction, with magnitude equal to the 2×2 determinant.
- **Part (c):** The area is $\frac{1}{2}ab\sin(\phi-\theta)$. Maximize this with respect to $\phi - \theta$. Two points on an ellipse whose eccentric angles differ by $\frac{\pi}{2}$ define *conjugate semi-diameters*.

Solution 3.10

The Coordinate Geometry Baseline (Context)
As outlined in the historical context, the traditional syllabus method evaluates base PQ and altitude h. When multiplied, the bulky radical $\sqrt{a^2\sin^2\frac{\theta+\phi}{2} + b^2\cos^2\frac{\theta+\phi}{2}}$ cancels identically:

$$\text{Area}(\triangle OPQ) = \frac{1}{2}\cdot PQ \cdot h = \frac{1}{2}\left[2\sin\frac{\phi-\theta}{2}\sqrt{\cdots}\right]\left[\frac{ab\cos\frac{\phi-\theta}{2}}{\sqrt{\cdots}}\right] = \frac{1}{2}ab\sin(\phi-\theta).$$

(a) The Matrix Determinant Shortcut (Two Short Lines)
Using the 3×3 coordinate determinant with vertices $O(0,0)$, P, and Q:

$$\begin{aligned}\text{Area}(\triangle OPQ) &= \frac{1}{2}\left|\det\begin{bmatrix} 0 & 0 & 1 \\ a\cos\theta & b\sin\theta & 1 \\ a\cos\phi & b\sin\phi & 1\end{bmatrix}\right| = \frac{1}{2}\left|1\cdot\det\begin{bmatrix} a\cos\theta & b\sin\theta \\ a\cos\phi & b\sin\phi\end{bmatrix}\right| \\ &= \frac{1}{2}|ab\cos\theta\sin\phi - ab\sin\theta\cos\phi| = \frac{1}{2}ab|\sin\phi\cos\theta - \cos\phi\sin\theta| \\ &= \frac{1}{2}ab\sin(\phi-\theta).\end{aligned}$$

What required an entire page of intricate trigonometric expansions, line equations, and radical cancellations collapses into **two lines of elementary algebra**.

(b) The Vector Cross-Product Bridge
Embedding the vectors in $\mathbb{R}^3$ with $z = 0$: $\mathbf{p} = \begin{bmatrix} a\cos\theta \\ b\sin\theta \\ 0\end{bmatrix}$ and $\mathbf{q} = \begin{bmatrix} a\cos\phi \\ b\sin\phi \\ 0\end{bmatrix}$.
Their cross product points purely along the normal $\mathbf{k}$:

$$\mathbf{p}\times\mathbf{q} = \begin{vmatrix} \mathbf{i} & \mathbf{j} & \mathbf{k} \\ a\cos\theta & b\sin\theta & 0 \\ a\cos\phi & b\sin\phi & 0\end{vmatrix} = \big(ab\cos\theta\sin\phi - ab\sin\theta\cos\phi\big)\mathbf{k} = ab\sin(\phi-\theta)\mathbf{k}.$$

The triangle area is $\frac{1}{2}\|\mathbf{p}\times\mathbf{q}\| = \frac{1}{2}ab\sin(\phi-\theta)$. This is mathematically identical to the matrix determinant because the $\mathbf{k}$-component of $\mathbf{p}\times\mathbf{q}$ is precisely the 2×2 determinant formed by the x and y coordinates.

(c) Geometric Optimization
Since $a, b > 0$, the maximum area occurs when $\sin(\phi-\theta) = 1 \implies \phi - \theta = \frac{\pi}{2}$ (or $90°$):

$$\text{Area}_{\max} = \frac{1}{2}ab.$$

Geometrically, two points on an ellipse whose eccentric angles differ by $\frac{\pi}{2}$ define *conjugate semi-diameters*. Under the horizontal/vertical scaling transformation $x' = x/a, y' = y/b$, the ellipse becomes a unit circle of area π. The conjugate radii map to perpendicular radii forming a right-angled triangle of area $\frac{1}{2}(1)(1) = \frac{1}{2}$. Scaling back by the determinant factor $\det(M) = ab$ yields the maximal area $\frac{1}{2}ab$.

Takeaways 3.10

- **Algebraic Muscle vs Mathematical Insight:** The coordinate geometry approach forces students to compute intermediate radicals (base length and perpendicular height) that exist solely to cancel out. The matrix determinant bypasses these intermediate expressions by evaluating the signed area invariant directly.
- **Matrices as a Universal Bridge:** This problem unites coordinate geometry, trigonometry, matrix algebra, and vector cross products.
- **Affine Invariance:** Scaling coordinates by $1/a$ and $1/b$ transforms the ellipse into a circle. Because determinants preserve area scaling ratios, optimization on an ellipse reduces directly to optimization on a circle.

Problem 3.11: NSW HSC Heritage (1975 & 1976)

In the 1975 and 1976 NSW HSC 4 Unit Mathematics examinations (Question 6 and Question 9 respectively), students were tested on the geometry of linear transformations and matrix powers.

(a) The Reflection Matrix (HSC 1975 Q6(i)): Let L be the straight line through the origin making an angle α with the positive x-axis. By considering the reflections of the unit basis vectors $\mathbf{i} = \begin{bmatrix}1\\0\end{bmatrix}$ and $\mathbf{j} = \begin{bmatrix}0\\1\end{bmatrix}$ across L, prove that the transformation matrix is:

$$R_\alpha = \begin{bmatrix}\cos 2\alpha & \sin 2\alpha \\ \sin 2\alpha & -\cos 2\alpha\end{bmatrix}.$$

(b) Composition of Reflections (HSC 1975 Q6(ii) & 1976 Q9): Let R_α and R_β denote reflections across lines making angles α and β with the positive x-axis. Prove by direct matrix multiplication that the product $R_\beta R_\alpha$ represents a counter-clockwise rotation about the origin through an angle $2(\beta - \alpha)$.

(c) Matrix Powers (HSC 1975 Q6(iii)): Given the matrix $A = \begin{bmatrix}1 & -1\\ 1 & 1\end{bmatrix}$, show that A can be written in polar-rotation form as $\sqrt{2}\,\mathrm{Rot}(45°)$. Hence, for any integer n, compute the exact elements of the matrix power A^{8n+4}.

Hint:

- **Part (a):** The vector $\mathbf{i}$ makes an angle 0 with the x-axis, so its angular displacement from L is α. Its reflected image makes an angle $\alpha + \alpha = 2\alpha$ with the x-axis. Similarly, $\mathbf{j}$ is at angle $\frac{\pi}{2}$, so its reflected image is at angle $2\alpha - \frac{\pi}{2}$.
- **Part (b):** Compute the product $R_\beta R_\alpha$. Apply the compound angle identities: $\cos(A-B) = \cos A\cos B + \sin A\sin B$ and $\sin(A-B) = \sin A\cos B - \cos A\sin B$.
- **Part (c):** Factor out $\sqrt{2}$ so that $A = \sqrt{2}\begin{bmatrix}\cos(\pi/4) & -\sin(\pi/4)\\ \sin(\pi/4) & \cos(\pi/4)\end{bmatrix}$. Then $A^k = 2^{k/2}\mathrm{Rot}(k\pi/4)$. Evaluate the angle for $k = 8n+4$.

Solution 3.11

(a) Derivation of the Reflection Matrix

The columns of a transformation matrix are the images of the basis vectors $\mathbf{i}$ and $\mathbf{j}$.

- The unit vector $\mathbf{i} = (1, 0)$ lies at angle 0. Reflection across the line inclined at angle α produces a unit vector at angle $2\alpha - 0 = 2\alpha$:
$$R_\alpha \mathbf{i} = \begin{bmatrix} \cos 2\alpha \\ \sin 2\alpha \end{bmatrix}.$$
- The unit vector $\mathbf{j} = (0, 1)$ lies at angle $\frac{\pi}{2}$. Reflection across the line at angle α produces a unit vector at angle $2\alpha - \frac{\pi}{2}$:
$$R_\alpha \mathbf{j} = \begin{bmatrix} \cos(2\alpha - \frac{\pi}{2}) \\ \sin(2\alpha - \frac{\pi}{2}) \end{bmatrix} = \begin{bmatrix} \sin 2\alpha \\ -\cos 2\alpha \end{bmatrix}.$$

Combining these two column vectors gives:
$$R_\alpha = \begin{bmatrix} \cos 2\alpha & \sin 2\alpha \\ \sin 2\alpha & -\cos 2\alpha \end{bmatrix}.$$

(b) Product of Two Reflections as a Rotation

Multiplying R_β and R_α:
$$\begin{aligned} R_\beta R_\alpha &= \begin{bmatrix} \cos 2\beta & \sin 2\beta \\ \sin 2\beta & -\cos 2\beta \end{bmatrix} \begin{bmatrix} \cos 2\alpha & \sin 2\alpha \\ \sin 2\alpha & -\cos 2\alpha \end{bmatrix} \\ &= \begin{bmatrix} \cos 2\beta \cos 2\alpha + \sin 2\beta \sin 2\alpha & \cos 2\beta \sin 2\alpha - \sin 2\beta \cos 2\alpha \\ \sin 2\beta \cos 2\alpha - \cos 2\beta \sin 2\alpha & \sin 2\beta \sin 2\alpha + \cos 2\beta \cos 2\alpha \end{bmatrix} \\ &= \begin{bmatrix} \cos(2\beta - 2\alpha) & -\sin(2\beta - 2\alpha) \\ \sin(2\beta - 2\alpha) & \cos(2\beta - 2\alpha) \end{bmatrix} = \mathrm{Rot}\big(2(\beta - \alpha)\big). \end{aligned}$$

Thus, the composite transformation is a rotation through angle $2(\beta - \alpha)$.

(c) Computing A^{8n+4}

Notice that $A = \begin{bmatrix} 1 & -1 \\ 1 & 1 \end{bmatrix} = \sqrt{2} \begin{bmatrix} \frac{1}{\sqrt{2}} & -\frac{1}{\sqrt{2}} \\ \frac{1}{\sqrt{2}} & \frac{1}{\sqrt{2}} \end{bmatrix} = \sqrt{2}\,\mathrm{Rot}\left(\frac{\pi}{4}\right)$.

Since rotation matrices compose by adding angles:
$$A^k = (\sqrt{2})^k \mathrm{Rot}\left(\frac{k\pi}{4}\right) = 2^{k/2} \begin{bmatrix} \cos\left(\frac{k\pi}{4}\right) & -\sin\left(\frac{k\pi}{4}\right) \\ \sin\left(\frac{k\pi}{4}\right) & \cos\left(\frac{k\pi}{4}\right) \end{bmatrix}.$$

Setting $k = 8n + 4$:
$$\frac{k\pi}{4} = \frac{(8n+4)\pi}{4} = 2n\pi + \pi.$$

Since $\cos(2n\pi + \pi) = -1$ and $\sin(2n\pi + \pi) = 0$:
$$\mathrm{Rot}(2n\pi + \pi) = \begin{bmatrix} -1 & 0 \\ 0 & -1 \end{bmatrix} = -I.$$

Furthermore, the scaling factor is $2^{k/2} = 2^{(8n+4)/2} = 2^{4n+2}$. Therefore:
$$A^{8n+4} = 2^{4n+2}(-I) = \begin{bmatrix} -2^{4n+2} & 0 \\ 0 & -2^{4n+2} \end{bmatrix}.$$

Takeaways 3.11

- **Reflections and Involutions:** Every reflection matrix R_α satisfies $R_\alpha^2 = I$ and $\det(R_\alpha) = -1$.
- **Reflections Generate Rotations:** Any rotation through θ can be decomposed into two reflections across lines separated by $\theta/2$.
- **Geometric Matrix Powers:** Factoring a transformation matrix into a scaling factor times an orthogonal rotation matrix renders high matrix powers trivial, avoiding messy matrix expansions.

4 Part 2: Practice and Challenge Bank

4.1 Warm-Up Drills

Problem 4.1: Matrix Systems and Polynomial Curve Fitting

A quadratic function has a rule of the form $f(x) = ax^2 + bx + c$. It is known that the points $(-1, 12)$ and $(1, 4)$ are on the graph of f, alongside a third point $(2, 9)$.

(a) By substituting the known points, write down three linear equations satisfied by a, b, and c. Formulate this information as a matrix equation of the form $M\mathbf{v} = \mathbf{k}$, where $\mathbf{v} = \begin{bmatrix} a \\ b \\ c \end{bmatrix}$.

(b) The inverse of the coefficient matrix M is given by $M^{-1} = \frac{1}{6}\begin{bmatrix} 1 & -3 & 2 \\ -3 & 3 & 0 \\ 2 & 6 & -2 \end{bmatrix}$. Use this inverse to solve the matrix equation and determine the exact values of a, b, and c.

(c) Suppose the y-coordinate of the third point is changed to a parameter p, such that the point is now $(2, p)$. Write down the new constant vector $\mathbf{k}$, and use M^{-1} to determine the coefficients a, b, and c in terms of p.

(d) Determine the specific value of p for which the quadratic function degenerates into a linear function (i.e., $a = 0$). Geometrically, what does this imply about the arrangement of the three points on the Cartesian plane?

Hint:

- **Part (a):** Substitute the x and y values of each coordinate pair into the quadratic equation to generate three distinct linear equations. The coefficients of a, b, and c will form your 3×3 matrix M.
- **Part (b):** To solve $M\mathbf{v} = \mathbf{k}$, pre-multiply both sides by the inverse matrix so that $\mathbf{v} = M^{-1}\mathbf{k}$.
- **Part (c):** Update the third entry of your $\mathbf{k}$ vector to p. Multiply the matrix M^{-1} by this new vector using standard row-by-column multiplication.
- **Part (d):** Take your parametric expression for a from part (c) and set it equal to 0. If a quadratic curve becomes a straight line, consider what that means for the points that define it.

Solution 4.1

(a)

Substituting $(-1, 12)$: $a(-1)^2 + b(-1) + c = 12 \implies a - b + c = 12$
Substituting $(1, 4)$: $a(1)^2 + b(1) + c = 4 \implies a + b + c = 4$
Substituting $(2, 9)$: $a(2)^2 + b(2) + c = 9 \implies 4a + 2b + c = 9$
Matrix equation: $\begin{bmatrix} 1 & -1 & 1 \\ 1 & 1 & 1 \\ 4 & 2 & 1 \end{bmatrix} \begin{bmatrix} a \\ b \\ c \end{bmatrix} = \begin{bmatrix} 12 \\ 4 \\ 9 \end{bmatrix}$.

(b)

$\mathbf{v} = M^{-1}\mathbf{k} = \frac{1}{6} \begin{bmatrix} 1 & -3 & 2 \\ -3 & 3 & 0 \\ 2 & 6 & -2 \end{bmatrix} \begin{bmatrix} 12 \\ 4 \\ 9 \end{bmatrix}$
$a = \frac{1}{6}(1(12) - 3(4) + 2(9)) = \frac{1}{6}(12 - 12 + 18) = 3$
$b = \frac{1}{6}(-3(12) + 3(4) + 0(9)) = \frac{1}{6}(-36 + 12) = -4$
$c = \frac{1}{6}(2(12) + 6(4) - 2(9)) = \frac{1}{6}(24 + 24 - 18) = 5$
Therefore, $a = 3$, $b = -4$, and $c = 5$.

(c)

With the new point $(2, p)$, the constant vector becomes $\mathbf{k} = \begin{bmatrix} 12 \\ 4 \\ p \end{bmatrix}$.

$\mathbf{v} = \frac{1}{6} \begin{bmatrix} 1 & -3 & 2 \\ -3 & 3 & 0 \\ 2 & 6 & -2 \end{bmatrix} \begin{bmatrix} 12 \\ 4 \\ p \end{bmatrix}$
$a = \frac{1}{6}(12 - 12 + 2p) = \frac{2p}{6} = \frac{p}{3}$
$b = \frac{1}{6}(-36 + 12 + 0) = -4$
$c = \frac{1}{6}(24 + 24 - 2p) = \frac{48-2p}{6} = 8 - \frac{p}{3}$.

(d)

For the function to be linear, $a = 0$.
$\frac{p}{3} = 0 \implies p = 0$.
If $a = 0$, the curve is a straight line $y = -4x + 8$. Geometrically, this implies that the three points $(-1, 12)$, $(1, 4)$, and $(2, 0)$ are perfectly collinear.

Takeaways 4.1

- **Vandermonde Matrices:** The coefficient matrix formed by fitting a polynomial to coordinate points is known as a Vandermonde matrix. Its determinant is always non-zero as long as the x-coordinates are distinct, guaranteeing a unique polynomial fit.
- **Parametric Matrix Operations:** Treating unknowns as parameters (like p) within a matrix equation allows you to generalize solutions for an entire family of curves at once, rather than re-solving the system from scratch.
- **Geometric Limits:** Setting the leading coefficient of a polynomial to zero algebraically represents flattening the curve's curvature, visually forcing the defining points into a straight line.

Problem 4.2: Matrix Systems and Intersecting Planes

Consider the following system of homogeneous linear equations, which represent three planes passing through the origin:

$$\begin{aligned} x - y - z &= 0 \\ 2x + 4y - z &= 0 \\ 3x + 3y + kz &= 0 \end{aligned}$$

where k is a real constant.

(a) Write the system in the matrix form $AX = O$, where O is the zero vector. Find the determinant of the coefficient matrix A in terms of k, and determine the specific value of k for which A is a singular matrix.

(b) For the singular value of k found in part (a), show algebraically that the system is consistent. Let $y = -t$ and solve the simultaneous equations for x and z strictly in terms of the parameter t.

(c) Hence, write the vector equation of the line of intersection of these planes in the form $\mathbf{r} = t(a\mathbf{i} + b\mathbf{j} + c\mathbf{k})$. Briefly explain how this line relates to the intersection of just the first two planes in the system.

(d) If $k = 4$, determine the unique solution to the system algebraically, and explain what this result means about the geometric arrangement of the three planes.

Hint:

- **Part (a):** The coefficient matrix A consists of the multipliers of x, y, and z. Calculate the determinant of this 3×3 matrix by expanding across the top row. A matrix is singular when its determinant is exactly zero.
- **Part (b):** Substitute the value of k into the third equation. To prove consistency, look for a linear combination of the first two equations (e.g., adding them together) that perfectly matches the third equation. Then, substitute $y = -t$ into the first two equations and solve the resulting system for x and z.
- **Part (c):** Construct a position vector $\mathbf{r} = x\mathbf{i} + y\mathbf{j} + z\mathbf{k}$ using your parametric expressions from part (b), and factor out t.
- **Part (d):** If $k = 4$, calculate $\det(A)$ to confirm it is non-zero (meaning the matrix is invertible). Since the constant vector is the zero vector, multiplying it by any inverse matrix will always yield the zero vector.

Solution 4.2

(a)

The matrix equation is $\begin{bmatrix} 1 & -1 & -1 \\ 2 & 4 & -1 \\ 3 & 3 & k \end{bmatrix} \begin{bmatrix} x \\ y \\ z \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix}$.

$\det(A) = 1(4k - (-3)) - (-1)(2k - (-3)) + (-1)(6 - 12)$

$\det(A) = (4k + 3) + (2k + 3) - (-6) = 6k + 12.$

For A to be singular, $\det(A) = 0 \implies 6k + 12 = 0 \implies k = -2$.

(b)

Substitute $k = -2$: the third equation is $3x + 3y - 2z = 0$.

Adding Equation 1 and Equation 2 gives:

$(x - y - z) + (2x + 4y - z) = 3x + 3y - 2z = 0.$

Since this perfectly matches the third equation, the system is consistent (the third plane is dependent on the first two).

Let $y = -t$.

Equation 1 becomes: $x - (-t) - z = 0 \implies x - z = -t$.

Equation 2 becomes: $2x + 4(-t) - z = 0 \implies 2x - z = 4t$.

Subtracting Equation 1 from Equation 2: $(2x - z) - (x - z) = 4t - (-t) \implies x = 5t$.

Substituting x back into Equation 1: $5t - z = -t \implies z = 6t$.

The solution is $x = 5t, y = -t, z = 6t$.

(c)

The vector equation is $\mathbf{r} = 5t\mathbf{i} - t\mathbf{j} + 6t\mathbf{k} = t(5\mathbf{i} - \mathbf{j} + 6\mathbf{k})$.

Because the third plane is simply a linear combination of the first two, it contains their line of intersection. Therefore, this vector equation represents the straight line intersection of the planes $x - y - z = 0$ and $2x + 4y - z = 0$ through the origin.

(d)

If $k = 4$, $\det(A) = 6(4) + 12 = 36$.

Since $\det(A) \neq 0$, matrix A is invertible.

$X = A^{-1}O = O \implies x = 0, y = 0, z = 0$.

Geometrically, this means the three planes intersect at a single, unique point, which is the origin $(0, 0, 0)$.

Takeaways 4.2

- **Homogeneous Systems:** A system where all constants are zero always has at least one solution (the trivial solution: the origin). Whether it has *infinite* solutions depends on singularity.
- **Singularity and Intersection:** If $\det A = 0$, the homogeneous system has non-trivial solutions. If $\text{rank}(A) = 2$, the solution set is a line; lower rank gives a higher-dimensional solution set.
- **Parametric Conversions:** Expressing an infinite solution set algebraically using a parameter (t) maps directly to the Cartesian components of a 3D vector line equation.

4.2 Stretch Problems

Problem 4.3: Leslie Matrices and Parametric Population Modeling

Based on standard Leslie matrix population modeling principles, consider a modified species of female insect with a maximum life span of 3 months. The population is divided into three age groups of 1 month each.
The Leslie matrix, L, for this population is given by:

$$L = \begin{bmatrix} 0 & k & 4 \\ 0.5 & 0 & 0 \\ 0 & 0.2 & 0 \end{bmatrix}$$

where k represents the unknown fecundity (birth rate) of the second age group. Assume that each of the three age groups initially consists of 400 females, represented by the initial population state matrix P_0.

(a) Express the population column matrix after 1 month, P_1, strictly in terms of k.

(b) Given that the total population of female insects after 1 month is exactly 2680, algebraically determine the value of k.

(c) Using your value of k from part (b), an environmental pesticide is introduced that immediately halves the survival rates of all age groups moving forward. Construct the new Leslie matrix, L_{new}, and calculate the exact population matrix for month 2, P_2.

(d) Evaluate the determinant of the original matrix L. Deduce whether the original population transition model is mathematically reversible (invertible), and algebraically explain why the value of the determinant is completely independent of the fecundity parameter k.

Hint:

- **Part (a):** The initial population matrix is a 3×1 column vector containing the value 400 in each row. Multiply the 3×3 Leslie matrix by this vector using standard row-by-column multiplication.
- **Part (b):** Sum the three row elements of your P_1 matrix to find an expression for the total population in terms of k. Set this expression equal to 2680 and solve the resulting linear equation.
- **Part (c):** The survival rates are located on the sub-diagonal of the Leslie matrix (the 0.5 and 0.2). Divide these by 2 to create L_{new}. To find P_2, multiply L_{new} by the numerical P_1 matrix you found in part (b).
- **Part (d):** Calculate the determinant of the 3×3 matrix L by expanding along the first column or the third row, as they contain multiple zeros. A matrix model is reversible (has an inverse) if its determinant is non-zero.

Solution 4.3

(a)

$$P_0 = \begin{bmatrix} 400 \\ 400 \\ 400 \end{bmatrix}$$

$$P_1 = LP_0 = \begin{bmatrix} 0 & k & 4 \\ 0.5 & 0 & 0 \\ 0 & 0.2 & 0 \end{bmatrix} \begin{bmatrix} 400 \\ 400 \\ 400 \end{bmatrix}$$

$$P_1 = \begin{bmatrix} 0(400) + k(400) + 4(400) \\ 0.5(400) + 0 + 0 \\ 0 + 0.2(400) + 0 \end{bmatrix} = \begin{bmatrix} 400k + 1600 \\ 200 \\ 80 \end{bmatrix}.$$

(b)

The total population is the sum of the elements in P_1:
Total $= (400k + 1600) + 200 + 80 = 400k + 1880$.
We are given that the total population is 2680.
$400k + 1880 = 2680 \implies 400k = 800 \implies k = 2$.

(c)

With $k = 2$, $P_1 = \begin{bmatrix} 2400 \\ 200 \\ 80 \end{bmatrix}$.

The pesticide halves the survival rates ($0.5 \to 0.25$ and $0.2 \to 0.1$). The fecundities in the top row remain unchanged.

$$L_{\text{new}} = \begin{bmatrix} 0 & 2 & 4 \\ 0.25 & 0 & 0 \\ 0 & 0.1 & 0 \end{bmatrix}$$

$$P_2 = L_{\text{new}} P_1 = \begin{bmatrix} 0 & 2 & 4 \\ 0.25 & 0 & 0 \\ 0 & 0.1 & 0 \end{bmatrix} \begin{bmatrix} 2400 \\ 200 \\ 80 \end{bmatrix}$$

$$P_2 = \begin{bmatrix} 0 + 2(200) + 4(80) \\ 0.25(2400) + 0 + 0 \\ 0 + 0.1(200) + 0 \end{bmatrix} = \begin{bmatrix} 400 + 320 \\ 600 \\ 20 \end{bmatrix} = \begin{bmatrix} 720 \\ 600 \\ 20 \end{bmatrix}.$$

After 2 months, there are 720 insects in age group 1, 600 in age group 2, and 20 in age group 3.

(d)

Expanding the determinant of L along the first column:

$$\det(L) = 0 - 0.5 \det \begin{bmatrix} k & 4 \\ 0.2 & 0 \end{bmatrix} + 0$$

$\det(L) = -0.5\,((k)(0) - (4)(0.2)) = -0.5(0 - 0.8) = 0.4$.
Since $\det(L) = 0.4 \neq 0$, the matrix L is invertible, meaning the population transition can be mathematically reversed to find past population states.
The value of k disappears during the determinant calculation because expanding the 2×2 minor matrix forces k to be multiplied by 0 (the bottom right element of the Leslie matrix). In a standard Leslie matrix, the determinant depends strictly on the product of the survival rates and the final age group's fecundity.

Takeaways 4.3

- **Parametric Modeling:** Using matrices to solve for unknown biological parameters (like fecundity k) demonstrates how models are calibrated using empirical field data (e.g., total population counts).
- **Dynamic Matrices:** Real-world matrix models are rarely static. Understanding how to manually manipulate specific elements (like survival rates) allows for the simulation of environmental impacts over discrete time steps.
- **Biological Meaning of Determinants:** The algebraic structure of a Leslie matrix dictates that its determinant is independent of intermediate birth rates. A singular Leslie matrix (determinant of zero) only occurs if an age group has a zero survival rate or if the oldest age group becomes sterile, mathematically erasing historical population data.

Problem 4.4: Leslie Matrices and Stable Populations

Consider a certain species of fish divided into three age groups, each one year in length. The fish do not reproduce in their first year, produce an average of 1.5 offspring in their second year, and produce an unknown average of k offspring in their third year, after which they die. Assume that 40% of the fish survive their first year, and 50% of those survivors make it to their third year. The initial population consists of 1000 newborns and zero older fish.

(a) Construct the Leslie matrix, L, for this fish population and calculate the population state column matrix after 2 years, P_2.

(b) Determine an algebraic expression for the number of newborns present in year 3 in terms of k.

(c) For the population to eventually reach a strict long-term stable size (a long-term growth rate of exactly 1), determine the required exact value of the parameter k.

(d) Using the value of k determined in part (c), find the long-term stable age distribution of the fish population, expressing the number of fish in each age group as a simplified fraction of the total population.

Hint:

- **Part (a):** The Leslie matrix is a 3×3 matrix where the first row contains the fecundities (birth rates) and the sub-diagonal contains the survival rates. To find P_2, multiply the initial population column matrix by L twice (or by L^2).
- **Part (b):** Multiply your result for P_2 by the Leslie matrix L to find P_3. The top element of this resulting matrix represents the newborns.
- **Part (c):** A long-term growth rate of 1 means the population state eventually satisfies the equation $LP = P$, where $P = \begin{bmatrix} x \\ y \\ z \end{bmatrix}$. Expand this matrix equation into a system of simultaneous linear equations and solve for k.
- **Part (d):** Using the system of equations from part (c), express the number of fish in each age group (x, y, and z) in terms of a single variable, such as y. Sum these to find the total population, and then write each age group as a fraction of that total.

Solution 4.4

(a)

The Leslie matrix is $L = \begin{bmatrix} 0 & 1.5 & k \\ 0.4 & 0 & 0 \\ 0 & 0.5 & 0 \end{bmatrix}$.

The initial population is $P_0 = \begin{bmatrix} 1000 \\ 0 \\ 0 \end{bmatrix}$.

$$P_1 = LP_0 = \begin{bmatrix} 0 & 1.5 & k \\ 0.4 & 0 & 0 \\ 0 & 0.5 & 0 \end{bmatrix} \begin{bmatrix} 1000 \\ 0 \\ 0 \end{bmatrix} = \begin{bmatrix} 0 \\ 400 \\ 0 \end{bmatrix}.$$

$$P_2 = LP_1 = \begin{bmatrix} 0 & 1.5 & k \\ 0.4 & 0 & 0 \\ 0 & 0.5 & 0 \end{bmatrix} \begin{bmatrix} 0 \\ 400 \\ 0 \end{bmatrix} = \begin{bmatrix} 600 \\ 0 \\ 200 \end{bmatrix}.$$

(b)

$$P_3 = LP_2 = \begin{bmatrix} 0 & 1.5 & k \\ 0.4 & 0 & 0 \\ 0 & 0.5 & 0 \end{bmatrix} \begin{bmatrix} 600 \\ 0 \\ 200 \end{bmatrix} = \begin{bmatrix} 0(600) + 1.5(0) + k(200) \\ 0.4(600) + 0 + 0 \\ 0 + 0.5(0) + 0 \end{bmatrix} = \begin{bmatrix} 200k \\ 240 \\ 0 \end{bmatrix}.$$

The number of newborns in year 3 is $200k$.

(c)

For a long-term growth rate of 1, the stable state satisfies $LP = P$:

$$\begin{bmatrix} 0 & 1.5 & k \\ 0.4 & 0 & 0 \\ 0 & 0.5 & 0 \end{bmatrix} \begin{bmatrix} x \\ y \\ z \end{bmatrix} = \begin{bmatrix} x \\ y \\ z \end{bmatrix}$$

This yields the system of equations:

(1) $1.5y + kz = x$

(2) $0.4x = y \implies x = 2.5y$

(3) $0.5y = z$

Substitute (2) and (3) into (1):

$1.5y + k(0.5y) = 2.5y$

Assuming a non-zero population, divide by y:

$1.5 + 0.5k = 2.5 \implies 0.5k = 1 \implies k = 2.$

(d)

With $k = 2$, the age group relations are $x = 2.5y$ and $z = 0.5y$.

Total population $T = x + y + z = 2.5y + y + 0.5y = 4y$.

Proportion of age group 1: $\frac{x}{T} = \frac{2.5y}{4y} = \frac{2.5}{4} = \frac{5}{8}$.

Proportion of age group 2: $\frac{y}{T} = \frac{1y}{4y} = \frac{1}{4} = \frac{2}{8}$.

Proportion of age group 3: $\frac{z}{T} = \frac{0.5y}{4y} = \frac{0.5}{4} = \frac{1}{8}$.

The stable age distribution is $\frac{5}{8}$ newborns, $\frac{1}{4}$ one-year-olds, and $\frac{1}{8}$ two-year-olds.

Takeaways 4.4

- **Eigenvalues in Context:** A long-term growth rate of 1 implies that the dominant eigenvalue of the Leslie matrix is $\lambda = 1$. Setting $LP = 1P$ allows you to bypass the characteristic polynomial and directly solve for matrix parameters.
- **Stable Age Distributions:** Once a population reaches its long-term growth trajectory (determined by the dominant eigenvalue), the *ratio* of the age groups remains constant year over year, representing the corresponding eigenvector.
- **Normalising an Eigenvector:** A stable age distribution is a ratio, so an eigenvector is only determined up to scale. After solving $LP = P$, divide each component by the sum of all components to convert the eigenvector into population proportions.

Problem 4.5: Leslie Matrices and Periodic Populations

A particular species of insect has a maximum life span of 3 months. The female insects reproduce only in their third month of life. When the female population is divided into three age groups, each of length one month, the Leslie matrix takes the form:

$$L = \begin{bmatrix} 0 & 0 & 8 \\ 0.5 & 0 & 0 \\ 0 & 0.25 & 0 \end{bmatrix}$$

(a) Evaluate L^3. Hence, deduce the long-term growth rate of this population and describe the cyclical nature of the population sizes every 3 months.

(b) Determine the stable age distribution for this population by finding the vector $\mathbf{v} = \begin{bmatrix} 1 \\ c \\ d \end{bmatrix}$ such that $L\mathbf{v} = \mathbf{v}$. Express this stable distribution as fractions of the total population.

(c) Suppose the initial female population consists purely of newborns: $P_0 = \begin{bmatrix} 1200 \\ 0 \\ 0 \end{bmatrix}$. Calculate the population matrices P_1, P_2, and P_3. Explain why this specific population does not appear to exhibit a stable age distribution over these initial months.

(d) To stabilize the population immediately, a conservationist proposes adding a specific number of 2nd-month and 3rd-month insects to the initial population P_0. Determine the exact number of insects that must be added to age groups 2 and 3 so that the new initial population P_{new} satisfies $LP_{\text{new}} = P_{\text{new}}$.

Hint:

- **Part (a):** Multiply L by itself to find L^2, and then multiply by L again to find L^3. Think about what $P_3 = L^3P_0$ means if L^3 simplifies to a scalar multiple of the identity matrix.
- **Part (b):** Multiply matrix L by the column vector $\mathbf{v}$ and set the result equal to $\mathbf{v}$. This will give you a system of linear equations to solve for c and d. Sum the components to find the fractional distribution.
- **Part (c):** Use the recursive formula $P_n = LP_{n-1}$. Observe where the non-zero values shift each month.
- **Part (d):** For P_{new} to be perfectly stable, it must be a scalar multiple of the stable age vector $\mathbf{v}$ you found in part (b). Since the number of newborns is fixed at 1200, use this to find the scalar multiplier.

Solution 4.5

(a)

$$L^2 = \begin{bmatrix} 0 & 0 & 8 \\ 0.5 & 0 & 0 \\ 0 & 0.25 & 0 \end{bmatrix} \begin{bmatrix} 0 & 0 & 8 \\ 0.5 & 0 & 0 \\ 0 & 0.25 & 0 \end{bmatrix} = \begin{bmatrix} 0 & 2 & 0 \\ 0 & 0 & 4 \\ 0.125 & 0 & 0 \end{bmatrix}.$$

$$L^3 = L^2 L = \begin{bmatrix} 0 & 2 & 0 \\ 0 & 0 & 4 \\ 0.125 & 0 & 0 \end{bmatrix} \begin{bmatrix} 0 & 0 & 8 \\ 0.5 & 0 & 0 \\ 0 & 0.25 & 0 \end{bmatrix} = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix} = I.$$

Since $L^3 = I$, applying the matrix 3 times leaves the population unchanged ($P_3 = IP_0 = P_0$). The long-term growth rate is 1. The population perfectly cycles every 3 months.

(b)

Setting $L\mathbf{v} = \mathbf{v}$:

$$\begin{bmatrix} 0 & 0 & 8 \\ 0.5 & 0 & 0 \\ 0 & 0.25 & 0 \end{bmatrix} \begin{bmatrix} 1 \\ c \\ d \end{bmatrix} = \begin{bmatrix} 1 \\ c \\ d \end{bmatrix} \implies \begin{bmatrix} 8d \\ 0.5 \\ 0.25c \end{bmatrix} = \begin{bmatrix} 1 \\ c \\ d \end{bmatrix}.$$

From row 2: $c = 0.5 = 1/2$.

From row 1: $8d = 1 \implies d = 1/8$.

The vector is $\begin{bmatrix} 1 \\ 1/2 \\ 1/8 \end{bmatrix}$. The total parts are $1 + \frac{1}{2} + \frac{1}{8} = \frac{13}{8}$.

Fractions: Age 1 is $\frac{1}{13/8} = \frac{8}{13}$, Age 2 is $\frac{1/2}{13/8} = \frac{4}{13}$, Age 3 is $\frac{1/8}{13/8} = \frac{1}{13}$.

(c)

$$P_1 = LP_0 = \begin{bmatrix} 0 & 0 & 8 \\ 0.5 & 0 & 0 \\ 0 & 0.25 & 0 \end{bmatrix} \begin{bmatrix} 1200 \\ 0 \\ 0 \end{bmatrix} = \begin{bmatrix} 0 \\ 600 \\ 0 \end{bmatrix}.$$

$$P_2 = LP_1 = \begin{bmatrix} 0 & 0 & 8 \\ 0.5 & 0 & 0 \\ 0 & 0.25 & 0 \end{bmatrix} \begin{bmatrix} 0 \\ 600 \\ 0 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 150 \end{bmatrix}.$$

$$P_3 = LP_2 = \begin{bmatrix} 0 & 0 & 8 \\ 0.5 & 0 & 0 \\ 0 & 0.25 & 0 \end{bmatrix} \begin{bmatrix} 0 \\ 0 \\ 150 \end{bmatrix} = \begin{bmatrix} 1200 \\ 0 \\ 0 \end{bmatrix}.$$

The population is not in a stable age distribution because the initial state consisted only of one age group. In a species that only reproduces in its final life stage, a cohort gap creates a traveling demographic wave that never smooths out unless multiple generations overlap initially.

(d)

We need $P_{\text{new}} = \begin{bmatrix} 1200 \\ y \\ z \end{bmatrix}$ to be a scalar multiple of $\mathbf{v} = \begin{bmatrix} 1 \\ 1/2 \\ 1/8 \end{bmatrix}$.

The scalar multiplier is 1200 (since $1 \times 1200 = 1200$).

$y = 1200 \times \frac{1}{2} = 600$.

$z = 1200 \times \frac{1}{8} = 150$.

Therefore, the conservationist must add 600 2nd-month insects and 150 3rd-month insects.

Takeaways 4.5

- **Imprimitive Matrices:** Leslie matrices where reproduction only occurs in a single, final age group are "imprimitive." They do not naturally converge to a stable age distribution from an arbitrary starting point; instead, they oscillate cyclically.
- **Eigenvectors as Proportions:** The steady-state vector (corresponding to a growth multiplier of 1) represents the exact population ratio required to maintain a static, unchanging demographic structure year over year.
- **Matrix Exponents:** Discovering that $L^k = I$ is a powerful algebraic shortcut. It immediately proves that the system is periodic with period k and that the long-term growth multiplier is exactly 1.

4.3 Challenge Corner

Problem 4.6: Parametric Simultaneous Equations and Matrices

Consider the simultaneous linear equations involving real constants m and n:

$$mx + 2y = 4$$
$$8x + my = n$$

(a) Express this system as a matrix equation of the form $AX = B$. Hence, determine the inverse matrix A^{-1} in terms of m, stating the specific values of m for which A^{-1} is undefined.

(b) Determine the values of m for which the system does *not* have a unique solution.

(c) Given that m takes the positive value found in part (b), determine the exact value of n that results in an infinite set of solutions. Furthermore, provide a geometric interpretation of the system if $n = 10$ instead.

(d) Suppose $n = 2m$. Assuming a unique solution exists, use the matrix inverse method to solve for x and y in terms of m, expressing your final answers in their simplest form.

Hint:

- **Part (a):** Set up the coefficients of x and y into a 2×2 matrix. The formula for the inverse includes dividing by the determinant, $ad - bc$. The inverse is undefined when this determinant is equal to zero.
- **Part (b):** A unique solution exists if and only if the coefficient matrix A is invertible. Therefore, a unique solution does *not* exist when the determinant you found in part (a) is zero.
- **Part (c):** Substitute the positive value of m into the original equations. For infinite solutions, the two line equations must be mathematically equivalent (one is a scalar multiple of the other). If the lines have the same gradient but different y-intercepts, what does that mean geometrically?
- **Part (d):** Substitute $n = 2m$ into your column matrix B. Multiply A^{-1} by B using standard row-by-column multiplication. Simplify the resulting algebraic expressions for x and y by factoring out common terms.

Solution 4.6

(a)

Matrix equation: $\begin{bmatrix} m & 2 \\ 8 & m \end{bmatrix} \begin{bmatrix} x \\ y \end{bmatrix} = \begin{bmatrix} 4 \\ n \end{bmatrix}$.

The determinant is $\det(A) = (m)(m) - (2)(8) = m^2 - 16$.

$A^{-1} = \frac{1}{m^2-16} \begin{bmatrix} m & -2 \\ -8 & m \end{bmatrix}$.

The inverse is undefined when $m^2 - 16 = 0$, which occurs when $m = 4$ or $m = -4$.

(b)

The system does not have a unique solution when the coefficient matrix is singular (i.e., $\det(A) = 0$). From part (a), this occurs when $m = 4$ or $m = -4$.

(c)

Let $m = 4$. The system becomes:

(1) $4x + 2y = 4 \implies 2x + y = 2$

(2) $8x + 4y = n \implies 2x + y = \frac{n}{4}$

For an infinite set of solutions, the equations must represent the same line. Therefore, $2 = \frac{n}{4} \implies n = 8$. If $n = 10$, the equations are $2x + y = 2$ and $2x + y = 2.5$. These equations represent distinct, parallel lines that never intersect, meaning there is no solution.

(d)

Let $n = 2m$. The matrix B becomes $\begin{bmatrix} 4 \\ 2m \end{bmatrix}$.

$X = A^{-1}B = \frac{1}{m^2-16} \begin{bmatrix} m & -2 \\ -8 & m \end{bmatrix} \begin{bmatrix} 4 \\ 2m \end{bmatrix}$

$X = \frac{1}{m^2-16} \begin{bmatrix} (m)(4) + (-2)(2m) \\ (-8)(4) + (m)(2m) \end{bmatrix}$

$X = \frac{1}{m^2-16} \begin{bmatrix} 4m - 4m \\ -32 + 2m^2 \end{bmatrix} = \frac{1}{m^2-16} \begin{bmatrix} 0 \\ 2(m^2 - 16) \end{bmatrix}$

$X = \begin{bmatrix} 0 \\ 2 \end{bmatrix}$.

The unique solution simplifies elegantly to $x = 0$ and $y = 2$ (for $m \neq \pm 4$).

Takeaways 4.6

- **Parametric Determinants:** Using parameters in the coefficient matrix directly links algebraic singularity (determinant $= 0$) to geometric relationships (parallel or coincident lines).
- **Infinite vs. No Solution:** Once a matrix is singular, comparing the constant terms (the B matrix) dictates whether the lines are coincident (infinite solutions) or strictly parallel (no solution).
- **Power of Matrix Algebra:** Part (d) demonstrates how matrix inverse multiplication can elegantly simplify complex parametric systems into clean integer solutions through algebraic factorization.

Problem 4.7: Matrix Systems and Geometric Transformations

Consider the following system of linear equations:

$$\sqrt{3}x - y = 4\sqrt{3}$$
$$x + \sqrt{3}y = 4$$

(a) Write this system in the matrix form $AX = B$, and determine the exact value of $\det(A)$.

(b) Determine A^{-1}, and hence solve the system of equations algebraically for $X = \begin{bmatrix} x \\ y \end{bmatrix}$.

(c) Express the coefficient matrix A in the form $k \begin{bmatrix} \cos\theta & -\sin\theta \\ \sin\theta & \cos\theta \end{bmatrix}$, where $k > 0$ and $0 \leq \theta \leq \frac{\pi}{2}$. Hence, describe the two sequential geometric transformations represented by the matrix A.

(d) The matrix equation $AX = B$ can be interpreted geometrically as finding a pre-image point (x, y) that maps to the image point $(4\sqrt{3}, 4)$ under transformation A. Using your geometric description from part (c), deduce the coordinates of (x, y) using polar coordinates and transformations, verifying your algebraic answer from part (b).

Hint:

- **Part (a):** Extract the coefficients of x and y carefully. The determinant is calculated as $ad - bc$. Be mindful of squaring surds and negative signs.
- **Part (b):** The inverse of a 2×2 matrix is $\frac{1}{\det(A)} \begin{bmatrix} d & -b \\ -c & a \end{bmatrix}$. To solve for X, pre-multiply the constant matrix B by A^{-1}.
- **Part (c):** Notice that the determinant of a pure rotation matrix is always 1. The determinant of A is not 1, which implies a dilation is present. The dilation factor k is equal to $\sqrt{\det(A)}$. Factor this k out of matrix A to reveal the standard rotation matrix.
- **Part (d):** Convert the image point $(4\sqrt{3}, 4)$ into polar form (r, ϕ). Since applying A dilates and rotates the pre-image *forward*, applying A^{-1} to find the pre-image must do the exact opposite: divide the radius by k and rotate *backwards* by θ.

Solution 4.7

(a)

Matrix form: $\begin{bmatrix} \sqrt{3} & -1 \\ 1 & \sqrt{3} \end{bmatrix} \begin{bmatrix} x \\ y \end{bmatrix} = \begin{bmatrix} 4\sqrt{3} \\ 4 \end{bmatrix}$

$\det(A) = (\sqrt{3})(\sqrt{3}) - (-1)(1) = 3 - (-1) = 4.$

(b)

$A^{-1} = \frac{1}{4} \begin{bmatrix} \sqrt{3} & 1 \\ -1 & \sqrt{3} \end{bmatrix}.$

$X = A^{-1}B = \frac{1}{4} \begin{bmatrix} \sqrt{3} & 1 \\ -1 & \sqrt{3} \end{bmatrix} \begin{bmatrix} 4\sqrt{3} \\ 4 \end{bmatrix}$

$X = \frac{1}{4} \begin{bmatrix} \sqrt{3}(4\sqrt{3}) + 1(4) \\ -1(4\sqrt{3}) + \sqrt{3}(4) \end{bmatrix} = \frac{1}{4} \begin{bmatrix} 12 + 4 \\ -4\sqrt{3} + 4\sqrt{3} \end{bmatrix} = \frac{1}{4} \begin{bmatrix} 16 \\ 0 \end{bmatrix} = \begin{bmatrix} 4 \\ 0 \end{bmatrix}.$

Therefore, $x = 4$ and $y = 0$.

(c)

Since $\det(A) = 4$, the dilation factor is $k = \sqrt{4} = 2$.

Factoring out 2: $A = 2 \begin{bmatrix} \frac{\sqrt{3}}{2} & -\frac{1}{2} \\ \frac{1}{2} & \frac{\sqrt{3}}{2} \end{bmatrix}.$

We need $\cos\theta = \frac{\sqrt{3}}{2}$ and $\sin\theta = \frac{1}{2}$, which gives $\theta = \frac{\pi}{6}$ (or $30°$).

Thus, $A = 2 \begin{bmatrix} \cos(\frac{\pi}{6}) & -\sin(\frac{\pi}{6}) \\ \sin(\frac{\pi}{6}) & \cos(\frac{\pi}{6}) \end{bmatrix}.$

Geometrically, matrix A represents a **dilation by a factor of 2** from the origin, followed by an **anti-clockwise rotation of $\frac{\pi}{6}$ radians** about the origin.

(d)

The image point is $(4\sqrt{3}, 4)$. In polar coordinates:

$r = \sqrt{(4\sqrt{3})^2 + 4^2} = \sqrt{48 + 16} = \sqrt{64} = 8.$

Angle $\phi = \tan^{-1}\left(\frac{4}{4\sqrt{3}}\right) = \tan^{-1}\left(\frac{1}{\sqrt{3}}\right) = \frac{\pi}{6}.$

So the image point is $(8, \frac{\pi}{6})$ in polar.

The inverse transformation A^{-1} reverses the effects of A: it dilates by $\frac{1}{2}$ and rotates clockwise by $\frac{\pi}{6}$.

New radius $r' = 8 \times \frac{1}{2} = 4$.

New angle $\phi' = \frac{\pi}{6} - \frac{\pi}{6} = 0$.

The pre-image in polar coordinates is $(4, 0)$, which corresponds to the Cartesian coordinates $(4, 0)$. This verifies the algebraic solution $x = 4, y = 0$.

Takeaways 4.7

- **Algebra Meets Geometry:** Solving a system of linear equations is algebraically identical to finding the pre-image of a point under a 2D matrix transformation.
- **Identifying Transformations:** Any matrix of the form $\begin{bmatrix} a & -b \\ b & a \end{bmatrix}$ represents a combined rotation and dilation. The dilation factor is $\sqrt{a^2+b^2}$ (or $\sqrt{\det(A)}$), and the rotation angle θ is found using right-triangle trigonometry with a and b.
- **Inverse Transformations:** If matrix A scales by k and rotates by θ, its inverse A^{-1} scales by $\frac{1}{k}$ and rotates by $-\theta$.

Problem 4.8: Matrix Determinants and System Solutions

Consider the following system of simultaneous linear equations involving the real parameters m and p:

$$\begin{aligned} mx + 2y &= p \\ 12x + (m-5)y &= 6 \end{aligned}$$

(a) Express this system in the matrix form $AX = B$. Determine the values of m for which matrix A is not an invertible matrix.

(b) Let m take the positive value found in part (a). Determine the exact value of p required for the system to have an infinite number of solution pairs. Describe the geometric relationship between the two linear equations in this specific case.

(c) Let m take the negative value found in part (a). Determine the restriction on p such that the system of equations has no solutions. Describe the geometric relationship between the two linear equations in this case.

(d) Suppose $p = 4$, and assume that m is chosen such that a unique solution exists. Use the matrix inverse method to solve for the x-coordinate of the solution in terms of m, expressing your final answer as a simplified rational fraction.

Hint:

- **Part (a):** A matrix is not invertible (it is singular) when its determinant equals zero. Extract the coefficients, calculate the determinant using $ad - bc$, set it to zero, and factor the resulting quadratic equation.
- **Part (b):** Substitute your positive m value into the original system. For a linear system to have infinite solutions, the two equations must be scalar multiples of one another (representing coincident lines).
- **Part (c):** Substitute your negative m value into the original system. For a linear system to have no solutions, the lines must have identical gradients but different y-intercepts. What value must p *not* equal to prevent the lines from overlapping?
- **Part (d):** The inverse matrix is $A^{-1} = \frac{1}{\det(A)} \begin{bmatrix} d & -c \\ -b & a \end{bmatrix}$. To find x, you only need to calculate the top row of the matrix multiplication $A^{-1}B$. Look for common factors in the numerator and denominator to simplify.

Solution 4.8

(a)

Matrix equation: $\begin{bmatrix} m & 2 \\ 12 & m-5 \end{bmatrix} \begin{bmatrix} x \\ y \end{bmatrix} = \begin{bmatrix} p \\ 6 \end{bmatrix}$.

Matrix A is not invertible when $\det(A) = 0$.

$\det(A) = (m)(m-5) - (2)(12) = m^2 - 5m - 24$.

Setting to zero: $m^2 - 5m - 24 = 0 \implies (m-8)(m+3) = 0$.

Therefore, A is singular when $m = 8$ or $m = -3$.

(b)

Let $m = 8$. The system becomes:

(1) $8x + 2y = p \implies 4x + y = \frac{p}{2}$

(2) $12x + 3y = 6 \implies 4x + y = 2$

For infinite solutions, the equations must be equivalent.

Therefore, $\frac{p}{2} = 2 \implies p = 4$.

Geometrically, the equations represent **coincident (identical) straight lines**.

(c)

Let $m = -3$. The system becomes:

(1) $-3x + 2y = p$

(2) $12x - 8y = 6 \implies -3x + 2y = -1.5$

For the system to have no solutions, the equations must be mathematically inconsistent.

Therefore, p can be any real number except -1.5 (i.e., $p \neq -1.5$).

Geometrically, the equations represent **distinct parallel lines** that never intersect.

(d)

Let $p = 4$. For a unique solution, $\det(A) = (m-8)(m+3) \neq 0$.

$X = A^{-1}B = \frac{1}{(m-8)(m+3)} \begin{bmatrix} m-5 & -2 \\ -12 & m \end{bmatrix} \begin{bmatrix} 4 \\ 6 \end{bmatrix}$.

We only need to evaluate the top row for x:

$x = \frac{4(m-5)+6(-2)}{(m-8)(m+3)} = \frac{4m-20-12}{(m-8)(m+3)} = \frac{4m-32}{(m-8)(m+3)}$.

Factoring the numerator:

$x = \frac{4(m-8)}{(m-8)(m+3)}$.

Since $m \neq 8$, we can cancel the common factor, yielding the simplified solution:

$x = \frac{4}{m+3}$.

Takeaways 4.8

- **The Zero Determinant:** When $\det(A) = 0$, a unique solution does not exist. This algebraically signals that the lines are either parallel or coincident.
- **The Role of Constants:** Once singularity is established by the coefficient matrix A, the constant matrix B dictates whether the system collapses into an infinite solution set or an empty set (no solutions).
- **Cancellation Does Not Change the Domain:** Although

$$\frac{4(m-8)}{(m-8)(m+3)}$$

simplifies to $\frac{4}{m+3}$, the original inverse required $m \neq 8, -3$. A cancelled factor does not make a singular parameter value valid again.

Problem 4.9: Determinants, Area and Orientation

Consider a triangle T in the Cartesian plane with vertices at the origin $O(0,0)$, $A(2,0)$, and $B(1,3)$. A linear transformation is applied to the plane, defined by the matrix $M_k = \begin{bmatrix} k & 1 \\ 2 & 3 \end{bmatrix}$, where k is a real parameter.

(a) Determine the coordinates of the transformed vertices O', A', and B' in terms of k.

(b) Calculate the exact area of the original triangle T, and show that the area of the transformed image T' is given by $|9k - 6|$.

(c) Verify algebraically that $\text{Area}(T') = |\det(M_k)| \times \text{Area}(T)$.

(d) Find the value of k for which the matrix M_k is singular. Describe the geometric effect this specific transformation has on the triangle T.

(e) Determine the range of values for k for which the transformation reverses the orientation of the triangle (i.e., changes the vertex order from counter-clockwise to clockwise).

Hint:

- **Part (a):** Multiply the matrix M_k by the position vectors of each vertex. The origin remains $(0, 0)$.
- **Part (b):** The original area is easily found using $\frac{1}{2} \times \text{base} \times \text{height}$. For the transformed area, you can use the shoelace formula or determinant formula for the area of a triangle given its vertices.
- **Part (c):** Calculate the determinant of M_k, multiply its absolute value by the area from part (b), and compare it to your algebraic expression for $\text{Area}(T')$.
- **Part (d):** A matrix is singular when its determinant is zero. If the area of the image is zero, what must the vertices form?
- **Part (e):** Orientation is reversed when the determinant of the transformation matrix is strictly negative.

Solution 4.9

(a)

$$O' = \begin{bmatrix} k & 1 \\ 2 & 3 \end{bmatrix} \begin{bmatrix} 0 \\ 0 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \end{bmatrix}.$$

$$A' = \begin{bmatrix} k & 1 \\ 2 & 3 \end{bmatrix} \begin{bmatrix} 2 \\ 0 \end{bmatrix} = \begin{bmatrix} 2k \\ 4 \end{bmatrix}.$$

$$B' = \begin{bmatrix} k & 1 \\ 2 & 3 \end{bmatrix} \begin{bmatrix} 1 \\ 3 \end{bmatrix} = \begin{bmatrix} k+3 \\ 11 \end{bmatrix}.$$

The transformed vertices are $O'(0,0)$, $A'(2k,4)$, and $B'(k+3,11)$.

(b)

The original triangle has base 2 (from $(0,0)$ to $(2,0)$) and height 3 (the y-coordinate of B).
$\text{Area}(T) = \frac{1}{2} \times 2 \times 3 = 3$.
For the area of the transformed triangle T' with a vertex at the origin, we can use the formula $\frac{1}{2}|x_1y_2 - x_2y_1|$:
$\text{Area}(T') = \frac{1}{2}|(2k)(11) - (k+3)(4)| = \frac{1}{2}|22k - 4k - 12| = \frac{1}{2}|18k - 12| = |9k - 6|$.

(c)

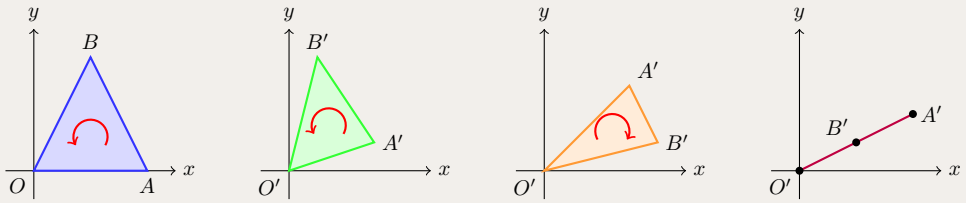
The determinant of M_k is:
$\det(M_k) = (k)(3) - (1)(2) = 3k - 2$.
Evaluating $|\det(M_k)| \times \text{Area}(T)$:
$|3k - 2| \times 3 = |3(3k - 2)| = |9k - 6|$.
This matches the exact area of T' calculated in part (b), verifying the relationship $\text{Area}(T') = |\det(M_k)| \times \text{Area}(T)$.

(d)

The matrix is singular when $\det(M_k) = 0$.
$3k - 2 = 0 \implies k = \frac{2}{3}$.
Geometrically, when $k = \frac{2}{3}$, the area of the transformed image becomes $|9(\frac{2}{3}) - 6| = 0$. The 2D triangle collapses into a 1D line segment; the points O', A', and B' become collinear.

(e)

A linear transformation reverses orientation if and only if the determinant of the transformation matrix is negative.
$\det(M_k) < 0 \implies 3k - 2 < 0 \implies k < \frac{2}{3}$.



Original T (CCW) | $\det(M_k) > 0$ (Preserves orientation) | $\det(M_k) < 0$ (Reverses orientation) | $\det(M_k) = 0$ (Collapses to 1D)

Takeaways 4.9

- **Geometric Determinant:** The absolute value of the determinant gives the area scaling factor of a linear transformation. This demonstrates that determinants are not just tools for solving equations, but fundamental geometric multipliers.
- **Singular Matrices:** A singular matrix (det = 0) squashes higher-dimensional space into lower-dimensional space (e.g., a 2D plane collapses onto a 1D line), which destroys information and explains why singular matrices cannot be inverted.
- **Orientation and Sign:** The sign of the determinant indicates orientation. A positive determinant preserves the "handedness" or cyclic order of vertices, while a negative determinant flips it (like a reflection).

Problem 4.10: Matrix Algebra and Transformations

Consider the 3×3 matrix $A = \begin{bmatrix} 2 & 0 & 0 \\ 0 & 1 & \sqrt{3} \\ 0 & \sqrt{3} & -1 \end{bmatrix}$.

(a) Determine A^2 and express your result in the form kI, where I is the 3×3 identity matrix and k is a real constant.

(b) Hence, algebraically deduce the inverse matrix A^{-1} without calculating the determinant.

(c) A 3D linear transformation represented by matrix A maps a coordinate point $P(x, y, z)$ to the image point $P'(4, -2, 2\sqrt{3})$. Use your result from part (b) to determine the exact coordinates of the original point P.

(d) Using the properties established in part (a), evaluate A^3 and hence determine an expression for A^5 strictly in terms of A.

Hint:

- **Part (a):** Perform standard 3×3 matrix multiplication $(A \times A)$. Be careful when multiplying the surds in the second and third rows.
- **Part (b):** The definition of an inverse matrix is $AA^{-1} = I$. Take your equation from part (a), $A^2 = kI$, and algebraically manipulate it to isolate I on one side of the equation.
- **Part (c):** The transformation is defined by the equation $AP = P'$. To solve for the pre-image P, you must pre-multiply both sides of the equation by A^{-1}.
- **Part (d):** Do not calculate the matrices manually. Use exponent rules for matrices: $A^3 = A^2 \times A$. Substitute the identity relationship you found in part (a) to simplify the powers.

Solution 4.10

(a)

$$A^2 = \begin{bmatrix} 2 & 0 & 0 \\ 0 & 1 & \sqrt{3} \\ 0 & \sqrt{3} & -1 \end{bmatrix} \begin{bmatrix} 2 & 0 & 0 \\ 0 & 1 & \sqrt{3} \\ 0 & \sqrt{3} & -1 \end{bmatrix}$$

$$A^2 = \begin{bmatrix} (2)(2)+0+0 & 0+0+0 & 0+0+0 \\ 0+0+0 & (1)(1)+(\sqrt{3})(\sqrt{3})+0 & (1)(\sqrt{3})+(\sqrt{3})(-1)+0 \\ 0+0+0 & (\sqrt{3})(1)+(-1)(\sqrt{3})+0 & (\sqrt{3})(\sqrt{3})+(-1)(-1)+0 \end{bmatrix}$$

$$A^2 = \begin{bmatrix} 4 & 0 & 0 \\ 0 & 1+3 & \sqrt{3}-\sqrt{3} \\ 0 & \sqrt{3}-\sqrt{3} & 3+1 \end{bmatrix} = \begin{bmatrix} 4 & 0 & 0 \\ 0 & 4 & 0 \\ 0 & 0 & 4 \end{bmatrix}$$

$$A^2 = 4 \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix} = 4I.$$

Therefore, $k = 4$.

(b)

From part (a), $A^2 = 4I$.

$A \cdot A = 4I$

Dividing both sides by the scalar 4:

$A \cdot \left(\frac{1}{4}A\right) = I$

Since $A \cdot A^{-1} = I$, it follows that $A^{-1} = \frac{1}{4}A$.

$$A^{-1} = \begin{bmatrix} 1/2 & 0 & 0 \\ 0 & 1/4 & \sqrt{3}/4 \\ 0 & \sqrt{3}/4 & -1/4 \end{bmatrix}.$$

(c)

Given $AP = P'$, we have $P = A^{-1}P'$.

Using $A^{-1} = \frac{1}{4}A$:

$$P = \frac{1}{4} \begin{bmatrix} 2 & 0 & 0 \\ 0 & 1 & \sqrt{3} \\ 0 & \sqrt{3} & -1 \end{bmatrix} \begin{bmatrix} 4 \\ -2 \\ 2\sqrt{3} \end{bmatrix}$$

$$P = \frac{1}{4} \begin{bmatrix} (2)(4)+0+0 \\ 0+(1)(-2)+(\sqrt{3})(2\sqrt{3}) \\ 0+(\sqrt{3})(-2)+(-1)(2\sqrt{3}) \end{bmatrix}$$

$$P = \frac{1}{4} \begin{bmatrix} 8 \\ -2+6 \\ -2\sqrt{3}-2\sqrt{3} \end{bmatrix} = \frac{1}{4} \begin{bmatrix} 8 \\ 4 \\ -4\sqrt{3} \end{bmatrix} = \begin{bmatrix} 2 \\ 1 \\ -\sqrt{3} \end{bmatrix}.$$

The coordinates of P are $(2, 1, -\sqrt{3})$.

(d)

$A^3 = A^2 \cdot A = (4I)A = 4A$.

$A^5 = A^3 \cdot A^2 = (4A)(4I) = 16A$.

Alternatively, $A^5 = A(A^2)^2 = A(4I)^2 = 16A$.

Takeaways 4.10

- **The "Hence" Command:** In mathematics, the word "hence" dictates that you must use the result of the previous section. Finding the inverse via $A^2 = kI$ is often faster and less prone to arithmetic error than manually calculating a 3×3 determinant and adjugate matrix.
- **Power Shortcut:** If $A^2 = kI$, then for $k \neq 0$,

$$A^{-1} = \frac{1}{k}A, \qquad A^{2n} = k^n I, \qquad A^{2n+1} = k^n A.$$

  High matrix powers can therefore be reduced without repeated multiplication.
- **Geometric Inverses:** Algebraically isolating a coordinate vector by multiplying by an inverse matrix geometrically represents reversing the original 3D transformation to find the starting position.

Problem 4.11: Systems of Linear Equations and 3D Geometry

Consider the following system of simultaneous linear equations representing three planes in 3D space, where k is a real constant:

$$\begin{aligned} x + 2y - 3z &= 4 \\ x + y + z &= 6 \\ 2x + 3y + kz &= 10 \end{aligned}$$

(a) Express this system in the matrix form $AX = B$. Calculate the determinant of the 3×3 coefficient matrix A in terms of k, and hence determine the value of k for which the matrix is singular.

(b) For the value of k found in part (a), demonstrate algebraically that the system is consistent (has solutions) rather than inconsistent (no solutions).

(c) Given that the system is consistent and has infinitely many solutions, let $z = \lambda$. Solve the equations to express x and y in terms of the parameter λ.

(d) Express the general solution from part (c) as a vector equation of a line in the form $\mathbf{r} = \mathbf{a} + \lambda\mathbf{d}$, and briefly describe the geometric arrangement of the three planes.

Hint:

- **Part (a):** The determinant of a 3×3 matrix can be found by expanding across the top row. A matrix is singular when its determinant is equal to zero.
- **Part (b):** Look for a linear combination of the first two equations that perfectly yields the third equation. If the left-hand side and right-hand side both match, the system is consistent.
- **Part (c):** Subtract the second equation from the first to eliminate x and find y in terms of z. Then substitute this back into the second equation to find x. Finally, replace z with λ.
- **Part (d):** Separate the constant terms from the terms containing λ to form your position vector **a** and direction vector **d**.

Solution 4.11

(a)

Matrix form: $\begin{bmatrix} 1 & 2 & -3 \\ 1 & 1 & 1 \\ 2 & 3 & k \end{bmatrix} \begin{bmatrix} x \\ y \\ z \end{bmatrix} = \begin{bmatrix} 4 \\ 6 \\ 10 \end{bmatrix}$.

$\det(A) = 1(1(k) - 1(3)) - 2(1(k) - 1(2)) + (-3)(1(3) - 1(2))$

$\det(A) = (k - 3) - 2(k - 2) - 3(1)$

$\det(A) = k - 3 - 2k + 4 - 3 = -k - 2.$

For A to be singular, $\det(A) = 0 \implies -k - 2 = 0 \implies k = -2$.

(b)

Substitute $k = -2$ into the third equation: $2x + 3y - 2z = 10$.

Let Equation 1 be $x + 2y - 3z = 4$ and Equation 2 be $x + y + z = 6$.

Adding Equation 1 and Equation 2 gives:

$(x + x) + (2y + y) + (-3z + z) = 4 + 6$

$2x + 3y - 2z = 10.$

Since this perfectly matches the third equation, the third plane contains the line of intersection of the first two planes. The system is consistent.

(c)

Let $z = \lambda$.

Subtracting Eq 2 from Eq 1:

$(x + 2y - 3z) - (x + y + z) = 4 - 6$

$y - 4z = -2 \implies y = 4z - 2 \implies y = 4\lambda - 2.$

Substitute $y = 4\lambda - 2$ into Eq 2:

$x + (4\lambda - 2) + \lambda = 6$

$x + 5\lambda - 2 = 6 \implies x = 8 - 5\lambda.$

(d)

From part (c), we have $x = 8 - 5\lambda$, $y = -2 + 4\lambda$, $z = 0 + 1\lambda$.

In vector form:

$\begin{bmatrix} x \\ y \\ z \end{bmatrix} = \begin{bmatrix} 8 \\ -2 \\ 0 \end{bmatrix} + \lambda \begin{bmatrix} -5 \\ 4 \\ 1 \end{bmatrix}$.

Geometrically, the three planes intersect along a single common line (forming a sheaf of planes).

Takeaways 4.11

- **Rank and Degrees of Freedom:** For a consistent system in three variables, rank 3 gives a single point, rank 2 gives a line with one free parameter, and rank 1 gives a plane with two free parameters.
- **Singularity and Consistency:** A determinant of zero only tells you a unique solution does not exist. You must verify algebraically whether the planes are parallel/form a prism (inconsistent) or intersect along a line (consistent).
- **Parametric Lines:** The process of setting a free variable to λ is exactly how the Cartesian equations of intersecting planes are converted into the vector equation of a 3D line.

Problem 4.12: Augmented Matrices and Parametric Solutions

Consider the following system of linear equations involving four variables (x, y, z, w) and a real constant k:

$$\begin{aligned} x + 2y - z + w &= 5 \\ 2x - y + 3z - 2w &= 0 \\ 3x + y + 2z - w &= k \end{aligned}$$

(a) Set up the augmented matrix for this system. By applying elementary row operations (specifically, examining $R_1 + R_2$), determine the exact value of k for which this system is consistent (has solutions).

(b) Given that the system is consistent, it will have a doubly-infinite family of solutions. Let $z = \lambda$ and $w = \mu$. Express x and y strictly in terms of the parameters λ and μ.

(c) A new constraint is added to the system such that $x - y = 3$. Determine the linear relationship that must exist between the parameters λ and μ to satisfy this constraint.

(d) Using your result from part (c), determine the unique solution for the coordinates (x, y, z, w) when $w = 2$.

Hint:

- **Part (a):** Write the system as an augmented matrix $[A|B]$. Notice that adding the first two rows together creates a row that has the exact same variable coefficients as the third row. For the system to be consistent, the constant terms must also match.
- **Part (b):** Since the third equation is redundant when consistent, use only the first two equations. Substitute $z = \lambda$ and $w = \mu$, moving them to the right-hand side. Then use simultaneous elimination on x and y.
- **Part (c):** Substitute your parametric expressions for x and y from part (b) into the equation $x - y = 3$ and group the like parameter terms together.
- **Part (d):** Substitute $w = \mu = 2$ into your relationship from part (c) to find the value of λ (which represents z). Then, plug both parameter values back into your expressions for x and y.

Solution 4.12

(a)
The augmented matrix is:
$$\left[\begin{array}{cccc|c} 1 & 2 & -1 & 1 & 5 \\ 2 & -1 & 3 & -2 & 0 \\ 3 & 1 & 2 & -1 & k \end{array}\right]$$
Performing the row operation $R_1 + R_2 \to R_{new}$:
$(1+2)x + (2-1)y + (-1+3)z + (1-2)w = 5 + 0 \implies 3x + y + 2z - w = 5$.
Comparing this to Row 3 ($3x + y + 2z - w = k$), we can see that for the system to be mathematically consistent, $k = 5$.

(b)
Let $z = \lambda$ and $w = \mu$. The system reduces to the first two equations:
(1) $x + 2y = 5 + \lambda - \mu$
(2) $2x - y = -3\lambda + 2\mu$
Multiply (2) by 2: $4x - 2y = -6\lambda + 4\mu$.
Add to (1): $5x = 5 - 5\lambda + 3\mu \implies x = 1 - \lambda + \frac{3}{5}\mu$.
Multiply (1) by 2: $2x + 4y = 10 + 2\lambda - 2\mu$.
Subtract (2) from this: $5y = 10 + 5\lambda - 4\mu \implies y = 2 + \lambda - \frac{4}{5}\mu$.

(c)
Apply the constraint: $x - y = 3$.
$\left(1 - \lambda + \frac{3}{5}\mu\right) - \left(2 + \lambda - \frac{4}{5}\mu\right) = 3$
$-1 - 2\lambda + \frac{7}{5}\mu = 3$
$-2\lambda + \frac{7}{5}\mu = 4$.
Multiplying by 5 yields the linear relationship: $-10\lambda + 7\mu = 20$.

(d)
Given $w = 2$, we have $\mu = 2$.
Substitute into the relationship from (c):
$-10\lambda + 7(2) = 20 \implies -10\lambda + 14 = 20 \implies -10\lambda = 6 \implies \lambda = -\frac{3}{5}$.
So, $z = -0.6$.
Find x: $x = 1 - \left(-\frac{3}{5}\right) + \frac{3}{5}(2) = 1 + \frac{3}{5} + \frac{6}{5} = \frac{14}{5} = 2.8$.
Find y: $y = 2 + \left(-\frac{3}{5}\right) - \frac{4}{5}(2) = 2 - \frac{3}{5} - \frac{8}{5} = -\frac{1}{5} = -0.2$.
The unique solution is $(x, y, z, w) = \left(\frac{14}{5}, -\frac{1}{5}, -\frac{3}{5}, 2\right)$ or $(2.8, -0.2, -0.6, 2)$.

Takeaways 4.12

- **Underdetermined Does Not Automatically Mean Infinite:** Having fewer independent equations than variables gives free parameters only when the system is consistent. For a consistent linear system,

$$\text{number of free variables} = n - \text{rank}(A).$$

- **Row Operations for Consistency:** Elementary row operations on augmented matrices quickly reveal inconsistencies (e.g., $0 = c$ where $c \neq 0$). If a row reduces entirely to zero on the left, the right side must also be zero for a solution to exist.
- **Applying Constraints:** Adding constraints to a parametric solution family reduces its degrees of freedom. Each new valid equation removes one parameter, eventually leading to a single unique point if enough constraints are provided.

Problem 4.13: Advanced Matrix Products and Inverses

Let $C = \begin{bmatrix} 2 & -1 & 1 \\ 1 & 0 & 2 \\ -1 & 1 & -1 \end{bmatrix}$ and $D = \begin{bmatrix} -2 & 0 & -2 \\ -1 & -1 & -3 \\ 1 & -1 & 1 \end{bmatrix}$.

(a) Evaluate the matrix product CD. Hence, determine the inverse matrices C^{-1} and D^{-1} in terms of C and D.

(b) A system of simultaneous linear equations is given by:

$$\begin{aligned} -2x - 2z &= 4k \\ -x - y - 3z &= 2 \\ x - y + z &= -2k \end{aligned}$$

Express this system in the form $DX = V$, and hence solve for x, y, and z in terms of the real constant k.

(c) Determine the specific value of k such that the solution point (x, y, z) lies on the plane defined by $x + z = 4$.

(d) Using the inverse properties established in part (a), solve the matrix equation $CYD = I$ for the 3×3 matrix Y, where I is the identity matrix.

Hint:

- **Part (a):** Multiply rows of C by columns of D. The result will be a scalar multiple of the identity matrix, mI. Use the definition of an inverse ($MM^{-1} = I$) to isolate I.
- **Part (b):** The coefficient matrix of the system matches matrix D. To solve $DX = V$, pre-multiply both sides by D^{-1}. Substitute your expression for D^{-1} from part (a) to avoid recalculating an inverse from scratch.
- **Part (c):** Substitute the expressions for x and z (which you found in terms of k in part b) into the plane equation and solve for k.
- **Part (d):** Do not attempt to define Y with 9 unknown variables. Instead, isolate Y algebraically by pre-multiplying by C^{-1} and post-multiplying by D^{-1}. Remember that a matrix commutes with its own inverse (e.g., $CD = mI \implies DC = mI$).

Solution 4.13

(a)

$$CD = \begin{bmatrix} 2 & -1 & 1 \\ 1 & 0 & 2 \\ -1 & 1 & -1 \end{bmatrix} \begin{bmatrix} -2 & 0 & -2 \\ -1 & -1 & -3 \\ 1 & -1 & 1 \end{bmatrix} = \begin{bmatrix} -4+1+1 & 0+1-1 & -4+3+1 \\ -2+0+2 & 0+0-2 & -2+0+2 \\ 2-1-1 & 0-1+1 & 2-3-1 \end{bmatrix}$$

$$CD = \begin{bmatrix} -2 & 0 & 0 \\ 0 & -2 & 0 \\ 0 & 0 & -2 \end{bmatrix} = -2I.$$

Since $CD = -2I$, we have $C\left(-\frac{1}{2}D\right) = I$, so $C^{-1} = -\frac{1}{2}D$.
Similarly, $\left(-\frac{1}{2}C\right)D = I$, so $D^{-1} = -\frac{1}{2}C$.

(b)

The system is $D\begin{bmatrix} x \\ y \\ z \end{bmatrix} = \begin{bmatrix} 4k \\ 2 \\ -2k \end{bmatrix}$.

$$X = D^{-1}V = -\tfrac{1}{2}CV = -\tfrac{1}{2}\begin{bmatrix} 2 & -1 & 1 \\ 1 & 0 & 2 \\ -1 & 1 & -1 \end{bmatrix}\begin{bmatrix} 4k \\ 2 \\ -2k \end{bmatrix}$$

$$X = -\tfrac{1}{2}\begin{bmatrix} 2(4k) - 1(2) + 1(-2k) \\ 1(4k) + 0(2) + 2(-2k) \\ -1(4k) + 1(2) - 1(-2k) \end{bmatrix} = -\tfrac{1}{2}\begin{bmatrix} 6k - 2 \\ 0 \\ -2k + 2 \end{bmatrix} = \begin{bmatrix} 1 - 3k \\ 0 \\ k - 1 \end{bmatrix}.$$

Thus, $x = 1 - 3k$, $y = 0$, and $z = k - 1$.

(c)

We require $x + z = 4$.
Substituting the results from (b):
$(1 - 3k) + (k - 1) = 4$
$-2k = 4 \implies k = -2$.

(d)

Given $CYD = I$.
Pre-multiply by C^{-1} and post-multiply by D^{-1}:
$Y = C^{-1}ID^{-1} = C^{-1}D^{-1}$.
Substitute the inverses from part (a):
$Y = \left(-\frac{1}{2}D\right)\left(-\frac{1}{2}C\right) = \frac{1}{4}DC$.
Since $CD = -2I$, it follows that $DC = -2I$ (a matrix and its inverse commute).

$$Y = \tfrac{1}{4}(-2I) = -\tfrac{1}{2}I = \begin{bmatrix} -0.5 & 0 & 0 \\ 0 & -0.5 & 0 \\ 0 & 0 & -0.5 \end{bmatrix}.$$

Takeaways 4.13

- **The "Hence" Method:** Calculating 3×3 inverses from scratch using determinants and adjugates is tedious. Finding a product $AB = mI$ is a standard shortcut to define $A^{-1} = \frac{1}{m}B$.
- **Parametric Systems:** Matrices handle parametric constants (like k) effortlessly because matrix multiplication distributes linearly over expressions.
- **Matrix Algebra vs. Arithmetic:** Part (d) demonstrates the power of abstract matrix algebra. Treating matrices as single algebraic entities (C^{-1}, D^{-1}) avoids massive blocks of arithmetic calculation.

Problem 4.14: Matrix Systems and Planes in 3D Space

Consider the points $A(1, 2, -1)$, $B(2, 3, 1)$, and $C_k(3, -1, k)$, where k is a real constant. A plane containing these three points can be modeled by the equation $ax + by + cz = 1$, provided the plane does not pass through the origin.

(a) Substitute the coordinates of the three points into the plane equation to formulate a system of three linear equations in terms of a, b, and c. Express this system as a matrix equation $M\mathbf{v} = \mathbf{u}$, and find $\det(M)$ in terms of k.

(b) Determine the value of k for which the matrix M is singular. Explain why the matrix equation $M\mathbf{v} = \mathbf{u}$ has no solution for this value of k, and what this implies about the geometric position of the plane containing A, B, and C_k relative to the origin.

(c) Let $k = 2$, corresponding to the specific point $C_2(3, -1, 2)$. Solve the system using Gaussian elimination (row reduction) to find the values of a, b, and c, and hence state the unique equation of the plane in the form $ax + by + cz + d = 0$ with integer coefficients.

(d) For the singular case where $k = 18$, the plane must pass through the origin. Set up a new homogeneous matrix equation for the plane $a'x + b'y + c'z = 0$, and find the specific Cartesian equation of this plane by letting $a' = 5$.

Hint:

- **Part (a):** Replace x, y, and z with the coordinates of each point to create three distinct equations. Extract the coefficients of a, b, and c to form the 3×3 matrix M. Calculate the determinant by expanding across the top row.
- **Part (b):** A matrix is singular when its determinant equals zero. If setting $ax + by + cz = 1$ yields a singular matrix and an inconsistent system, it means no such plane exists that misses the origin.
- **Part (c):** Form the augmented matrix $[M|\mathbf{u}]$ and use elementary row operations to reach upper triangular form. Back-substitute to find c, b, and a. Multiply the entire plane equation by the lowest common denominator to remove fractions.
- **Part (d):** A plane through the origin equates to 0 rather than 1. Substitute the points A, B, and C_{18} to form a homogeneous system $M\mathbf{v} = \mathbf{0}$. Use the first two equations to express b' and c' strictly in terms of a', then substitute $a' = 5$.

Solution 4.14

(a)

Substituting the points into $ax + by + cz = 1$:

$A(1, 2, -1) \implies a + 2b - c = 1$

$B(2, 3, 1) \implies 2a + 3b + c = 1$

$C_k(3, -1, k) \implies 3a - b + kc = 1$

Matrix equation: $\begin{bmatrix} 1 & 2 & -1 \\ 2 & 3 & 1 \\ 3 & -1 & k \end{bmatrix} \begin{bmatrix} a \\ b \\ c \end{bmatrix} = \begin{bmatrix} 1 \\ 1 \\ 1 \end{bmatrix}$.

$\det(M) = 1(3k - (-1)) - 2(2k - 3) + (-1)(-2 - 9) = 3k + 1 - 4k + 6 + 11 = 18 - k$.

(b)

Matrix M is singular when $\det(M) = 0 \implies 18 - k = 0 \implies k = 18$.

When $k = 18$, applying the row operation $11R_1 - 7R_2 + R_3$ to the augmented matrix gives the row $\begin{bmatrix} 0 & 0 & 0 & | & 5 \end{bmatrix}$, hence $0 = 5$. This contradiction means the system has **no solution**. Geometrically, this implies the plane containing A, B, and C_{18} must pass exactly through the origin, meaning its true equation is of the form $ax + by + cz = 0$, not 1.

(c)

Let $k = 2$. The augmented matrix is:

$$\left[\begin{array}{ccc|c} 1 & 2 & -1 & 1 \\ 2 & 3 & 1 & 1 \\ 3 & -1 & 2 & 1 \end{array}\right]$$

$$R_2 \to R_2 - 2R_1 \implies \left[\begin{array}{ccc|c} 1 & 2 & -1 & 1 \\ 0 & -1 & 3 & -1 \\ 3 & -1 & 2 & 1 \end{array}\right]$$

$$R_3 \to R_3 - 3R_1 \implies \left[\begin{array}{ccc|c} 1 & 2 & -1 & 1 \\ 0 & -1 & 3 & -1 \\ 0 & -7 & 5 & -2 \end{array}\right]$$

$$R_3 \to R_3 - 7R_2 \implies \left[\begin{array}{ccc|c} 1 & 2 & -1 & 1 \\ 0 & -1 & 3 & -1 \\ 0 & 0 & -16 & 5 \end{array}\right]$$

From R_3: $-16c = 5 \implies c = -\frac{5}{16}$.

From R_2: $-b + 3(-\frac{5}{16}) = -1 \implies b = \frac{1}{16}$.

From R_1: $a + 2(\frac{1}{16}) - (-\frac{5}{16}) = 1 \implies a = \frac{9}{16}$.

The equation is $\frac{9}{16}x + \frac{1}{16}y - \frac{5}{16}z = 1$, which simplifies to $9x + y - 5z - 16 = 0$.

(d)

For $k = 18$, the homogeneous system is:

(1) $a' + 2b' - c' = 0$

(2) $2a' + 3b' + c' = 0$

(3) $3a' - b' + 18c' = 0$

Add (1) and (2) to eliminate c': $3a' + 5b' = 0 \implies b' = -\frac{3}{5}a'$.

Let $a' = 5 \implies b' = -3$.

Substitute into (1): $5 + 2(-3) - c' = 0 \implies -1 - c' = 0 \implies c' = -1$.

The equation of the plane is $5x - 3y - z = 0$.

Takeaways 4.14

- **Non-Homogeneous vs Homogeneous Form:** Setting a plane's constant d to a non-zero value allows for a quick 3×3 matrix setup, but it operates on the assumption that the plane does not cross the origin.
- **What Singularity Actually Tells You:** $\det M = 0$ only tells you that $M\mathbf{v} = \mathbf{1}$ has no unique solution. You must check consistency. In this problem the system is inconsistent at $k = 18$, showing that the normalisation $ax+by+cz = 1$ fails; the required plane instead passes through the origin and must be written in homogeneous form.
- **Gaussian Elimination:** Row reduction is a powerful tool for solving 3×3 systems without needing to calculate cumbersome matrix inverses or adjugates manually.

Problem 4.15: Matrix Polynomials and System Solutions

Let $A = \begin{bmatrix} 3 & -2 & -2 \\ 3 & -2 & -3 \\ 1 & -1 & 0 \end{bmatrix}$.

(a) Evaluate A^2 to show that A is an involutory matrix (i.e., a matrix where $A^2 = I$).

(b) Let $E = \frac{1}{2}(I + A)$. Using matrix algebra and your result from part (a), prove that E is an idempotent matrix (i.e., $E^2 = E$).

(c) Consider the matrix $M = I + kA$, where k is a real constant. By expanding the algebraic product $M(I - kA)$, determine the exact values of k for which the matrix M does not have an inverse.

(d) Let $k = 2$. Use your algebraic result from part (c) to determine M^{-1}, and hence solve the following system of linear equations:

$$\begin{aligned} 7x - 4y - 4z &= 3 \\ 6x - 3y - 6z &= -3 \\ 2x - 2y + z &= 6 \end{aligned}$$

Hint:

- **Part (a):** Multiply A by itself using standard row-by-column multiplication. Verify that the main diagonal consists of 1s and all other entries are 0.
- **Part (b):** Square the algebraic expression for E. Remember that scalar multiples can be factored out, and matrix multiplication distributes over addition. Substitute $A^2 = I$ when simplifying.
- **Part (c):** Expand $(I + kA)(I - kA)$ as you would a difference of perfect squares, keeping in mind that I commutes with any matrix. For M to be singular, the scalar coefficient of I in your resulting expression must equal zero.
- **Part (d):** Notice that the coefficient matrix of the system is precisely M when $k = 2$. Calculate M^{-1} by setting $k = 2$ in the identity from part (c), then pre-multiply the constant column vector by M^{-1}.

Solution 4.15

(a)

$$A^2 = \begin{bmatrix} 3 & -2 & -2 \\ 3 & -2 & -3 \\ 1 & -1 & 0 \end{bmatrix} \begin{bmatrix} 3 & -2 & -2 \\ 3 & -2 & -3 \\ 1 & -1 & 0 \end{bmatrix}$$

$$A^2 = \begin{bmatrix} 9-6-2 & -6+4+2 & -6+6+0 \\ 9-6-3 & -6+4+3 & -6+6+0 \\ 3-3+0 & -2+2+0 & -2+3+0 \end{bmatrix} = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix} = I.$$

(b)

$E^2 = \left(\frac{1}{2}(I+A)\right)^2 = \frac{1}{4}(I+A)(I+A)$
$E^2 = \frac{1}{4}(I^2 + IA + AI + A^2)$.
Since $I^2 = I$, $IA = AI = A$, and $A^2 = I$:
$E^2 = \frac{1}{4}(I + A + A + I) = \frac{1}{4}(2I + 2A) = \frac{1}{2}(I + A) = E$.

(c)

$M(I - kA) = (I + kA)(I - kA) = I^2 - kIA + kAI - k^2A^2$.
Because $IA = AI = A$, the middle terms cancel:
$M(I - kA) = I - k^2A^2$.
Substitute $A^2 = I$:
$M(I - kA) = I - k^2I = (1 - k^2)I$.
For $k \neq \pm 1$, rearranging gives $M\left[\frac{1}{1-k^2}(I - kA)\right] = I$, proving M is invertible.
At $k = 1$, $M = I + A$ and $(I + A)(I - A) = 0$. Since $I - A \neq 0$, $I + A$ cannot be invertible.
At $k = -1$, $M = I - A$ and $(I - A)(I + A) = 0$. Since $I + A \neq 0$, $I - A$ cannot be invertible.
Thus, M is singular if and only if $k = \pm 1$.

(d)

Let $k = 2$. The system's coefficient matrix is $M = I + 2A = \begin{bmatrix} 7 & -4 & -4 \\ 6 & -3 & -6 \\ 2 & -2 & 1 \end{bmatrix}$.
From (c), $M^{-1} = \frac{1}{1-2^2}(I - 2A) = -\frac{1}{3}(I - 2A)$.

$$I - 2A = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix} - \begin{bmatrix} 6 & -4 & -4 \\ 6 & -4 & -6 \\ 2 & -2 & 0 \end{bmatrix} = \begin{bmatrix} -5 & 4 & 4 \\ -6 & 5 & 6 \\ -2 & 2 & 1 \end{bmatrix}.$$

$$M^{-1} = -\frac{1}{3}\begin{bmatrix} -5 & 4 & 4 \\ -6 & 5 & 6 \\ -2 & 2 & 1 \end{bmatrix}.$$

$$X = M^{-1}B = -\frac{1}{3}\begin{bmatrix} -5 & 4 & 4 \\ -6 & 5 & 6 \\ -2 & 2 & 1 \end{bmatrix}\begin{bmatrix} 3 \\ -3 \\ 6 \end{bmatrix}$$

$$X = -\frac{1}{3}\begin{bmatrix} -15-12+24 \\ -18-15+36 \\ -6-6+6 \end{bmatrix} = -\frac{1}{3}\begin{bmatrix} -3 \\ 3 \\ -6 \end{bmatrix} = \begin{bmatrix} 1 \\ -1 \\ 2 \end{bmatrix}.$$

Therefore, $x = 1$, $y = -1$, and $z = 2$.

Takeaways 4.15

- **Abstract Matrix Algebra:** Abstract algebra simplifies complex computational problems. Establishing $A^2 = I$ creates a generalized shortcut for inverting any matrix of the parametric form $I + kA$.
- **Inverse from a Polynomial Identity:** Since $A^2 = I$,

$$(I + kA)(I - kA) = (1 - k^2)I.$$

  Therefore, for $k \neq \pm 1$,

$$(I + kA)^{-1} = \frac{I - kA}{1 - k^2}.$$

- **Idempotent and Involutory Links:** Idempotent matrices ($E^2 = E$, which geometrically represent projections) and involutory matrices have eigenvalues ± 1, with reflections being an important special case, and are fundamentally linked via simple algebraic transformations like $E = \frac{1}{2}(I + A)$.

Problem 4.16: The Geometry of Mirrors

Matrices provide a powerful algebraic way to represent geometric transformations in 2D space. The general matrix representing a reflection across a line passing through the origin at an angle θ to the positive x-axis is given by:

$$M_\theta = \begin{bmatrix} \cos 2\theta & \sin 2\theta \\ \sin 2\theta & -\cos 2\theta \end{bmatrix}$$

(i) Let A be the matrix representing a reflection in the x-axis, and B be the matrix representing a reflection in the line $y = x$. Write down matrices A and B. Then, calculate the matrix product BA and describe the single geometric transformation this product represents.
(ii) Find the determinant of the general reflection matrix M_θ. Geometrically, what does the sign of this determinant tell us about the orientation of a shape after it undergoes a single reflection?
(iii) Consider two distinct lines, L_1 and L_2, passing through the origin at angles α and β respectively. A vector is first reflected across L_1 and then reflected across L_2. Using the general matrix M_θ and trigonometric compound angle identities, derive the single matrix R that represents this combined transformation ($R = M_\beta M_\alpha$).
(iv) Based on your derived matrix in part (iii), state the general geometric relationship between the two intersecting lines of reflection and the resulting single transformation.
Application: In a 2D physics simulation, a graphic needs to be rotated counter-clockwise by exactly 150°. The engine is temporarily restricted and can *only* perform reflections. If the first reflection must be across the line $y = \frac{1}{\sqrt{3}}x$, determine the exact equation of the line for the second reflection.

Hint:

- **Part (i):** Remember that matrix multiplication is read from right to left: BA means "apply A first, then B". To find A and B, look at where the basis vectors $\begin{bmatrix}1\\0\end{bmatrix}$ and $\begin{bmatrix}0\\1\end{bmatrix}$ end up, or use the M_θ formula with $\theta = 0°$ and $\theta = 45°$.
- **Part (ii):** Use the Pythagorean trigonometric identity $\cos^2 x + \sin^2 x = 1$. Think about how a clock face looks in a mirror (standard vs. inverted orientation).
- **Part (iii):** Write out the full 2×2 matrix multiplication carefully. You will need the identities for $\cos(A \pm B)$ and $\sin(A \pm B)$ to simplify the four resulting elements.
- **Part (iv):** Compare your simplified matrix from part (iii) to the standard rotation matrix $\begin{bmatrix}\cos\phi & -\sin\phi\\ \sin\phi & \cos\phi\end{bmatrix}$. What is ϕ in terms of α and β? For the application, use $m = \tan\theta$ to find the angle of the first line.

Solution 4.16

(i)

A (reflection in x-axis, $\theta = 0°$): $A = \begin{bmatrix} 1 & 0 \\ 0 & -1 \end{bmatrix}$

B (reflection in $y = x$, $\theta = 45°$): $B = \begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix}$

Product $BA = \begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix} \begin{bmatrix} 1 & 0 \\ 0 & -1 \end{bmatrix} = \begin{bmatrix} 0 & -1 \\ 1 & 0 \end{bmatrix}$.

Geometrically, this is a 90° **counter-clockwise rotation** about the origin.

(ii)

$\det(M_\theta) = (\cos 2\theta)(-\cos 2\theta) - (\sin 2\theta)(\sin 2\theta) = -(\cos^2 2\theta + \sin^2 2\theta) = -1$.

A determinant of -1 indicates that the transformation preserves area but **reverses orientation** (it flips the "handedness" of the object, like a mirror image).

(iii)

$M_\beta M_\alpha = \begin{bmatrix} \cos 2\beta & \sin 2\beta \\ \sin 2\beta & -\cos 2\beta \end{bmatrix} \begin{bmatrix} \cos 2\alpha & \sin 2\alpha \\ \sin 2\alpha & -\cos 2\alpha \end{bmatrix}$

Row 1, Col 1: $\cos 2\beta \cos 2\alpha + \sin 2\beta \sin 2\alpha = \cos(2\beta - 2\alpha)$

Row 1, Col 2: $\cos 2\beta \sin 2\alpha - \sin 2\beta \cos 2\alpha = -\sin(2\beta - 2\alpha)$

Row 2, Col 1: $\sin 2\beta \cos 2\alpha - \cos 2\beta \sin 2\alpha = \sin(2\beta - 2\alpha)$

Row 2, Col 2: $\sin 2\beta \sin 2\alpha + (-\cos 2\beta)(-\cos 2\alpha) = \cos(2\beta - 2\alpha)$

Result:

$$R = \begin{bmatrix} \cos(2(\beta - \alpha)) & -\sin(2(\beta - \alpha)) \\ \sin(2(\beta - \alpha)) & \cos(2(\beta - \alpha)) \end{bmatrix}$$

(iv)

Geometric Relationship: The composition of two reflections is always a rotation. The angle of rotation is exactly **twice the angle between the two intersecting lines of reflection** ($2(\beta - \alpha)$). *(This is the "aha!" moment: a fundamental theorem of geometry proven with pure algebra).*

Application:

We need a rotation of 150°, so $2(\beta - \alpha) = 150° \implies \beta - \alpha = 75°$.

The first line is $y = \frac{1}{\sqrt{3}}x$. Since $m = \tan\alpha = \frac{1}{\sqrt{3}}$, we know $\alpha = 30°$.

Therefore, $\beta = 30° + 75° = 105°$.

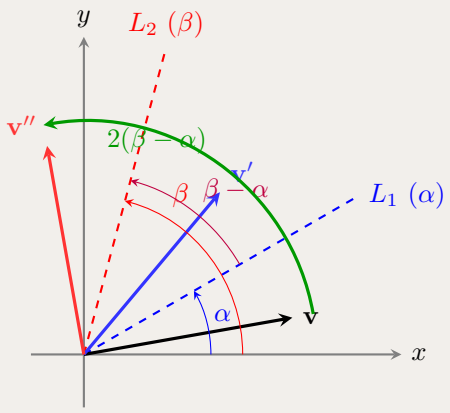
The gradient of the second line is $m_2 = \tan(105°)$.

Using exact values:

$$\tan(60° + 45°) = \frac{\tan 60° + \tan 45°}{1 - \tan 60° \tan 45°} = \frac{\sqrt{3} + 1}{1 - \sqrt{3}(1)}$$

Rationalising the denominator yields $-(2 + \sqrt{3})$.

Equation of the second line: $y = -(2 + \sqrt{3})x$.



Takeaways 4.16

- **Matrix Order Matters:** $BA \neq AB$. If you reversed the order of the reflections, the sign inside the sine functions would flip, resulting in a clockwise rotation instead of counter-clockwise.
- **Algebra Proves Geometry:** The messy trigonometric algebra in part (iii) collapses perfectly into the rotation matrix. This proves mathematically a beautiful visual truth: *any rotation can be constructed using just two mirrors.* Each reflection has determinant -1, while their product has determinant $+1$, consistent with a rotation.
- **Two-Reflection Formula:**

$$M_\beta M_\alpha = R(2(\beta - \alpha)) .$$

Thus two reflections produce a rotation through twice the directed angle between the mirror lines. Reversing the order reverses the rotation direction.

Problem 4.17: The Hidden Bridge

Any complex number $z = a + bi$ can be uniquely mapped to a 2×2 matrix of the form $M_z = \begin{bmatrix} a & -b \\ b & a \end{bmatrix}$, where $a, b \in \mathbb{R}$.

(i) Let $w = 1 + i$ and $v = \sqrt{3} - i$. Find the matrices M_w and M_v. Calculate the matrix product $M_w M_v$, and show that this product is exactly equal to the matrix representation of the complex product wv.

(ii) For a general, non-zero complex number $z = a + bi$:

a) Prove that the determinant of its matrix representation, $\det(M_z)$, is equal to $|z|^2$.

b) Find the matrix inverse $(M_z)^{-1}$ using the standard 2×2 inverse formula, and show that it maps perfectly to the complex reciprocal z^{-1}.

(iii) Consider a complex number on the unit circle, $z = \cos\theta + i\sin\theta$. Write down its matrix representation M_z. By interpreting M_z as a geometric transformation, explain why $(M_z)^n$ must equal M_{z^n} for any positive integer n. What famous theorem does this matrix equivalence prove?

(iv) The matrix $T = \begin{bmatrix} \sqrt{3} & -1 \\ 1 & \sqrt{3} \end{bmatrix}$ represents a transformation in 2D space. A vector $\mathbf{u} = \begin{bmatrix} 2 \\ 5 \end{bmatrix}$ is transformed by T repeatedly, 12 times in total.

By leveraging the connection between matrices and complex numbers explored in the previous parts, evaluate $T^{12}\mathbf{u}$ exactly, without performing manual matrix multiplications.

Hint:

- **Part (i):** Multiply the complex numbers w and v as usual (remember $i^2 = -1$). Then, perform standard row-by-column matrix multiplication for $M_w M_v$. The resulting matrix should have the real part of wv on the main diagonal and the imaginary part on the anti-diagonal.
- **Part (ii):** Recall that $|z| = \sqrt{a^2 + b^2}$, and $z^{-1} = \frac{\bar{z}}{|z|^2}$ where $\bar{z} = a - bi$. Compare this to the matrix inverse formula: $\frac{1}{ad-bc}\begin{bmatrix} d & -b \\ -c & a \end{bmatrix}$.
- **Part (iii):** What geometric transformation does the matrix $\begin{bmatrix} \cos\theta & -\sin\theta \\ \sin\theta & \cos\theta \end{bmatrix}$ represent? If you apply this transformation n times, what is the net geometric effect? Compare this to raising $z = \text{cis}(\theta)$ to the power of n.
- **Part (iv):** Matrix T is in the exact form of M_z. What is the complex number z that corresponds to T? Convert that complex number to polar form ($r\text{cis}\theta$). Use De Moivre's Theorem to evaluate z^{12}, map the result *back* to a matrix, and then multiply by the vector $\mathbf{u}$.

Solution 4.17

(i)

$w = 1 + i \implies M_w = \begin{bmatrix} 1 & -1 \\ 1 & 1 \end{bmatrix}$

$v = \sqrt{3} - i \implies M_v = \begin{bmatrix} \sqrt{3} & 1 \\ -1 & \sqrt{3} \end{bmatrix}$

$wv = (1+i)(\sqrt{3}-i) = (\sqrt{3}+1) + (\sqrt{3}-1)i.$

$M_w M_v = \begin{bmatrix} 1 & -1 \\ 1 & 1 \end{bmatrix} \begin{bmatrix} \sqrt{3} & 1 \\ -1 & \sqrt{3} \end{bmatrix} = \begin{bmatrix} \sqrt{3}+1 & -(\sqrt{3}-1) \\ \sqrt{3}-1 & \sqrt{3}+1 \end{bmatrix}.$

This exactly matches M_{wv}, proving the mapping holds for multiplication.

(ii)

a) $\det(M_z) = (a)(a) - (-b)(b) = a^2 + b^2$. By definition, $|z| = \sqrt{a^2+b^2}$, so $\det(M_z) = |z|^2$.

b) $(M_z)^{-1} = \frac{1}{a^2+b^2} \begin{bmatrix} a & b \\ -b & a \end{bmatrix}$. The corresponding complex number is $\frac{a}{a^2+b^2} - \frac{b}{a^2+b^2}i$, which is exactly $\frac{\bar{z}}{|z|^2} = z^{-1}$.

(iii)

$M_z = \begin{bmatrix} \cos\theta & -\sin\theta \\ \sin\theta & \cos\theta \end{bmatrix}$. This is the standard 2D rotation matrix by angle θ counter-clockwise.

Applying this transformation n times ($(M_z)^n$) results in a rotation by $n\theta$, which is geometrically represented by the matrix $\begin{bmatrix} \cos n\theta & -\sin n\theta \\ \sin n\theta & \cos n\theta \end{bmatrix}$.

Because the matrices map uniquely to complex numbers, the complex numbers must match, beautifully proving **De Moivre's Theorem**: $(\cos\theta + i\sin\theta)^n = \cos n\theta + i\sin n\theta$.

(iv) *(The "Aha!" Moment)*

Instead of multiplying T by itself 12 times (which is practically impossible by hand), map T to the complex number $z = \sqrt{3} + i$.

Convert z to polar form: $r = \sqrt{3+1} = 2$. $\theta = \arctan(1/\sqrt{3}) = \pi/6$.

So, $z = 2\text{cis}(\pi/6)$.

We want T^{12}, which corresponds to z^{12}.

Using De Moivre's Theorem: $z^{12} = 2^{12}\text{cis}\left(12 \times \frac{\pi}{6}\right) = 4096\text{cis}(2\pi) = 4096(1 + 0i) = 4096$.

Map this back to a matrix: $T^{12} = M_{4096} = \begin{bmatrix} 4096 & 0 \\ 0 & 4096 \end{bmatrix}$.

Finally, transform the vector:

$$T^{12}\mathbf{u} = \begin{bmatrix} 4096 & 0 \\ 0 & 4096 \end{bmatrix} \begin{bmatrix} 2 \\ 5 \end{bmatrix} = \begin{bmatrix} 8192 \\ 20480 \end{bmatrix}$$

Takeaways 4.17

- **Complex–Matrix Dictionary:** Under

$$a + bi \longleftrightarrow \begin{bmatrix} a & -b \\ b & a \end{bmatrix},$$

multiplication is preserved and

$$\det(M_z) = a^2 + b^2 = |z|^2.$$

Hence M_z is invertible exactly when $z \neq 0$, and matrix inversion corresponds to taking the complex reciprocal $1/z$.

- **Multiple Lenses:** A difficult algebra problem (matrix powers) can become trivial when viewed through a geometric lens (polar complex numbers).
- **The "Why" of Matrices:** This explains *why* the rotation matrix formula looks the way it does. It isn't random; it is the exact matrix manifestation of multiplying by a complex number on the unit circle! $M_z^n = M_{z^n}$ is the matrix version of **De Moivre's theorem**.

Problem 4.18: Linearising Trigonometry

A rotation about the origin by an angle θ counter-clockwise is represented by the matrix:

$$R(\theta) = \begin{bmatrix} \cos\theta & -\sin\theta \\ \sin\theta & \cos\theta \end{bmatrix}$$

(i) Geometrically, rotating by β, and then rotating the result by α, is equivalent to a single rotation by $(\alpha + \beta)$. Therefore, $R(\alpha)R(\beta) = R(\alpha + \beta)$.
By expanding the matrix product $R(\alpha)R(\beta)$, derive the trigonometric identities for $\cos(\alpha + \beta)$ and $\sin(\alpha + \beta)$.
(ii) The inverse matrix $R(\theta)^{-1}$ "undoes" a rotation, meaning it geometrically represents a rotation by $-\theta$. Therefore, $R(\theta)^{-1} = R(-\theta)$.
Find the inverse of $R(\theta)$ algebraically using the standard 2×2 matrix inverse formula. By comparing your result to $R(-\theta)$, mathematically prove the even/odd symmetry identities: $\cos(-\theta) = \cos\theta$ and $\sin(-\theta) = -\sin\theta$.
(iii) Let I be the 2×2 identity matrix. Using your knowledge from part (ii), evaluate the matrix addition $R(\beta) + R(-\beta)$, and show that the result can be written as the scalar multiplication $(2\cos\beta)I$.
(iv) Consider the matrix expression $E = R(\alpha)\big[R(\beta) + R(-\beta)\big]$.
Evaluate E in two different ways:
1. **Algebraically:** Substitute your result from part (iii) into the bracket and perform the scalar multiplication.
2. **Geometrically/Distributively:** Expand the bracket first, and use the angle-addition rule from part (i).

By equating the corresponding elements of your two resulting matrices, deduce the "product-to-sum" trigonometric identities for $2\cos\alpha\cos\beta$ and $2\sin\alpha\cos\beta$.

Hint:

- **Part (i):** Multiply the two 2×2 matrices row-by-column. Then, look at the top-left element and the bottom-left element to find the formulas for $\cos(\alpha+\beta)$ and $\sin(\alpha+\beta)$.
- **Part (ii):** The inverse of $\begin{bmatrix} a & b \\ c & d \end{bmatrix}$ is $\frac{1}{ad-bc}\begin{bmatrix} d & -b \\ -c & a \end{bmatrix}$. Don't forget the Pythagorean identity $\cos^2\theta + \sin^2\theta = 1$ when finding the determinant.
- **Part (iii):** Add the matrices element-by-element. $R(-\beta)$ is the inverse matrix you found in part (ii). Notice how the sine terms cancel out.
- **Part (iv):** For Method 1, you are evaluating $R(\alpha) \times (2\cos\beta)I$. Because I is the identity matrix, multiplying it by $R(\alpha)$ leaves $R(\alpha)$ unchanged, you just multiply every term in $R(\alpha)$ by the scalar $2\cos\beta$. For Method 2, use $R(\alpha)R(-\beta) = R(\alpha-\beta)$.

Solution 4.18

(i)

$R(\alpha)R(\beta) = \begin{bmatrix}\cos\alpha & -\sin\alpha \\ \sin\alpha & \cos\alpha\end{bmatrix}\begin{bmatrix}\cos\beta & -\sin\beta \\ \sin\beta & \cos\beta\end{bmatrix}$

Top-left (Row 1 $\times$ Col 1): $\cos\alpha\cos\beta - \sin\alpha\sin\beta$.

Bottom-left (Row 2 $\times$ Col 1): $\sin\alpha\cos\beta + \cos\alpha\sin\beta$.

Equating this to $R(\alpha+\beta) = \begin{bmatrix}\cos(\alpha+\beta) & -\sin(\alpha+\beta) \\ \sin(\alpha+\beta) & \cos(\alpha+\beta)\end{bmatrix}$ yields:

$\cos(\alpha+\beta) = \cos\alpha\cos\beta - \sin\alpha\sin\beta$

$\sin(\alpha+\beta) = \sin\alpha\cos\beta + \cos\alpha\sin\beta$

(ii)

Determinant of $R(\theta) = (\cos\theta)(\cos\theta) - (-\sin\theta)(\sin\theta) = \cos^2\theta + \sin^2\theta = 1$.

$R(\theta)^{-1} = \frac{1}{1}\begin{bmatrix}\cos\theta & \sin\theta \\ -\sin\theta & \cos\theta\end{bmatrix}$.

We know this equals $R(-\theta) = \begin{bmatrix}\cos(-\theta) & -\sin(-\theta) \\ \sin(-\theta) & \cos(-\theta)\end{bmatrix}$.

Equating the corresponding elements proves that:

$\cos(-\theta) = \cos\theta$ (even function)

$-\sin(-\theta) = \sin\theta \implies \sin(-\theta) = -\sin\theta$ (odd function).

(iii)

$R(\beta) + R(-\beta) = \begin{bmatrix}\cos\beta & -\sin\beta \\ \sin\beta & \cos\beta\end{bmatrix} + \begin{bmatrix}\cos\beta & \sin\beta \\ -\sin\beta & \cos\beta\end{bmatrix}$

$R(\beta) + R(-\beta) = \begin{bmatrix}2\cos\beta & 0 \\ 0 & 2\cos\beta\end{bmatrix}$

Factoring out $2\cos\beta$ gives $2\cos\beta\begin{bmatrix}1 & 0 \\ 0 & 1\end{bmatrix} = (\mathbf{2}\cos\beta)\mathbf{I}$.

(iv) *(The "Aha!" Moment)*

Method 1 (Algebraic):

$E = R(\alpha)\big[(2\cos\beta)I\big] = (2\cos\beta)R(\alpha)$

$E = \begin{bmatrix}2\cos\beta\cos\alpha & -2\cos\beta\sin\alpha \\ 2\cos\beta\sin\alpha & 2\cos\beta\cos\alpha\end{bmatrix}$

Method 2 (Distributive/Geometric):

$E = R(\alpha)R(\beta) + R(\alpha)R(-\beta)$

Using the geometric angle-addition property from (i):

$E = R(\alpha+\beta) + R(\alpha-\beta)$

$E = \begin{bmatrix}\cos(\alpha+\beta)+\cos(\alpha-\beta) & -\sin(\alpha+\beta)-\sin(\alpha-\beta) \\ \sin(\alpha+\beta)+\sin(\alpha-\beta) & \cos(\alpha+\beta)+\cos(\alpha-\beta)\end{bmatrix}$

Equating the top-left elements of Method 1 and Method 2:

$2\cos\alpha\cos\beta = \cos(\alpha+\beta) + \cos(\alpha-\beta)$

Equating the bottom-left elements:

$2\sin\alpha\cos\beta = \sin(\alpha+\beta) + \sin(\alpha-\beta)$

Takeaways 4.18

- **Matrices encapsulate complex algebra:** The long, tedious proofs of trigonometric identities found in textbooks disappear when you use matrices. Matrix arithmetic automatically handles the complexities of geometry.
- **Matrices Generate Trigonometric Identities:** $R(\alpha)R(\beta) = R(\alpha+\beta)$ encodes the angle-addition formulas. Also,

$$2\cos\alpha\cos\beta = \cos(\alpha+\beta) + \cos(\alpha-\beta),$$

$$2\sin\alpha\cos\beta = \sin(\alpha+\beta) + \sin(\alpha-\beta).$$

- **Linear Transformations:** Trigonometric identities are not arbitrary rules about triangles; they are the fundamental algebraic properties of 2D space undergoing rotational transformations.

Problem 4.19: The Matrix Code of Fibonacci

The famous Fibonacci sequence is defined by the recursive relation $F_{n+1} = F_n + F_{n-1}$ for $n \geq 1$, with starting values $F_0 = 0$ and $F_1 = 1$. The first few terms are $0, 1, 1, 2, 3, 5, 8, \ldots$

(i) Write down a 2×2 matrix A such that:

$$\begin{bmatrix} F_{n+1} \\ F_n \end{bmatrix} = A \begin{bmatrix} F_n \\ F_{n-1} \end{bmatrix}$$

Calculate A^2 and A^3 using matrix multiplication, and verify that your matrix A^3 is exactly equal to $\begin{bmatrix} F_4 & F_3 \\ F_3 & F_2 \end{bmatrix}$.

(ii) Use the principle of mathematical induction to prove that for all integers $n \geq 1$:

$$A^n = \begin{bmatrix} F_{n+1} & F_n \\ F_n & F_{n-1} \end{bmatrix}$$

(iii) The determinant of a product of matrices is the product of their determinants, meaning $\det(A^n) = (\det A)^n$.

Evaluate the determinant of both sides of the result proven in part (ii) to derive *Cassini's Identity*: $F_{n+1}F_{n-1} - F_n^2 = (-1)^n$.

Briefly describe what this identity tells us about the square of any Fibonacci number compared to the product of its immediate neighbours.

(iv) To find an explicit formula for the n-th Fibonacci number, the matrix A can be decomposed (diagonalised) into $A = PDP^{-1}$, where:

$$P = \begin{bmatrix} \alpha & \beta \\ 1 & 1 \end{bmatrix} \quad \text{and} \quad D = \begin{bmatrix} \alpha & 0 \\ 0 & \beta \end{bmatrix}$$

Here, α and β are the roots of the quadratic equation $x^2 - x - 1 = 0$ (with $\alpha > \beta$).

Given that $A^n = PD^nP^{-1}$, calculate the top-right entry of PD^nP^{-1} and equate it to your result from part (ii) to discover **Binet's Formula** for F_n.

(Note: From the properties of roots, use $\alpha - \beta = \sqrt{5}$ and $\alpha\beta = -1$ to simplify your final algebra).

Hint:

- **Part (i):** Expand the matrix equation. The top row must give $F_{n+1} = 1 \cdot F_n + 1 \cdot F_{n-1}$. The bottom row must give $F_n = 1 \cdot F_n + 0 \cdot F_{n-1}$.
- **Part (ii):** Follow the standard induction structure. Your base case is $n = 1$. For the inductive step, assume true for $n = k$, and use the fact that $A^{k+1} = A^k A$ alongside the recursive rule $F_{k+1} + F_k = F_{k+2}$ to prove for $n = k + 1$.
- **Part (iii):** Find $\det(A)$ from part (i). Then find the determinant of the 2×2 matrix containing the Fibonacci terms. Equate them.
- **Part (iv):**
  1. First, find P^{-1} using the 2×2 inverse formula: $\frac{1}{ad-bc}\begin{bmatrix} d & -b \\ -c & a \end{bmatrix}$. The determinant of P is $\alpha - \beta = \sqrt{5}$.
  2. Write down $D^n = \begin{bmatrix} \alpha^n & 0 \\ 0 & \beta^n \end{bmatrix}$.
  3. Multiply PD^nP^{-1}. To save time, you *only* need to compute the element in Row 1, Column 2 of the final matrix, because you know from part (ii) that this specific element is equal to F_n.

Solution 4.19

(i)

From $F_{n+1} = 1F_n + 1F_{n-1}$ and $F_n = 1F_n + 0F_{n-1}$, we get $A = \begin{bmatrix} 1 & 1 \\ 1 & 0 \end{bmatrix}$.

$A^2 = \begin{bmatrix} 1 & 1 \\ 1 & 0 \end{bmatrix}\begin{bmatrix} 1 & 1 \\ 1 & 0 \end{bmatrix} = \begin{bmatrix} 2 & 1 \\ 1 & 1 \end{bmatrix}$.

$A^3 = A^2A = \begin{bmatrix} 2 & 1 \\ 1 & 1 \end{bmatrix}\begin{bmatrix} 1 & 1 \\ 1 & 0 \end{bmatrix} = \begin{bmatrix} 3 & 2 \\ 2 & 1 \end{bmatrix}$.

Checking the sequence $(0, 1, 1, 2, 3, 5\ldots)$, $F_4 = 3, F_3 = 2, F_2 = 1$. Thus, $A^3 = \begin{bmatrix} F_4 & F_3 \\ F_3 & F_2 \end{bmatrix}$.

(ii)

Base case $(n = 1)$**:** $A^1 = \begin{bmatrix} F_2 & F_1 \\ F_1 & F_0 \end{bmatrix} = \begin{bmatrix} 1 & 1 \\ 1 & 0 \end{bmatrix} = A$. True.

Assumption: Assume $A^k = \begin{bmatrix} F_{k+1} & F_k \\ F_k & F_{k-1} \end{bmatrix}$ for some integer $k \geq 1$.

Step $(n = k + 1)$**:**

$A^{k+1} = A^kA = \begin{bmatrix} F_{k+1} & F_k \\ F_k & F_{k-1} \end{bmatrix}\begin{bmatrix} 1 & 1 \\ 1 & 0 \end{bmatrix}$

$A^{k+1} = \begin{bmatrix} F_{k+1}(1) + F_k(1) & F_{k+1}(1) + F_k(0) \\ F_k(1) + F_{k-1}(1) & F_k(1) + F_{k-1}(0) \end{bmatrix}$

Using the recurrence relation $F_m + F_{m-1} = F_{m+1}$, this simplifies to:

$A^{k+1} = \begin{bmatrix} F_{k+2} & F_{k+1} \\ F_{k+1} & F_k \end{bmatrix}$.

This fits the required form, proving the statement by induction.

(iii) *(Aha! Moment 1)*

$\det(A) = (1)(0) - (1)(1) = -1$.

Therefore, $\det(A^n) = (-1)^n$.

From part (ii), the determinant of $\begin{bmatrix} F_{n+1} & F_n \\ F_n & F_{n-1} \end{bmatrix}$ is $F_{n+1}F_{n-1} - F_n^2$.

Equating them gives Cassini's Identity: $F_{n+1}F_{n-1} - F_n^2 = (-1)^n$.

Meaning: The square of any Fibonacci number is always exactly 1 more or 1 less than the product of the two numbers surrounding it (e.g., for 5: $5^2 = 25$, and $3 \times 8 = 24$).

(iv) *(Aha! Moment 2)*

First, find P^{-1}. $\det(P) = \alpha - \beta = \sqrt{5}$. So, $P^{-1} = \frac{1}{\sqrt{5}}\begin{bmatrix} 1 & -\beta \\ -1 & \alpha \end{bmatrix}$.

We need the top-right element of PD^nP^{-1}.

$PD^n = \begin{bmatrix} \alpha & \beta \\ 1 & 1 \end{bmatrix}\begin{bmatrix} \alpha^n & 0 \\ 0 & \beta^n \end{bmatrix} = \begin{bmatrix} \alpha^{n+1} & \beta^{n+1} \\ \alpha^n & \beta^n \end{bmatrix}$.

Now multiply by P^{-1}:

$PD^nP^{-1} = \frac{1}{\sqrt{5}}\begin{bmatrix} \alpha^{n+1} & \beta^{n+1} \\ \alpha^n & \beta^n \end{bmatrix}\begin{bmatrix} 1 & -\beta \\ -1 & \alpha \end{bmatrix}$.

The top-right element (Row 1 × Column 2) is:

$\frac{1}{\sqrt{5}}\big(\alpha^{n+1}(-\beta) + \beta^{n+1}(\alpha)\big) = \frac{1}{\sqrt{5}}\big(-\alpha^n(\alpha\beta) + \beta^n(\alpha\beta)\big)$.

Since $\alpha\beta = -1$ (product of roots of $x^2 - x - 1 = 0$), we substitute this in:

$= \frac{1}{\sqrt{5}}\big(-\alpha^n(-1) + \beta^n(-1)\big) = \frac{\alpha^n - \beta^n}{\sqrt{5}}$.

From part (ii), the top right element of A^n is F_n. Equating them yields:

$F_n = \frac{\alpha^n - \beta^n}{\sqrt{5}}$ *(where* $\alpha = \frac{1+\sqrt{5}}{2}$ *is the Golden Ratio).*

Takeaways 4.19

- **Matrices Encapsulate State:** A sequence like Fibonacci isn't just a list of numbers; it's a vector moving through space. A single matrix A acts as an engine that advances the sequence one step at a time.

$$A^n = \begin{bmatrix} F_{n+1} & F_n \\ F_n & F_{n-1} \end{bmatrix}.$$

- **Determinants Expose Number-Theoretic Identities:** What is ordinarily a grueling, messy algebraic proof (Cassini's identity) drops out instantly in one line simply because the determinant of a matrix product is the product of the determinants.

- **The Power of Diagonalisation:** By splitting a matrix into its eigenvalues (diagonal matrix D) and eigenvectors (matrix P), raising it to a massive power n becomes trivial. This fundamentally bridges discrete recursive algorithms to continuous explicit formulas:

$$F_n = \frac{\alpha^n - \beta^n}{\sqrt{5}}, \qquad \alpha = \frac{1+\sqrt{5}}{2}, \quad \beta = \frac{1-\sqrt{5}}{2}.$$

Problem 4.20: The Nilpotent Shortcut

Mathematical induction is a powerful tool for proving patterns in matrices, but sometimes a deeper structural understanding of the matrix can bypass the need for induction entirely. Consider the matrix $A = \begin{bmatrix} 2 & -1 \\ 1 & 0 \end{bmatrix}$.

(i) Using standard matrix multiplication, calculate A^2 and A^3. By observing the pattern in the elements, formulate a conjecture for the general matrix A^n in the form of a single 2×2 matrix containing the variable n.

(ii) Use the principle of mathematical induction to formally prove your conjecture for A^n for all integers $n \geq 1$.

(iii) Let I be the 2×2 identity matrix. The original matrix A can be decomposed into a sum: $A = I + N$.

Find the matrix N. Then, calculate N^2. What unique property does N possess when multiplied by itself?

(iv) In algebra, the Binomial Theorem states that $(x+y)^n = x^n + nx^{n-1}y + \frac{n(n-1)}{2}x^{n-2}y^2 + \dots$

This theorem can also be applied to matrices in the form $(I+N)^n$, because the identity matrix I commutes with any matrix.

Using the Binomial Theorem and your result from part (iii), expand the expression $(I+N)^n$. Show how this single line of algebraic expansion perfectly derives your conjectured matrix A^n from part (i), bypassing the entire inductive proof.

Hint:

- **Part (i):** Calculate $A^2 = A \times A$, and $A^3 = A^2 \times A$. Look closely at how the top-left and bottom-left numbers relate to the power n. Then, figure out how the right-hand column relates to n.
- **Part (ii):** Follow the standard three-step induction structure:
  1. Show true for $n = 1$.
  2. Assume true for $n = k$.
  3. Prove true for $n = k + 1$ by evaluating $A^{k+1} = A^k \times A$. You will need to group your k's to match the conjectured format $(k + 1)$.
- **Part (iii):** To find N, simply rearrange the equation to $N = A - I$. Subtract the identity matrix from A. Then multiply the resulting matrix by itself.
- **Part (iv):** Write out the first three terms of the binomial expansion for $(I + N)^n$. Remember that I raised to any power is just I, and $I \times N = N$. What happens to the third term (and all subsequent terms) based on your discovery in part (iii)?

Solution 4.20

(i)

$$A^2 = \begin{bmatrix} 2 & -1 \\ 1 & 0 \end{bmatrix}\begin{bmatrix} 2 & -1 \\ 1 & 0 \end{bmatrix} = \begin{bmatrix} 4-1 & -2+0 \\ 2+0 & -1+0 \end{bmatrix} = \begin{bmatrix} 3 & -2 \\ 2 & -1 \end{bmatrix}$$

$$A^3 = A^2A = \begin{bmatrix} 3 & -2 \\ 2 & -1 \end{bmatrix}\begin{bmatrix} 2 & -1 \\ 1 & 0 \end{bmatrix} = \begin{bmatrix} 6-2 & -3+0 \\ 4-1 & -2+0 \end{bmatrix} = \begin{bmatrix} 4 & -3 \\ 3 & -2 \end{bmatrix}$$

Conjecture: $A^n = \begin{bmatrix} n+1 & -n \\ n & 1-n \end{bmatrix}$

(ii)

Base case $(n = 1)$**:** $A^1 = \begin{bmatrix} 1+1 & -1 \\ 1 & 1-1 \end{bmatrix} = \begin{bmatrix} 2 & -1 \\ 1 & 0 \end{bmatrix} = A$. True.

Assume true for $n = k$**:** $A^k = \begin{bmatrix} k+1 & -k \\ k & 1-k \end{bmatrix}$ for some integer $k \geq 1$.

Prove for $n = k+1$**:**

$$A^{k+1} = A^kA = \begin{bmatrix} k+1 & -k \\ k & 1-k \end{bmatrix}\begin{bmatrix} 2 & -1 \\ 1 & 0 \end{bmatrix}$$

$$A^{k+1} = \begin{bmatrix} 2(k+1)-k & -(k+1)+0 \\ 2k+(1-k) & -k+0 \end{bmatrix} = \begin{bmatrix} k+2 & -(k+1) \\ k+1 & -k \end{bmatrix}$$

Rewriting to match the conjecture structure:

$$A^{k+1} = \begin{bmatrix} (k+1)+1 & -(k+1) \\ (k+1) & 1-(k+1) \end{bmatrix}.$$

This fits the assumption form perfectly. By mathematical induction, the conjecture is true for all $n \geq 1$.

(iii)

$$N = A - I = \begin{bmatrix} 2 & -1 \\ 1 & 0 \end{bmatrix} - \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} = \begin{bmatrix} 1 & -1 \\ 1 & -1 \end{bmatrix}.$$

$$N^2 = \begin{bmatrix} 1 & -1 \\ 1 & -1 \end{bmatrix}\begin{bmatrix} 1 & -1 \\ 1 & -1 \end{bmatrix} = \begin{bmatrix} 1-1 & -1+1 \\ 1-1 & -1+1 \end{bmatrix} = \begin{bmatrix} 0 & 0 \\ 0 & 0 \end{bmatrix}.$$

Property: N^2 is the zero matrix. (In linear algebra, a matrix that eventually becomes zero when raised to a power is called a *nilpotent* matrix).

(iv) *(The "Aha!" Moment)*

Expanding $(I+N)^n$ using the Binomial Theorem:

$(I+N)^n = I^n + \binom{n}{1}I^{n-1}N + \binom{n}{2}I^{n-2}N^2 + \binom{n}{3}I^{n-3}N^3 + \dots$

Because $N^2 = 0$, every term containing $N^2, N^3, \dots$ vanishes completely into the zero matrix!

The infinite expansion collapses into just two terms:

$(I+N)^n = I + nN$

Substituting our known matrices back in:

$$A^n = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} + n\begin{bmatrix} 1 & -1 \\ 1 & -1 \end{bmatrix} = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} + \begin{bmatrix} n & -n \\ n & -n \end{bmatrix} = \begin{bmatrix} n+1 & -n \\ n & 1-n \end{bmatrix}.$$

This exactly matches our conjecture, derived in seconds without induction!

Takeaways 4.20

- **Multiple Paths to Proof:** Mathematical induction is a reliable "brute force" method for proving sequences. However, discovering the underlying algebraic structure (like a binomial expansion) often provides a much faster and more elegant solution.
- **Binomial Theorem—Only When Matrices Commute:** The usual binomial expansion works here because I commutes with N. Since $N^2 = 0$,

$$(I + N)^n = I + nN.$$

  Do not apply the ordinary binomial theorem blindly to arbitrary non-commuting matrices.
- **The Power of Nilpotency:** Finding a matrix that squares to zero is like finding a mathematical wormhole. It truncates infinite series or complex powers into simple, linear equations.

Problem 4.21: Matrices and Probability: Predicting Changing Systems

In a simplified weather model, the weather each day is classified as either Sunny (S) or Rainy (R). Historical data shows that if it is sunny today, there is an 80% chance it will be sunny tomorrow and a 20% chance it will be rainy. If it is rainy today, there is a 40% chance it will be sunny tomorrow and a 60% chance it will be rainy.

This system can be modeled using a transition matrix P and a state vector $\mathbf{x}_n = \begin{bmatrix} s_n \\ r_n \end{bmatrix}$, where s_n and r_n are the probabilities of it being sunny or rainy on day n. The weather for the next day is given by the matrix equation $\mathbf{x}_{n+1} = P\mathbf{x}_n$.

(i) Write down the 2×2 transition matrix P. If today (Day 0) is known with absolute certainty to be sunny ($\mathbf{x}_0 = \begin{bmatrix} 1 \\ 0 \end{bmatrix}$), calculate the state vectors $\mathbf{x}_1$ and $\mathbf{x}_2$. What is the probability that it will be rainy the day after tomorrow?

(ii) Over a long period of time, such systems often settle into a *steady state* $\mathbf{q} = \begin{bmatrix} s \\ r \end{bmatrix}$, where the probabilities no longer change from day to day. This implies that $P\mathbf{q} = \mathbf{q}$.
Knowing that the sum of the probabilities must always equal 1 ($s + r = 1$), solve the matrix equation $P\mathbf{q} = \mathbf{q}$ algebraically to find the exact steady-state probabilities for sunny and rainy weather.

(iii) The equation $P\mathbf{q} = \mathbf{q}$ can be rewritten as $P\mathbf{q} = 1\mathbf{q}$. This means the steady state is actually an eigenvector corresponding to an eigenvalue of $\lambda = 1$.
Find the characteristic equation of P and solve it to verify that $\lambda = 1$ is an eigenvalue, and find the second eigenvalue, λ_2.

(iv) Since this P has two distinct eigenvalues, 1 and λ_2, it is diagonalisable as $P = MDM^{-1}$, where $D = \begin{bmatrix} 1 & 0 \\ 0 & \lambda_2 \end{bmatrix}$ is the diagonal matrix of eigenvalues, and M is the matrix of corresponding eigenvectors.
Given that $P^n = MD^nM^{-1}$, use your second eigenvalue λ_2 from part (iii) to evaluate the limit of the matrix D^n as $n \to \infty$.
Using this limit, mathematically prove that *no matter what* the initial weather state $\mathbf{x}_0 = \begin{bmatrix} s_0 \\ r_0 \end{bmatrix}$ is, the system will always eventually "forget" its initial state and arrive at the exact steady state you found in part (ii).

Hint:

- **Part (i):** The columns of the transition matrix represent "Today", and the rows represent "Tomorrow". Column 1 should be the probabilities if today is Sunny; Column 2 if today is Rainy.
- **Part (ii):** Expand the matrix equation $P\begin{bmatrix} s \\ r \end{bmatrix} = \begin{bmatrix} s \\ r \end{bmatrix}$. This will give you two simultaneous equations. You only need to use one of them, alongside the given equation $s + r = 1$, to solve for s and r.
- **Part (iii):** To find the eigenvalues, solve the equation $\det(P - \lambda I) = 0$. You should end up with a quadratic equation in terms of λ.
- **Part (iv):** Think about what happens to a fraction (or decimal between -1 and 1) when you raise it to a very large power. Because D is a diagonal matrix, $D^n = \begin{bmatrix} 1^n & 0 \\ 0 & \lambda_2^n \end{bmatrix}$. Write down the final matrix $MD^\infty M^{-1}$ and multiply it by $\begin{bmatrix} s_0 \\ r_0 \end{bmatrix}$. Remember that $s_0 + r_0 = 1$.

Solution 4.21

(i)

$P = \begin{bmatrix} 0.8 & 0.4 \\ 0.2 & 0.6 \end{bmatrix}$

$\mathbf{x}_1 = P\mathbf{x}_0 = \begin{bmatrix} 0.8 & 0.4 \\ 0.2 & 0.6 \end{bmatrix} \begin{bmatrix} 1 \\ 0 \end{bmatrix} = \begin{bmatrix} 0.8 \\ 0.2 \end{bmatrix}$.

$\mathbf{x}_2 = P\mathbf{x}_1 = \begin{bmatrix} 0.8 & 0.4 \\ 0.2 & 0.6 \end{bmatrix} \begin{bmatrix} 0.8 \\ 0.2 \end{bmatrix} = \begin{bmatrix} 0.64 + 0.08 \\ 0.16 + 0.12 \end{bmatrix} = \begin{bmatrix} 0.72 \\ 0.28 \end{bmatrix}$.

The probability it is rainy the day after tomorrow is 0.28 **(or** 28%).

(ii)

$P\mathbf{q} = \mathbf{q} \implies \begin{bmatrix} 0.8 & 0.4 \\ 0.2 & 0.6 \end{bmatrix} \begin{bmatrix} s \\ r \end{bmatrix} = \begin{bmatrix} s \\ r \end{bmatrix}$.

Top row gives: $0.8s + 0.4r = s \implies -0.2s + 0.4r = 0 \implies 0.2s = 0.4r \implies s = 2r$.

Substitute into $s + r = 1$:

$2r + r = 1 \implies 3r = 1 \implies r = \frac{1}{3}$.

Therefore, $s = \frac{2}{3}$.

The steady state is $\mathbf{q} = \begin{bmatrix} 2/3 \\ 1/3 \end{bmatrix}$.

(iii)

$\det(P - \lambda I) = \det \begin{bmatrix} 0.8 - \lambda & 0.4 \\ 0.2 & 0.6 - \lambda \end{bmatrix} = 0$

$(0.8 - \lambda)(0.6 - \lambda) - (0.4)(0.2) = 0$

$\lambda^2 - 1.4\lambda + 0.48 - 0.08 = 0$

$\lambda^2 - 1.4\lambda + 0.40 = 0$

$(\lambda - 1)(\lambda - 0.4) = 0.$

The eigenvalues are $\lambda_1 = 1$ and $\lambda_2 = 0.4$.

(iv) *(The "Aha!" Moment)*

For $P = MDM^{-1}$, we found $D = \begin{bmatrix} 1 & 0 \\ 0 & 0.4 \end{bmatrix}$.

$P^n = M \begin{bmatrix} 1^n & 0 \\ 0 & 0.4^n \end{bmatrix} M^{-1}$.

As $n \to \infty$, $0.4^n \to 0$. This is the crucial step: the second eigenvalue *vanishes*.

$D^\infty = \begin{bmatrix} 1 & 0 \\ 0 & 0 \end{bmatrix}$.

Let $M = \begin{bmatrix} 2 & 1 \\ 1 & -1 \end{bmatrix}$ (eigenvectors for $\lambda = 1, 0.4$) and $M^{-1} = \begin{bmatrix} 1/3 & 1/3 \\ 1/3 & -2/3 \end{bmatrix}$.

$P^\infty = \begin{bmatrix} 2 & 1 \\ 1 & -1 \end{bmatrix} \begin{bmatrix} 1 & 0 \\ 0 & 0 \end{bmatrix} \begin{bmatrix} 1/3 & 1/3 \\ 1/3 & -2/3 \end{bmatrix} = \begin{bmatrix} 2 & 0 \\ 1 & 0 \end{bmatrix} \begin{bmatrix} 1/3 & 1/3 \\ 1/3 & -2/3 \end{bmatrix} = \begin{bmatrix} 2/3 & 2/3 \\ 1/3 & 1/3 \end{bmatrix}$.

Applying this infinite matrix to *any* initial state $\mathbf{x}_0 = \begin{bmatrix} s_0 \\ r_0 \end{bmatrix}$:

$\mathbf{x}_\infty = \begin{bmatrix} 2/3 & 2/3 \\ 1/3 & 1/3 \end{bmatrix} \begin{bmatrix} s_0 \\ r_0 \end{bmatrix} = \begin{bmatrix} \frac{2}{3}s_0 + \frac{2}{3}r_0 \\ \frac{1}{3}s_0 + \frac{1}{3}r_0 \end{bmatrix} = \begin{bmatrix} \frac{2}{3}(s_0 + r_0) \\ \frac{1}{3}(s_0 + r_0) \end{bmatrix}$.

Because the initial probabilities must sum to 1 ($s_0 + r_0 = 1$), the state completely simplifies to $\begin{bmatrix} 2/3 \\ 1/3 \end{bmatrix}$.

The system mathematically overrides the initial conditions.

Takeaways 4.21

- **The Meaning of "1":** For this transition matrix, eigenvalue 1 gives the steady state, and because the other eigenvalue has $|0.4| < 1$, every probability state converges to it.
- **Eigenvalues Dictate Decay:** The other eigenvalue (0.4 in this case) acts as the system's "memory". Because its magnitude is less than 1, its powers shrink to zero. This algebraic decay is *why* this specific system settles into a predictable, long-term average.
- **Forgetting Is Asymptotic, Not Instant:** The effect of the initial state is multiplied by 0.4^n, which approaches 0 as $n \to \infty$. Different initial states are generally not exactly identical after any fixed finite number of days; they converge to the same steady state only in the limit.

Problem 4.22: The Algebra of Transformations

In 2D space, a point (x, y) on a curve can be transformed to a new point (x', y') using matrix multiplication: $\begin{bmatrix} x' \\ y' \end{bmatrix} = M \begin{bmatrix} x \\ y \end{bmatrix}$.

(i) Let $M_1 = \begin{bmatrix} 2 & 0 \\ 0 & 1 \end{bmatrix}$ (a horizontal stretch by a factor of 2) and $M_2 = \begin{bmatrix} 1 & 0 \\ 0 & -1 \end{bmatrix}$ (a reflection in the x-axis).
Find the single matrix $T = M_2 M_1$ that represents applying M_1 followed by M_2.
Apply T to a generic point (x, y) to find expressions for x' and y'. By making x and y the subjects, find the new Cartesian equation of the curve $y = e^x$ after this combined transformation.
(ii) Generalising the process from part (i), mathematically prove that if a curve undergoes a transformation matrix M, the algebraic substitution required to find the new Cartesian equation is given by $\begin{bmatrix} x \\ y \end{bmatrix} = M^{-1} \begin{bmatrix} x' \\ y' \end{bmatrix}$.
Calculate the inverse matrix T^{-1} from part (i) to verify it perfectly generates the algebraic substitutions you found earlier.
(iii) Consider the standard rational function $y = \frac{1}{x}$, which can be written as the rectangular hyperbola $xy = 1$.
Let R be a transformation matrix that rotates the plane by 45° ($\frac{\pi}{4}$ radians) *clockwise* about the origin. Write down the matrix R, and use the properties of rotation matrices to instantly write down its inverse, R^{-1}.
(iv) Use your findings from part (ii) and part (iii) to apply the 45° clockwise rotation R to the curve $xy = 1$.
Substitute the expressions for x and y into $xy = 1$, expand, and simplify completely.
What classic conic section equation does this produce? What does this reveal about the true geometric shape of the simple rational function $y = \frac{1}{x}$?

Hint:

- **Part (i):** Remember matrix multiplication order: "Apply M_1 then M_2" means calculating M_2M_1, reading right-to-left. Once you have $x' = 2x$ and $y' = -y$, solve for x and y before substituting them into $y = e^x$.
- **Part (ii):** Start with the matrix equation $\mathbf{v}' = M\mathbf{v}$. How do you isolate the vector $\mathbf{v}$ on one side? Recall the standard 2×2 inverse formula.
- **Part (iii):** The general rotation matrix for angle θ counter-clockwise is $\begin{bmatrix} \cos\theta & -\sin\theta \\ \sin\theta & \cos\theta \end{bmatrix}$. A clockwise rotation means $\theta = -45°$. To find the inverse of a rotation, simply rotate back in the opposite direction (i.e., $+45°$).
- **Part (iv):** When you multiply $R^{-1}\begin{bmatrix} x' \\ y' \end{bmatrix}$, you will get $x = \frac{x'-y'}{\sqrt{2}}$ and $y = \frac{x'+y'}{\sqrt{2}}$. Substitute these brackets directly into $(x)(y) = 1$. Use the "difference of two squares" expansion $(a-b)(a+b) = a^2 - b^2$ to simplify the numerator.

Solution 4.22

(i)

$T = M_2M_1 = \begin{bmatrix}1 & 0\\0 & -1\end{bmatrix}\begin{bmatrix}2 & 0\\0 & 1\end{bmatrix} = \begin{bmatrix}2 & 0\\0 & -1\end{bmatrix}$.

$\begin{bmatrix}x'\\y'\end{bmatrix} = \begin{bmatrix}2 & 0\\0 & -1\end{bmatrix}\begin{bmatrix}x\\y\end{bmatrix} = \begin{bmatrix}2x\\-y\end{bmatrix}$.

Therefore, $x' = 2x \implies x = \frac{x'}{2}$ and $y' = -y \implies y = -y'$.

Substituting into $y = e^x$:

$-y' = e^{(x'/2)} \implies \mathbf{y' = -e^{x'/2}}$.

(ii)

Start with $\begin{bmatrix}x'\\y'\end{bmatrix} = M\begin{bmatrix}x\\y\end{bmatrix}$.

Left-multiply both sides by M^{-1}:

$M^{-1}\begin{bmatrix}x'\\y'\end{bmatrix} = M^{-1}M\begin{bmatrix}x\\y\end{bmatrix}$.

Since $M^{-1}M = I$ (the identity matrix), we have $\begin{bmatrix}x\\y\end{bmatrix} = M^{-1}\begin{bmatrix}x'\\y'\end{bmatrix}$.

For T: $\det(T) = (2)(-1) - (0)(0) = -2$.

$T^{-1} = -\frac{1}{2}\begin{bmatrix}-1 & 0\\0 & 2\end{bmatrix} = \begin{bmatrix}\frac{1}{2} & 0\\0 & -1\end{bmatrix}$.

$T^{-1}\begin{bmatrix}x'\\y'\end{bmatrix} = \begin{bmatrix}\frac{1}{2} & 0\\0 & -1\end{bmatrix}\begin{bmatrix}x'\\y'\end{bmatrix} = \begin{bmatrix}x'/2\\-y'\end{bmatrix}$. This exactly matches the x and y substitutions found in part (i).

(iii)

For a clockwise rotation, $\theta = -45° = -\frac{\pi}{4}$.

$R = \begin{bmatrix}\cos(-\frac{\pi}{4}) & -\sin(-\frac{\pi}{4})\\\sin(-\frac{\pi}{4}) & \cos(-\frac{\pi}{4})\end{bmatrix} = \begin{bmatrix}\frac{1}{\sqrt{2}} & \frac{1}{\sqrt{2}}\\-\frac{1}{\sqrt{2}} & \frac{1}{\sqrt{2}}\end{bmatrix}$.

The inverse R^{-1} undoes this by rotating $+45°$:

$R^{-1} = \begin{bmatrix}\frac{1}{\sqrt{2}} & -\frac{1}{\sqrt{2}}\\\frac{1}{\sqrt{2}} & \frac{1}{\sqrt{2}}\end{bmatrix}$.

(iv) *(The "Aha!" Moment)*

Using the inverse matrix rule: $\begin{bmatrix}x\\y\end{bmatrix} = R^{-1}\begin{bmatrix}x'\\y'\end{bmatrix} = \begin{bmatrix}\frac{1}{\sqrt{2}} & -\frac{1}{\sqrt{2}}\\\frac{1}{\sqrt{2}} & \frac{1}{\sqrt{2}}\end{bmatrix}\begin{bmatrix}x'\\y'\end{bmatrix}$.

$x = \frac{x'-y'}{\sqrt{2}}$ and $y = \frac{x'+y'}{\sqrt{2}}$.

Substitute into $xy = 1$:

$\left(\frac{x'-y'}{\sqrt{2}}\right)\left(\frac{x'+y'}{\sqrt{2}}\right) = 1$

$\frac{(x')^2-(y')^2}{2} = 1$

$(x')^2 - (y')^2 = 2$

Geometric Revelation: The graph of $y = \frac{1}{x}$ is exactly the standard hyperbola $x^2 - y^2 = 2$, just tilted sideways by 45°! The coordinate axes x and y are actually the standard hyperbola's asymptotes $y = \pm x$ in disguise.

Takeaways 4.22

- **Inverse-Substitution Rule:** If

$$\mathbf{x}' = M\mathbf{x},$$

then the equation of the transformed curve is found by substituting

$$\mathbf{x} = M^{-1}\mathbf{x}'$$

into the original equation. This explains, for example, why a horizontal stretch by 2 produces an $x/2$ substitution in the transformed rule.

- **Algebra is Geometry:** An intimidating algebraic process (simultaneously substituting difference-of-squares fractions) is simply the machinery required to pivot the 2D plane.
- **Unifying the Syllabus:** Functions (Unit 1) and Conic Sections/Matrices (Unit 2) are not isolated topics. $y = 1/x$ and $x^2 - y^2 = 2$ are identically the same shape, just observed from a different angle.

Problem 4.23: The Calculus Matrix

A general cubic polynomial $P(x) = a_0 + a_1x + a_2x^2 + a_3x^3$ can be uniquely represented by a column vector containing its coefficients: $\mathbf{p} = \begin{bmatrix} a_0 \\ a_1 \\ a_2 \\ a_3 \end{bmatrix}$.

Rather than transforming points in space, matrices can act as *operators* that transform functions into new functions.

(i) The derivative of the polynomial is $P'(x) = a_1 + 2a_2x + 3a_3x^2$. Its coefficient vector is $\begin{bmatrix} a_1 \\ 2a_2 \\ 3a_3 \\ 0 \end{bmatrix}$.

Construct a 4×4 matrix D such that multiplying D by $\mathbf{p}$ produces this derivative vector (i.e., $D\mathbf{p} = \mathbf{p}'$).

Once you have D, calculate the matrix powers D^2, D^3, and D^4. What unique property does D possess, and what does the result of D^4 mean in terms of polynomial calculus?

(ii) Let T be a "shift" matrix such that multiplying T by $\mathbf{p}$ yields the coefficient vector of the horizontally shifted polynomial, $P(x + 1)$.

By expanding $P(x + 1) = a_0 + a_1(x + 1) + a_2(x + 1)^2 + a_3(x + 1)^3$ and grouping the like powers of x, find the 4×4 matrix T.

(iii) Consider the infinite matrix series $M = I + D + \frac{1}{2!}D^2 + \frac{1}{3!}D^3 + \frac{1}{4!}D^4 + \dots$ (where I is the 4×4 identity matrix).

Using your power calculations from part (i), simplify and evaluate the matrix M. Compare your final matrix M to your matrix T from part (ii).

(iv) Translate the matrix equivalence you discovered in part (iii) back into the language of polynomial functions.

By replacing $T\mathbf{p}$ and $D^k\mathbf{p}$ with their corresponding function notations, write down the algebraic formula you have just derived. What calculus expansion does this represent?

Hint:

- **Part (i):** To get a_1 in the top row of your answer, the top row of D must be $0, 1, 0, 0$. Continue this logic for the rest of the rows. When multiplying D by itself, notice how the non-zero numbers shift upwards and to the right.
- **Part (ii):** Expand $(x + 1)^2$ and $(x + 1)^3$ using the binomial theorem. Your goal is to write $P(x + 1)$ in the form $(\dots) + (\dots)x + (\dots)x^2 + (\dots)x^3$. The contents of those brackets will form the rows of T.
- **Part (iii):** Because D^4 is the zero matrix, what happens to D^5, D^6, and the rest of the infinite series? You only need to add four matrices together. Remember that $3! = 6$.
- **Part (iv):** You proved that $T = I + D + \frac{1}{2!}D^2 + \frac{1}{3!}D^3$. Multiply both sides of this equation by the vector $\mathbf{p}$. Remember that $T\mathbf{p}$ translates to $P(x+1)$, and $D\mathbf{p}$ translates to $P'(x)$. What does $D^2\mathbf{p}$ translate to?

Solution 4.23

(i)

To map $\begin{bmatrix} a_0 \\ a_1 \\ a_2 \\ a_3 \end{bmatrix}$ to $\begin{bmatrix} a_1 \\ 2a_2 \\ 3a_3 \\ 0 \end{bmatrix}$, the derivative matrix is:

$$D = \begin{bmatrix} 0 & 1 & 0 & 0 \\ 0 & 0 & 2 & 0 \\ 0 & 0 & 0 & 3 \\ 0 & 0 & 0 & 0 \end{bmatrix}.$$

$$D^2 = \begin{bmatrix} 0 & 0 & 2 & 0 \\ 0 & 0 & 0 & 6 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \end{bmatrix}, D^3 = \begin{bmatrix} 0 & 0 & 0 & 6 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \end{bmatrix}, D^4 = \begin{bmatrix} 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \end{bmatrix}.$$

D is a **nilpotent** matrix. The fact that $D^4 = \mathbf{0}$ mathematically represents that the 4th derivative of any cubic polynomial is identically zero.

(ii)

$P(x+1) = a_0 + a_1(x+1) + a_2(x^2+2x+1) + a_3(x^3+3x^2+3x+1)$.
Grouping by powers of x:
Constant: $a_0 + a_1 + a_2 + a_3$
x^1: $a_1 + 2a_2 + 3a_3$
x^2: $a_2 + 3a_3$
x^3: a_3
Putting these coefficients into a matrix yields:

$$T = \begin{bmatrix} 1 & 1 & 1 & 1 \\ 0 & 1 & 2 & 3 \\ 0 & 0 & 1 & 3 \\ 0 & 0 & 0 & 1 \end{bmatrix}.$$

(iii)

Since $D^4 = \mathbf{0}$, the infinite series truncates to exactly four terms:
$M = I + D + \frac{1}{2}D^2 + \frac{1}{6}D^3$

$$M = \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix} + \begin{bmatrix} 0 & 1 & 0 & 0 \\ 0 & 0 & 2 & 0 \\ 0 & 0 & 0 & 3 \\ 0 & 0 & 0 & 0 \end{bmatrix} + \begin{bmatrix} 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 3 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \end{bmatrix} + \begin{bmatrix} 0 & 0 & 0 & 1 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \end{bmatrix}$$

Adding these together gives $M = \begin{bmatrix} 1 & 1 & 1 & 1 \\ 0 & 1 & 2 & 3 \\ 0 & 0 & 1 & 3 \\ 0 & 0 & 0 & 1 \end{bmatrix}$.

Result: The infinite matrix series M is exactly identical to the translation matrix T.

(iv) *(The "Aha!" Moment)*
We have proven algebraically that $T = I + D + \frac{1}{2!}D^2 + \frac{1}{3!}D^3$.
Applying this matrix operator equation to our polynomial vector $\mathbf{p}$:
$T\mathbf{p} = \mathbf{p} + D\mathbf{p} + \frac{1}{2!}D^2\mathbf{p} + \frac{1}{3!}D^3\mathbf{p}$.
Translating this back into functions:
$P(x+1) = P(x) + P'(x) + \frac{P''(x)}{2!} + \frac{P'''(x)}{3!}$
This proves the exact Taylor expansion for cubic polynomials with step 1, a finite special case of Taylor's theorem. We have demonstrated that the geometric shift operator is actually the "matrix exponential" of the derivative operator!

Takeaways 4.23

- **Matrices are Operators:** We usually think of matrices moving geometric points in 2D or 3D space, but they can just as easily move *functions* through calculus operations.
- **Nilpotency acts as a limit:** The fact that a cubic's 4th derivative vanishes to zero directly causes the matrix sequence to terminate. An intimidating infinite series naturally collapses into a finite, exact matrix.
- **Shift Operator = Matrix Exponential:** Because $D^4 = 0$ for cubics,

$$T = e^D = I + D + \frac{D^2}{2!} + \frac{D^3}{3!}.$$

Applied to a polynomial, this gives exactly

$$P(x+1) = P(x) + P'(x) + \frac{P''(x)}{2!} + \frac{P'''(x)}{3!},$$

which is Taylor's formula for a unit shift.

Problem 4.24: The Commutator Conundrum

In matrix algebra, matrix multiplication is generally not commutative. The expression $C = AB - BA$ measures the "failure" of commutativity and is called the commutator of matrices A and B. This scaffolded investigation explores a surprising property of 2×2 commutators.

Part (i): Consider the matrices $A = \begin{pmatrix} a+1 & b \\ b & a \end{pmatrix}$ and $B = \begin{pmatrix} c & x \\ x & c \end{pmatrix}$. Evaluate their commutator $C = AB - BA$ and show that the sum of the elements on its main diagonal (known as the trace) is exactly zero.

Part (ii): Let $M = \begin{pmatrix} p & q \\ r & s \end{pmatrix}$ be a general 2×2 matrix. Prove that $M^2 - (p+s)M + (ps - qr)I_2 = 0_2$, where I_2 is the 2×2 identity matrix and 0_2 is the zero matrix.

Part (iii): Let A and B be *any* two arbitrary 2×2 matrices with real entries. Prove that the square of their commutator, $(AB - BA)^2$, is always a scalar multiple of the identity matrix I_2.

Part (iv): Suppose A and B are arbitrary 2×2 real matrices such that $(AB - BA)^n = I_2$ for some positive integer n. Using the discovery from Part (iii), prove that n must be an even number, and deduce that $(AB - BA)^4 = I_2$.

Hint:

- **Part (i):** Perform the matrix multiplication for AB and BA separately, then subtract them. The algebraic terms will simplify significantly.
- **Part (ii):** Expand M^2 directly by multiplying M by itself. Then subtract the scalar multiplication of $(p+s)M$ and observe what remains on the diagonals and off-diagonals.
- **Part (iii):** What is the trace of AB compared to the trace of BA? What does this imply about the trace of $C = AB - BA$? Once you know the trace of C, apply the identity proven in Part (ii) by substituting $M = C$.
- **Part (iv):** Let $C = AB - BA$. From Part (iii), you know $C^2 = kI_2$ for some real number k. Consider two cases: n is odd ($n = 2m+1$) and n is even ($n = 2m$). If n is odd, can C be anything other than the zero matrix?

Solution 4.24

Part (i)

$$AB = \begin{pmatrix} a+1 & b \\ b & a \end{pmatrix}\begin{pmatrix} c & x \\ x & c \end{pmatrix} = \begin{pmatrix} (a+1)c+bx & (a+1)x+bc \\ bc+ax & bx+ac \end{pmatrix}$$

$$BA = \begin{pmatrix} c & x \\ x & c \end{pmatrix}\begin{pmatrix} a+1 & b \\ b & a \end{pmatrix} = \begin{pmatrix} c(a+1)+xb & cb+xa \\ x(a+1)+cb & xb+ca \end{pmatrix}$$

Subtracting BA from AB gives:

$$C = AB - BA = \begin{pmatrix} 0 & x \\ -x & 0 \end{pmatrix}$$

The trace is the sum of the main diagonal: $0 + 0 = 0$.

Part (ii)

$$M^2 = \begin{pmatrix} p & q \\ r & s \end{pmatrix}\begin{pmatrix} p & q \\ r & s \end{pmatrix} = \begin{pmatrix} p^2+qr & pq+qs \\ pr+sr & qr+s^2 \end{pmatrix}$$

$$(p+s)M = \begin{pmatrix} p^2+ps & pq+qs \\ pr+sr & ps+s^2 \end{pmatrix}$$

Subtracting the two matrices yields:

$$M^2 - (p+s)M = \begin{pmatrix} qr-ps & 0 \\ 0 & qr-ps \end{pmatrix} = -(ps-qr)\begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix} = -(ps-qr)I_2$$

Therefore, $M^2 - (p+s)M + (ps-qr)I_2 = 0_2$.

Part (iii) Let $A = \begin{pmatrix} a_{11} & a_{12} \\ a_{21} & a_{22} \end{pmatrix}$ and $B = \begin{pmatrix} b_{11} & b_{12} \\ b_{21} & b_{22} \end{pmatrix}$. The trace of AB is $a_{11}b_{11}+a_{12}b_{21}+a_{21}b_{12}+a_{22}b_{22}$, which is strictly identical to the trace of BA. Thus, $\operatorname{Tr}(AB - BA) = \operatorname{Tr}(AB) - \operatorname{Tr}(BA) = 0$. Let $C = AB - BA = \begin{pmatrix} p & q \\ r & s \end{pmatrix}$. Since $\operatorname{Tr}(C) = p + s = 0$, we substitute $M = C$ into the identity from Part (ii):

$$C^2 - (0)C + (ps-qr)I_2 = 0_2 \implies C^2 = -(ps-qr)I_2$$

Thus, $C^2 = kI_2$ where $k = -(ps-qr)$ is a real scalar.

Part (iv) Let $C = AB - BA$. We are given $C^n = I_2$. From Part (iii), $C^2 = kI_2$ for a real scalar $k = -\det(C)$. Assume n is odd, so $n = 2m + 1$.

$$C^{2m+1} = C(C^2)^m = C(kI_2)^m = k^m C$$

Since $C^n = I_2$, we have $k^m C = I_2 \implies C = \frac{1}{k^m} I_2$. This means C is a scalar multiple of the identity matrix, which would imply $\operatorname{Tr}(C) = \frac{2}{k^m}$. But from Part (iii), $\operatorname{Tr}(C) = 0$. This implies $\frac{2}{k^m} = 0$, which is impossible. Thus, n cannot be odd. Therefore, n must be even, so $n = 2m$.

$$C^{2m} = (C^2)^m = (kI_2)^m = k^m I_2 = I_2$$

This requires $k^m = 1$. Since A and B are real matrices, $k = -\det(C)$ is a real number. The only real solutions to $k^m = 1$ are $k = 1$ (if m is even or odd) or $k = -1$ (if m is even). In either case, $k^2 = 1$. Therefore, $(AB - BA)^4 = C^4 = (C^2)^2 = (kI_2)^2 = k^2 I_2 = I_2$.

Takeaways 4.24

- **Linearity of the Trace:** The sum of diagonal elements (trace) holds the elegant property $\text{Tr}(AB) = \text{Tr}(BA)$, rendering the trace of any commutator identically zero.
- **Cayley-Hamilton Theorem:** Part (ii) serves as a gentle algebraic introduction to the Cayley-Hamilton theorem for 2×2 matrices, proving that a matrix satisfies its own characteristic polynomial.
- **Logical Proofs in Algebra:** Part (iv) highlights the power of contradiction. By assuming an odd power could yield the identity matrix, students apply properties of scalar matrices to force an impossible conclusion, seamlessly connecting polynomials, invariants (trace/determinant), and matrix algebra.

Problem 4.25: The Net Strength Theorem

Context:
The matrix M provided records the results of a 5-team round-robin netball tournament. In this system, 2 points are awarded for a win, 1 point for a draw, and 0 points for a loss. The entry m_{ij} represents the points team i scored in their match against team j. Let J be a 5×5 matrix where every entry is 1, I be the 5×5 identity matrix, and c be a 5×1 column vector of ones.

Part (i): The Transpose Property
Evaluate the matrix addition $S = M + M^T$. Express S entirely in terms of J and I. Briefly explain the logical reason why this algebraic relationship must hold true for *any* round-robin tournament that uses this 2-1-0 point system.

Part (ii): Attack vs Defence
The column vector $p = Mc$ calculates the total points scored by each team. The vector $a = M^T c$ calculates the total points scored *against* each team (points conceded). Using your matrix equation from Part (i), algebraically prove that $a = 8c - p$. What does this imply about a team's defensive record compared to their attacking record in this specific scoring system?

Part (iii): The Net Performance Vector
In tournaments, a secondary ranking vector $M^2c = M(Mc) = Mp$ is often used to award secondary points to teams for performing well against strong opponents. Similarly, $M^T p$ calculates the secondary points a team *conceded* to strong opponents. Let $N = Mp - M^T p$ be the "Net Performance Vector", representing the difference between secondary points earned and secondary points conceded. Using the relationship established in Part (i), prove that $N = 2M^2c + 2Mc - 40c$. *(Note: You will need to determine the numerical value of the vector Jp to complete this proof).*

Part (iv): The "Aha!" Moment
The original problem uses the model $M + M^2 + M^3$ to rank the teams. Consider a simpler second-order ranking model, $R_2 = M^2c + Mc$, which ranks teams based on the sum of their primary and secondary points. Using your result from Part (iii), explain why ranking the teams from highest to lowest using the complex Net Performance Vector (N) will always result in the exact same tournament ranking as using the standard second-order model (R_2).

Hint:

- **Part (i):** Consider a single match between team i and team j. What is the sum of $m_{ij} + m_{ji}$? Does this sum ever change regardless of whether the match is a win, loss, or draw? Note the main diagonal where $i = j$.
- **Part (ii):** Take the equation $M + M^T = 2J - 2I$ and post-multiply both sides by the column vector c. Think about what the vector Jc represents: multiplying a 5×5 matrix of ones by a 5×1 vector of ones.
- **Part (iii):** Substitute M^T out of the equation using $M^T = 2J - 2I - M$. When expanding, you will get the term Jp. Remember that p is the vector of total points for all 5 teams. If there are 5 teams playing a round-robin, how many games are played in total, and how many total points are distributed across the whole tournament?
- **Part (iv):** Look closely at the equation $N = 2R_2 - 40c$. If you have a set of scores, and you double them all, then subtract 40 from each of them, does the highest score remain the highest?

Solution 4.25

Part (i)

In a 2-1-0 scoring system, the total points awarded in any match between team i and team j is always 2 (either 2+0, 1+1, or 0+2). Therefore, $m_{ij} + m_{ji} = 2$ for all $i \neq j$. Since a team cannot play itself, the main diagonal consists of zeros, so $m_{ii} = 0$. Thus, $M + M^T$ results in a matrix with 0s on the diagonal and 2s everywhere else. This can be written as: $M + M^T = 2J - 2I$.

Part (ii)

We start with $M + M^T = 2J - 2I$ and post-multiply by c:

$$Mc + M^T c = 2Jc - 2Ic$$

Substitute $p = Mc$ and $a = M^T c$:

$$p + a = 2Jc - 2c$$

The vector c is a 5×1 column of 1s. Therefore, Jc is a 5×1 vector where each entry is 5.

$$p + a = 2(5c) - 2c = 10c - 2c = 8c$$

$$a = 8c - p$$

This implies that a team's defensive record (points conceded) is perfectly inversely proportional to their attacking record. Every point a team fails to score out of their maximum possible 8 points is a point conceded to an opponent.

Part (iii)

From Part (i), $M^T = 2J - 2I - M$. Substitute this into the Net Performance Vector N:

$$N = Mp - M^T p = Mp - (2J - 2I - M)p$$

$$N = Mp - 2Jp + 2Ip + Mp$$

$$N = 2Mp - 2Jp + 2p$$

We know $p = Mc$. The term Jp represents multiplying the all-ones matrix J by the points vector p. The result is a column vector where each entry is the sum of all elements in p. In a 5-team round robin, there are 10 matches. At 2 points per match, the total points in the tournament is 20. Therefore, the sum of p is 20, making $Jp = 20c$.

$$N = 2M(Mc) - 2(20c) + 2(Mc)$$

$$N = 2M^2 c + 2Mc - 40c$$

Part (iv)

From Part (iii), we can factor out the 2:

$$N = 2(M^2 c + Mc) - 40c$$

$$N = 2R_2 - 40c$$

The Net Performance Vector N is simply the second-order model R_2 multiplied by a positive scalar (2) and shifted by a constant vector ($40c$). Linear transformations of this nature (where $y = mx + c$ and $m > 0$) strictly preserve the numerical hierarchy. Therefore, ranking teams based on N will yield the exact same order as ranking them by R_2.

Takeaways 4.25

- **Matrix Symmetries:** The 2-1-0 scoring system is not just convenient for counting; it creates a beautiful algebraic symmetry ($M + M^T = 2J - 2I$) that allows us to interrelate points scored and points conceded without looking at individual matches.
- **The Power of Constants:** Recognizing that Jp evaluates to a constant vector representing the total points in the tournament ($20c$) is a crucial leap. It transforms a matrix multiplication into a scalar shift, demonstrating how macroeconomic tournament rules (total games played) constrain microeconomic matrix calculations.
- **The "Aha" Insight:** Standard tournament models like $M + M^2$ can seem arbitrary—why just add M^2? The proof in Part (iv) validates the model. It proves that calculating a simple R_2 ranking inherently calculates the much more complex concept of "Net Strength" (points gained from strong teams minus points leaked to strong teams).

Problem 4.26: A Hidden Pattern in Matrix Powers

Let

$$A = \begin{pmatrix} 1 & 2 & 2 \\ 2 & 1 & 1 \\ 2 & 2 & 1 \end{pmatrix},$$

and let I denote the 3×3 identity matrix.

(i) Calculate A^2 and A^3. Without calculating the whole matrix A^4, determine only the following three entries:

$$(A^4)_{11}, \qquad (A^4)_{12}, \qquad (A^4)_{23}.$$

(ii) Suppose

$$A^4 = pA^3 + qA^2 + rA + 3I,$$

where p, q, r are real numbers. Using the entries found in part (i), determine p, q and r. Hence write A^4 entirely in terms of A^3, A^2, A and I.

(iii) A student proposes calculating A^5 and A^6 by repeated matrix multiplication. Using your result from part (ii) instead, express each of

$$A^5 \quad \text{and} \quad A^6$$

in the form

$$aA^3 + bA^2 + cA + dI.$$

Do not calculate A^5 or A^6 directly.

(iv) For $n \geq 0$, let u_n denote the entry in row 1, column 3 of A^n, where $A^0 = I$. Use the matrix identity from part (ii) to show that, for $n \geq 4$,

$$u_n = 2u_{n-1} + 10u_{n-2} + 10u_{n-3} + 3u_{n-4}.$$

Hence determine the $(1, 3)$ entry of A^7 without calculating A^7. Briefly explain why the same idea could be used to find any particular entry of a very large power such as A^{100}.

Hint:

(i) You do not need all of A^4. Since $A^4 = A^3 A$, calculate a required entry by taking the appropriate row of A^3 and column of A. For example,

$$(A^4)_{11} = (\text{row 1 of } A^3) \cdot (\text{column 1 of } A).$$

(ii) Equality of matrices means corresponding entries are equal. Apply the equation to the three entries from part (i) to obtain three simultaneous equations in p, q, r.

(iii) Multiply the identity from part (ii) by A. Whenever A^4 appears again, replace it using your known identity.

(iv) Multiply the identity in part (ii) by A^{n-4}. Then look only at the $(1, 3)$ entry.

Solution 4.26

(i) Direct multiplication gives

$$A^2 = \begin{pmatrix} 9 & 8 & 6 \\ 6 & 7 & 6 \\ 8 & 8 & 7 \end{pmatrix}$$

and

$$A^3 = \begin{pmatrix} 37 & 38 & 32 \\ 32 & 31 & 25 \\ 38 & 38 & 31 \end{pmatrix}.$$

Since $A^4 = A^3 A$,

$$(A^4)_{11} = 37(1) + 38(2) + 32(2) = 177.$$

Similarly,

$$(A^4)_{12} = 176, \qquad (A^4)_{23} = 120.$$

Thus

$$\boxed{(A^4)_{11} = 177, \qquad (A^4)_{12} = 176, \qquad (A^4)_{23} = 120.}$$

(ii) Using the $(1,1)$ entry,

$$177 = 37p + 9q + r + 3,$$

so

$$37p + 9q + r = 174. \tag{1}$$

Using the $(1,2)$ entry,

$$176 = 38p + 8q + 2r. \tag{2}$$

Using the $(2,3)$ entry,

$$120 = 25p + 6q + r. \tag{3}$$

Solving (1)–(3) gives

$$\boxed{p = 2, \qquad q = 10, \qquad r = 10}.$$

Hence

$$\boxed{A^4 = 2A^3 + 10A^2 + 10A + 3I.}$$

(iii) Multiplying the identity from part (ii) by A,

$$A^5 = 2A^4 + 10A^3 + 10A^2 + 3A.$$

Substituting for A^4,

$$\begin{aligned} A^5 &= 2(2A^3 + 10A^2 + 10A + 3I) + 10A^3 + 10A^2 + 3A \\ &= \boxed{14A^3 + 30A^2 + 23A + 6I}. \end{aligned}$$

Similarly,

$$A^6 = 14A^4 + 30A^3 + 23A^2 + 6A,$$

so

$$\boxed{A^6 = 58A^3 + 163A^2 + 146A + 42I.}$$

(iv) From

$$A^4 = 2A^3 + 10A^2 + 10A + 3I,$$

multiply by A^{n-4} to obtain

$$A^n = 2A^{n-1} + 10A^{n-2} + 10A^{n-3} + 3A^{n-4}.$$

Therefore the $(1,3)$ entries satisfy

$$\boxed{u_n = 2u_{n-1} + 10u_{n-2} + 10u_{n-3} + 3u_{n-4}.}$$

The initial values are

$$u_0 = 0, \qquad u_1 = 2, \qquad u_2 = 6, \qquad u_3 = 32.$$

Hence

$$u_4 = 2(32) + 10(6) + 10(2) + 3(0) = 144,$$

$$u_5 = 2(144) + 10(32) + 10(6) + 3(2) = 674,$$

$$u_6 = 2(674) + 10(144) + 10(32) + 3(6) = 3126,$$

and

$$u_7 = 2(3126) + 10(674) + 10(144) + 3(32)$$

Takeaways 4.26

- If only a few entries of a matrix product are required, calculate only those entries using the relevant row–column products.
- A matrix identity can provide a powerful shortcut for calculating higher powers. Once
$$A^4 = 2A^3 + 10A^2 + 10A + 3I$$
is known, higher powers can be reduced using algebra rather than repeated matrix multiplication.
- A relationship between whole matrices automatically produces the same recurrence relation for each corresponding entry of their powers.
- Large powers such as A^{20} or A^{100} may therefore be much easier to analyse than they first appear.
- **Further question:** If another matrix B satisfies a different relation involving B^4, B^3, B^2, B and I, what recurrence would the entries of B^n satisfy?

Problem 4.27: Esports Dominance Networks

Six esports teams — **Apex (A), Brawlers (B), Cyphers (C), Drifters (D), Echo (E), and Falcons (F)** — competed in a preliminary regional tournament. Due to time constraints, not every team played every other team.
The results were:

- Brawlers defeated Drifters, Echo, and Falcons.
- Cyphers defeated Drifters, Echo, and Falcons.
- Apex defeated Brawlers and Cyphers.
- Drifters defeated Apex.
- Echo and Falcons did not win any matches.

The tournament organisers use a **dominance matrix** N, where

$$N_{ij} = \begin{cases} 1, & \text{if team } i \text{ defeated team } j, \\ 0, & \text{otherwise.} \end{cases}$$

They initially rank the teams using the model

$$R = N + N^2.$$

The score of a team is the sum of the entries in its corresponding row of R.
(i) Construct the 6×6 dominance matrix N, using the order A, B, C, D, E, F.
(ii) Calculate N^2, and hence determine the ranking scores obtained from $R = N + N^2$. Rank the six teams from highest to lowest score.
(iii) Brawlers and Cyphers complain:

> "We each won three matches, while Apex won only two. How can Apex be ranked above us?"

Use your results from part (ii) to explain mathematically why Apex receives the highest score. In your explanation, interpret the entries of N^2 as **two-step dominance paths**. In particular, explain why some entries in N^2 can be greater than 1.
(iv) The organisers decide that a direct victory should be worth more than a two-step dominance path. They replace the model by

$$R_k = N + kN^2, \qquad 0 < k < 1.$$

1. Find Apex's score in terms of k.
2. Hence find Apex's score when $k = \frac{1}{2}$.

Hint:Part (i)
Place the winning team on the row and the defeated team on the column. For example, because Brawlers defeated Drifters, $N_{BD} = 1$.
Part (ii)
Recall that $N^2 = N \times N$. An entry $(N^2)_{ij}$ counts the number of ways to travel from team i to team j through exactly one intermediate team. For example, $A \to B \to D$ is one two-step path from Apex to Drifters.
Part (iii)
Compare the row sums of N and N^2. Ask:

- How many direct wins does Apex have?
- Which teams did Apex defeat?
- How many wins do those teams themselves have?

Also investigate why $(N^2)_{AD} = 2$. Can you find the two different paths?
Part (iv)
If $R_k = N + kN^2$, then Apex's score is (row sum of Apex in N) $+k$(row sum of Apex in N^2). Use your results from part (ii).

Solution 4.27

(i)

Using the order A, B, C, D, E, F, the dominance matrix is

$$N = \begin{pmatrix} 0 & 1 & 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 1 & 1 & 1 \\ 0 & 0 & 0 & 1 & 1 & 1 \\ 1 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 \end{pmatrix}.$$

(ii)

Multiplying,

$$N^2 = \begin{pmatrix} 0 & 0 & 0 & 2 & 2 & 2 \\ 1 & 0 & 0 & 0 & 0 & 0 \\ 1 & 0 & 0 & 0 & 0 & 0 \\ 0 & 1 & 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 \end{pmatrix}.$$

Therefore the row sums of $R = N + N^2$ are

Team	Score
A	8
B	4
C	4
D	3
E	0
F	0

Hence the ranking is

$$\boxed{A > B = C > D > E = F}$$

so Apex is first, Brawlers and Cyphers are equal second, Drifters is fourth, and Echo and Falcons are equal fifth.

(iii)

Apex has only two direct victories: $A \to B$ and $A \to C$. Thus its row sum in N is 2. However, both Brawlers and Cyphers defeated Drifters, Echo, and Falcons. For example, Apex has two different two-step paths to Drifters: $A \to B \to D$ and $A \to C \to D$. Therefore $(N^2)_{AD} = 2$. Similarly, $(N^2)_{AE} = 2$ and $(N^2)_{AF} = 2$. Hence Apex's row sum in N^2 is $2 + 2 + 2 = 6$. Its total score is therefore $2 + 6 = 8$.

The important point is that N^2 does **not merely record whether an indirect dominance exists**. It counts the number of distinct two-step paths. Thus defeating opponents who themselves defeated several other teams can contribute many additional two-step paths. This is why Apex ranks above Brawlers and Cyphers despite having fewer direct victories.

(iv)

For $R_k = N + kN^2$, Apex has 2 direct-win points and 6 two-step-path contributions. Therefore

$$\boxed{S_A = 2 + 6k}.$$

When $k = \frac{1}{2}$,

$$S_A = 2 + 6\left(\frac{1}{2}\right) = 2 + 3 = \boxed{5}.$$

Takeaways 4.27

- **Matrix powers count paths.** In a dominance network, the entry $(N^2)_{ij}$ counts the number of distinct two-step paths from team i to team j. This is why entries of N^2 can be greater than 1.
- **Do not interpret N^2 as simply "indirect wins".** Apex indirectly reaches only three teams — Drifters, Echo, and Falcons — but it has six two-step paths to them because each can be reached through either Brawlers or Cyphers.
- **Matrix multiplication contains information about structure, not just arithmetic.** The product N^2 reveals how the direct relationships represented by N combine to form longer chains.
- **A ranking model can reward the quality of opponents defeated.** Apex defeats two teams that themselves have many victories, so those wins generate many two-step paths.
- **Mathematical models involve choices.** The model $N + N^2$ gives direct wins and two-step paths equal weight. A weighted model $N + kN^2, 0 < k < 1$, gives greater importance to direct victories.
- **A useful modelling question remains:** What value of k should organisers choose? Different values of k may produce different rankings, showing that the ranking system depends not only on the match results but also on the assumptions built into the mathematical model.

Problem 4.28: Matrix Powers and Convergence

Consider the matrix

$$A = \begin{pmatrix} 0 & \frac{1}{2} \\ \frac{1}{4} & 0 \end{pmatrix}.$$

Repeated multiplication by A defines a discrete system

$$\mathbf{x}_{n+1} = A\mathbf{x}_n.$$

We investigate what happens after many repeated applications of A.

(a) Characteristic roots

The characteristic equation of A is

$$\det(A - \lambda I) = 0.$$

Show that the characteristic equation is

$$\lambda^2 - \frac{1}{8} = 0,$$

and hence find its two roots. Verify that both roots satisfy $|\lambda| < 1$.

(b) A surprising pattern in the powers

Calculate A^2. Hence show that, for every non-negative integer n,

$$A^{2n} = \left(\frac{1}{8}\right)^n I \quad \text{and} \quad A^{2n+1} = \left(\frac{1}{8}\right)^n A.$$

Use your result to calculate A^{10} without multiplying A by itself ten times.

(c) What happens after infinitely many steps?

Using part (b), determine $\lim_{n\to\infty} A^n$. Now let $\mathbf{x}_{n+1} = A\mathbf{x}_n$ for an arbitrary starting vector $\mathbf{x}_0$. Explain why $\mathbf{x}_n \to \mathbf{0}$ regardless of the initial vector. How does this result relate to the fact found in part (a) that both characteristic roots have absolute value less than 1?

(d) The system no longer dies out

Suppose that at each step a constant input is added:

$$\mathbf{x}_{n+1} = A\mathbf{x}_n + \begin{pmatrix} 10 \\ 8 \end{pmatrix}.$$

An equilibrium vector $\mathbf{x}^*$ is a vector that remains unchanged from one step to the next.

(i) Find $\mathbf{x}^*$ by solving

$$\mathbf{x}^* = A\mathbf{x}^* + \begin{pmatrix} 10 \\ 8 \end{pmatrix}.$$

(ii) Define $\mathbf{y}_n = \mathbf{x}_n - \mathbf{x}^*$. Show that $\mathbf{y}_{n+1} = A\mathbf{y}_n$. Hence prove that, for every starting vector $\mathbf{x}_0$,

$$\boxed{\mathbf{x}_n \to \mathbf{x}^*}.$$

Hint:(a)
Write $A-\lambda I=\begin{pmatrix}-\lambda & \frac{1}{2}\\ \frac{1}{4} & -\lambda\end{pmatrix}$ and evaluate its determinant.

(b)
Do not begin by calculating $A^3, A^4, \ldots$. Calculate A^2 first. You should obtain a scalar multiple of I. Then use $A^{2n}=(A^2)^n$.

(c)
Treat even and odd powers separately using the formulas from part (b). Remember that $\left(\frac{1}{8}\right)^n \to 0$. Also, $\mathbf{x}_n = A^n\mathbf{x}_0$.

(d)
Let $\mathbf{x}^*=\begin{pmatrix}p\\ q\end{pmatrix}$. The equilibrium condition gives two simultaneous equations. For part (ii), subtract the equilibrium equation $\mathbf{x}^*=A\mathbf{x}^*+\mathbf{b}$ from $\mathbf{x}_{n+1}=A\mathbf{x}_n+\mathbf{b}$. The constant vector $\mathbf{b}$ will disappear.

Solution 4.28

(a)

$$\det(A-\lambda I)=\begin{vmatrix}-\lambda & \frac{1}{2}\\ \frac{1}{4} & -\lambda\end{vmatrix}=\lambda^2-\frac{1}{8}.$$

Hence $\lambda^2-\frac{1}{8}=0$, so $\lambda=\pm\sqrt{\frac{1}{8}}=\pm\frac{\sqrt{2}}{4}$. Since $\frac{\sqrt{2}}{4}<1$, both characteristic roots have absolute value less than 1.

(b)

$$A^2=\begin{pmatrix}0 & \frac{1}{2}\\ \frac{1}{4} & 0\end{pmatrix}^2=\begin{pmatrix}\frac{1}{8} & 0\\ 0 & \frac{1}{8}\end{pmatrix}=\frac{1}{8}I.$$

Hence $A^{2n}=(A^2)^n=\left(\frac{1}{8}\right)^n I$, and $A^{2n+1}=A^{2n}A=\left(\frac{1}{8}\right)^n A$. Therefore $A^{10}=(A^2)^5=\left(\frac{1}{8}\right)^5 I=\frac{1}{32768}I$.

(c)

For even powers, $A^{2n}=\left(\frac{1}{8}\right)^n I\to 0$. For odd powers, $A^{2n+1}=\left(\frac{1}{8}\right)^n A\to 0$. Therefore $\boxed{A^n\to 0}$. Since $\mathbf{x}_n=A^n\mathbf{x}_0$, it follows that $\boxed{\mathbf{x}_n\to\mathbf{0}}$. The characteristic roots in part (a) both have magnitude less than 1, and repeated powers of these numbers become smaller and smaller. The behaviour of A^n reflects the same decay.

(d)

Let $\mathbf{x}^*=\begin{pmatrix}p\\ q\end{pmatrix}$. Then

$$\begin{pmatrix}p\\ q\end{pmatrix}=\begin{pmatrix}\frac{1}{2}q\\ \frac{1}{4}p\end{pmatrix}+\begin{pmatrix}10\\ 8\end{pmatrix}.$$

Thus $p=\frac{1}{2}q+10$ and $q=\frac{1}{4}p+8$. Solving gives $p=16, q=12$. Hence $\boxed{\mathbf{x}^*=\begin{pmatrix}16\\ 12\end{pmatrix}}$. Now define $\mathbf{y}_n=\mathbf{x}_n-\mathbf{x}^*$. Then

$$\begin{aligned}\mathbf{y}_{n+1}&=\mathbf{x}_{n+1}-\mathbf{x}^*\\ &=(A\mathbf{x}_n+\mathbf{b})-(A\mathbf{x}^*+\mathbf{b})\\ &=A(\mathbf{x}_n-\mathbf{x}^*)\\ &=A\mathbf{y}_n.\end{aligned}$$

Therefore $\mathbf{y}_n=A^n\mathbf{y}_0$. From part (c), $A^n\to 0$, so $\mathbf{y}_n\to\mathbf{0}$. Thus $\boxed{\mathbf{x}_n\to\begin{pmatrix}16\\ 12\end{pmatrix}}$. The constant input changes the destination, but not the mechanism that makes the system settle down.

Takeaways 4.28

- **Look for structure before multiplying repeatedly.** In this problem, $A^2 = \frac{1}{8}I$, which makes all higher powers straightforward.
- **Characteristic roots can give insight into long-term behaviour.** Here both roots satisfy $|\lambda| < 1$, while repeated application of the matrix causes the system to decay towards zero.
- **Even and odd powers may have different forms.** Splitting A^n into A^{2n} and A^{2n+1} is useful whenever A^2 has a particularly simple form.
- **A constant input can produce a non-zero equilibrium.** The recurrence $\mathbf{x}_{n+1} = A\mathbf{x}_n + \mathbf{b}$ approaches the fixed point satisfying $\mathbf{x}^* = A\mathbf{x}^* + \mathbf{b}$.
- **The substitution $\mathbf{y}_n = \mathbf{x}_n - \mathbf{x}^*$** is a useful technique because it converts the recurrence back into $\mathbf{y}_{n+1} = A\mathbf{y}_n$.
- **Question to think about:** What changes if a characteristic root has absolute value greater than 1, or is exactly equal to 1?

Problem 4.29: A Coupled Recurrence and Rational Approximations to $\sqrt{2}$

Two sequences of positive integers (a_n) and (b_n) are defined by

$$a_1 = 2026, \qquad b_1 = 2027, \qquad a_{n+1} = a_n + 2b_n, \qquad b_{n+1} = a_n + b_n.$$

Let $x_n = \dfrac{a_n}{b_n}$ and

$$\mathbf{X}_n = \begin{pmatrix} a_n \\ b_n \end{pmatrix}.$$

(i) Express the recurrence in the form

$$\mathbf{X}_{n+1} = \mathbf{M}\mathbf{X}_n$$

and state the matrix $\mathbf{M}$. Given

$$\mathbf{Q} = \begin{pmatrix} 1 & -\sqrt{2} \\ 1 & \sqrt{2} \end{pmatrix}, \qquad \mathbf{D} = \begin{pmatrix} 1-\sqrt{2} & 0 \\ 0 & 1+\sqrt{2} \end{pmatrix},$$

find $\mathbf{Q}^{-1}$ and verify that

$$\mathbf{Q}\mathbf{M}\mathbf{Q}^{-1} = \mathbf{D}.$$

(ii) Define $\mathbf{Y}_n = \mathbf{Q}\mathbf{X}_n$. Show that

$$\mathbf{Y}_{n+1} = \mathbf{D}\mathbf{Y}_n$$

and use its first row to prove that, for all $n \in \mathbb{Z}^+$,

$$\left| x_{n+1} - \sqrt{2} \right| < \frac{1}{2} \left| x_n - \sqrt{2} \right|.$$

(iii) Use mathematical induction to prove that

$$\mathbf{M}^n = \mathbf{Q}^{-1}\mathbf{D}^n\mathbf{Q}$$

for all $n \in \mathbb{Z}^+$. Hence find a closed-form expression for $\mathbf{X}_n$ in terms of n, $\alpha = 1 - \sqrt{2}$, $\beta = 1 + \sqrt{2}$, $c_1 = a_1 - \sqrt{2}b_1$, and $c_2 = a_1 + \sqrt{2}b_1$.

(iv) Evaluate $\lim\limits_{n\to\infty} x_n$ in two ways:

1. by applying the inequality from part (ii);
2. by using the explicit formula from part (iii).

Briefly explain why this limit is independent of the choice of positive initial values a_1 and b_1.

Hint:(i) Use the inverse formula for a 2×2 matrix, or verify $\mathbf{QM}=\mathbf{DQ}$ and right-multiply by $\mathbf{Q}^{-1}$.
(ii) Write

$$\mathbf{Y}_n=\begin{pmatrix} a_n-\sqrt{2}b_n \\ a_n+\sqrt{2}b_n \end{pmatrix}.$$

Divide the first row equation by b_{n+1} and use $b_{n+1}=a_n+b_n>b_n>0$ and $\sqrt{2}<\frac{3}{2}$.
(iii) In the inductive step, substitute $\mathbf{M}=\mathbf{Q}^{-1}\mathbf{DQ}$ and use $\mathbf{QQ}^{-1}=\mathbf{I}$. Also, $\mathbf{QX}_1=(c_1,c_2)^{\mathsf{T}}$.
(iv) Iterate the inequality for $n\geq 2$. For the explicit formula, divide the numerator and denominator of a_n/b_n by β^{n-1}.

Solution 4.29: Brief

(i) The recurrence gives

$$\mathbf{M}=\begin{pmatrix} 1 & 2 \\ 1 & 1 \end{pmatrix}.$$

Since $\det(\mathbf{Q})=2\sqrt{2}$,

$$\mathbf{Q}^{-1}=\frac{1}{2\sqrt{2}}\begin{pmatrix} \sqrt{2} & \sqrt{2} \\ -1 & 1 \end{pmatrix}.$$

Direct multiplication gives $\mathbf{QM}=\mathbf{DQ}$, and hence $\mathbf{QMQ}^{-1}=\mathbf{D}$.
(ii) By part (i),

$$\mathbf{Y}_{n+1}=\mathbf{QX}_{n+1}=\mathbf{QMX}_n=\mathbf{DY}_n.$$

The first row yields

$$a_{n+1}-\sqrt{2}b_{n+1}=(1-\sqrt{2})(a_n-\sqrt{2}b_n),$$

so

$$x_{n+1}-\sqrt{2}=(1-\sqrt{2})\frac{b_n}{b_{n+1}}(x_n-\sqrt{2}).$$

The terms a_n, b_n stay positive integers, so $b_n/b_{n+1}<1$. Also, x_n is rational and therefore differs from $\sqrt{2}$. As $\sqrt{2}-1<\frac{1}{2}$, taking absolute values proves the required strict inequality.
(iii) For $n=1$, the identity follows from $\mathbf{M}=\mathbf{Q}^{-1}\mathbf{DQ}$. If it holds for $n=k$, then

$$\mathbf{M}^{k+1}=(\mathbf{Q}^{-1}\mathbf{D}^k\mathbf{Q})(\mathbf{Q}^{-1}\mathbf{DQ})=\mathbf{Q}^{-1}\mathbf{D}^{k+1}\mathbf{Q}.$$

Thus it holds for all positive integers n. Since $\mathbf{X}_n=\mathbf{M}^{n-1}\mathbf{X}_1$,

$$\mathbf{X}_n=\mathbf{Q}^{-1}\begin{pmatrix} \alpha^{n-1} & 0 \\ 0 & \beta^{n-1} \end{pmatrix}\begin{pmatrix} c_1 \\ c_2 \end{pmatrix}=\begin{pmatrix} \dfrac{c_1\alpha^{n-1}+c_2\beta^{n-1}}{2} \\ \dfrac{-c_1\alpha^{n-1}+c_2\beta^{n-1}}{2\sqrt{2}} \end{pmatrix}.$$

(iv) Iterating part (ii), for $n\geq 2$,

$$0\leq\left|x_n-\sqrt{2}\right|<\left(\frac{1}{2}\right)^{n-1}\left|x_1-\sqrt{2}\right|,$$

so $x_n\to\sqrt{2}$. Alternatively, part (iii) gives

$$x_n=\sqrt{2}\,\frac{c_1(\alpha/\beta)^{n-1}+c_2}{-c_1(\alpha/\beta)^{n-1}+c_2}.$$

Here $c_2>0$ and $|\alpha/\beta|<1$, so this expression also tends to $\sqrt{2}$. For any positive initial values, the coefficient $c_2=a_1+\sqrt{2}b_1$ of the dominant term is positive, while the other term decays relative to it; the limit is therefore unchanged.

Takeaways 4.29

- **Diagonalisation decouples a coupled recurrence.** The rows of $\mathbf{Q}$ transform the original state into two independent geometric sequences.
- **A contraction gives quantitative convergence.** The error from $\sqrt{2}$ is reduced by a factor of less than $\frac{1}{2}$ at every step.
- **Induction proves identities for matrix powers.** The diagonal form makes the closed formula for $\mathbf{X}_n$ accessible.
- **The dominant eigenvalue determines the limiting ratio.** The contribution from $\beta = 1 + \sqrt{2}$ dominates the decaying contribution from $\alpha = 1 - \sqrt{2}$.
- **Further exploration:** Replacing $\mathbf{M}$ by $\begin{pmatrix} m & k \\ 1 & m \end{pmatrix}$ for positive m, k leads to a related recurrence whose ratios converge to $\sqrt{k}$.

5 Matrix Toolkit (Quick Reference)

This section provides a quick summary of the key matrix definitions, operations, and formulas used throughout this booklet.

1. Notation and Dimensions

An $m \times n$ matrix A has m rows and n columns. The entry in the i-th row and j-th column is denoted a_{ij}.

$$A = \begin{pmatrix} a_{11} & a_{12} & \cdots & a_{1n} \\ a_{21} & a_{22} & \cdots & a_{2n} \\ \vdots & \vdots & \ddots & \vdots \\ a_{m1} & a_{m2} & \cdots & a_{mn} \end{pmatrix}$$

2. Matrix Arithmetic

Addition/Subtraction (same dimensions only):

$A \pm B$ adds/subtracts corresponding elements.

Scalar Multiplication:

kA multiplies every element of A by the scalar k.

3. Matrix Multiplication

For the product AB to be defined, the number of columns in A must equal the number of rows in B.

$$A_{m\times n} B_{n\times p} \to (AB)_{m\times p}$$

Note: Matrix multiplication is **not** commutative ($AB \neq BA$ in general).

4. Transpose

The transpose A^T is formed by swapping rows and columns.

$$(A^T)_{ij} = a_{ji}$$

Properties: $(A^T)^T = A$, and $(AB)^T = B^T A^T$.

5. Identity and Zero Matrices

Identity Matrix I: A square matrix with 1s on the main diagonal and 0s elsewhere.

$$IA = A \quad \text{and} \quad AI = A$$

Zero Matrix O: A matrix where all entries are 0.

$$A + O = A \quad \text{and} \quad AO = O$$

6. Determinant and Inverse (2×2)

For a 2×2 matrix $A = \begin{pmatrix} a & b \\ c & d \end{pmatrix}$:

- **Determinant:** $\det(A) = |A| = ad - bc$
- **Inverse:** If $\det(A) \neq 0$, the inverse exists:

$$A^{-1} = \frac{1}{ad - bc} \begin{pmatrix} d & -b \\ -c & a \end{pmatrix}$$

Properties: $AA^{-1} = A^{-1}A = I$, and $(AB)^{-1} = B^{-1}A^{-1}$.

7. Solving Matrix Equations

To solve a linear system $AX = B$:

$$AX = B \implies X = A^{-1}B \quad (\text{if } A^{-1} \text{ exists})$$

(Pre-multiply both sides by A^{-1}).

8. Geometric Area via Determinants

Triangle Area (with one vertex at the origin $O(0,0)$):

$$\text{Area}(\triangle OPQ) = \frac{1}{2}|x_1 y_2 - x_2 y_1| = \frac{1}{2}\left|\det \begin{pmatrix} x_1 & y_1 \\ x_2 & y_2 \end{pmatrix}\right|$$

General Triangle Area (vertices P_1, P_2, P_3):

$$\text{Area}(\triangle P_1P_2P_3) = \frac{1}{2}\left|\det \begin{pmatrix} x_1 & y_1 & 1 \\ x_2 & y_2 & 1 \\ x_3 & y_3 & 1 \end{pmatrix}\right| = \frac{1}{2}\left|\det \begin{pmatrix} x_2 - x_1 & y_2 - y_1 \\ x_3 - x_1 & y_3 - y_1 \end{pmatrix}\right|$$

Collinearity: Points P_1, P_2, P_3 are collinear if and only if $\det = 0$.

 Vu Hung Nguyen · CC BY-NC 4.0

6 Summary and Where Next

What You Have Learned

- ☐ matrix arithmetic
- ☐ $\det(A)$
- ☐ A^{-1}
- ☐ $Ax = b$

The Big Ideas

- matrices as data
- matrices as transformations
- matrices as systems
- matrices as dynamical models
- matrices as networks

Connections

$$\boxed{\text{Matrix}} \rightarrow \begin{cases} \text{linear systems} \\ \text{transformations} \\ \text{models} \\ \text{graphs} \\ \text{linear algebra} \end{cases}$$

Where to Go Next

Calculus → Linear Algebra → Numerical Methods → Data Science / Engineering / CS.

Contact Information:

LinkedIn: https://www.linkedin.com/in/nguyenvuhung/

GitHub: https://github.com/vuhung16au/

Repository: https://github.com/vuhung16au/math-olympiad-ml/tree/main/HSC-Matrix

A Appendix A: Bridge to University Linear Algebra

Enrichment Material

The topics in this appendix go **beyond** the Queensland (QCAA) Specialist Mathematics syllabus. They are included to give curious students a preview of the linear algebra they will meet in first- and second-year university courses in mathematics, engineering, computer science, data science, and AI. None of this material is examinable in QLD Year 12 assessments.

A.1 Linear Combinations and Span

A **linear combination** of a set of vectors is formed by multiplying each vector by a scalar constant and adding the results together. For example, if you have vectors $\mathbf{v}_1$ and $\mathbf{v}_2$, any vector $\mathbf{v}$ of the form:

$$\mathbf{v} = c_1\mathbf{v}_1 + c_2\mathbf{v}_2$$

is a linear combination of $\mathbf{v}_1$ and $\mathbf{v}_2$, where c_1 and c_2 are real numbers.

Span

The **span** of a set of vectors is the collection of *all possible* linear combinations of those vectors. Think of it as the entire space that can be reached by travelling along those vectors.

- If you have one non-zero vector in $\mathbb{R}^2$ or $\mathbb{R}^3$, its span is a **line** passing through the origin.
- If you have two vectors pointing in different directions, their span is a flat **plane**.
- If you have three vectors that do not lie on the same plane, their span is the entire 3D space ($\mathbb{R}^3$).

When we represent a system of equations as a matrix equation $A\mathbf{x} = \mathbf{b}$, we are actually asking a question about span: *"Is the vector* $\mathbf{b}$ *in the span of the columns of matrix A?"* If it is, a solution exists. If not, the system has no solution.

A.2 Linear Independence

In linear algebra, a set of vectors is **linearly independent** if no vector in the set can be written as a linear combination of the others. Conversely, if at least one vector can be formed by combining the others (using scalar multiplication and vector addition), the set is **linearly dependent**.

The Core Idea

Formally, a set of vectors $\{\mathbf{v}_1, \mathbf{v}_2, \ldots, \mathbf{v}_n\}$ is linearly independent if the only solution to the equation:

$$c_1\mathbf{v}_1 + c_2\mathbf{v}_2 + \cdots + c_n\mathbf{v}_n = \mathbf{0}$$

is $c_1 = 0, c_2 = 0, \ldots, c_n = 0$.

If there exists a solution where not all c_i are zero, then the vectors are linearly dependent. This means you can rearrange the equation to express one vector in terms of the others.

Geometric Intuition

Think about vectors in 2D or 3D space:

- **In 2D:** Two vectors are linearly independent if they don't point in the same (or exactly opposite) direction. If they point in different directions, they can reach any point in the 2D plane. If they point along the same line, they are linearly dependent.
- **In 3D:** Three vectors are linearly independent if they don't all lie on the same flat plane. If they do lie on the same plane, the third vector is just a combination of the first two, making the set linearly dependent.

Example

Consider the vectors in $\mathbb{R}^3$:

$$\mathbf{v}_1 = \begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix}, \quad \mathbf{v}_2 = \begin{bmatrix} 0 \\ 1 \\ 0 \end{bmatrix}, \quad \mathbf{v}_3 = \begin{bmatrix} 2 \\ 3 \\ 0 \end{bmatrix}$$

Notice that $\mathbf{v}_3 = 2\mathbf{v}_1 + 3\mathbf{v}_2$. Because we can write $\mathbf{v}_3$ as a combination of $\mathbf{v}_1$ and $\mathbf{v}_2$, the set $\{\mathbf{v}_1, \mathbf{v}_2, \mathbf{v}_3\}$ is linearly **dependent**. Geometrically, all three vectors lie in the xy-plane (where $z = 0$).

However, the set $\{\mathbf{v}_1, \mathbf{v}_2, \mathbf{v}_4\}$ where $\mathbf{v}_4 = \begin{bmatrix} 0 \\ 0 \\ 1 \end{bmatrix}$ is linearly **independent**. They point along the x, y, and z axes respectively, and no single vector can be made from the other two. These three vectors form a **basis** for $\mathbb{R}^3$, meaning their span covers the entire 3D space.

Why it matters

Linear independence is foundational for understanding the structure of mathematical spaces.

- **Machine Learning & Data Science:** Linearly dependent features in a dataset are redundant. Removing them reduces dimensionality without losing information (a core concept in Principal Component Analysis, PCA).
- **Engineering & Physics:** In structural analysis, linearly independent equations represent distinct physical constraints.
- **Linear Algebra Fundamentals:** It leads directly to the concepts of **span**, **basis**, and **dimension** (rank), which tell us the "true size" or information content of a matrix.

A.3 Basis and Dimension

Not all sets of vectors that span a space are equally useful. Sometimes, a set contains redundant vectors that aren't strictly necessary to reach every point in the space.

A **basis** for a vector space is a set of vectors that has two crucial properties:

1. It **spans** the space (you can reach every point).
2. It is **linearly independent** (there is no redundancy; no vector can be made from the others).

Think of a basis as the most efficient set of building blocks for a space. In $\mathbb{R}^3$, the standard basis consists of the unit vectors along the x, y, and z axes:

$$\mathbf{i} = \begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix}, \quad \mathbf{j} = \begin{bmatrix} 0 \\ 1 \\ 0 \end{bmatrix}, \quad \mathbf{k} = \begin{bmatrix} 0 \\ 0 \\ 1 \end{bmatrix}$$

Dimension

The **dimension** of a vector space is simply the number of vectors in its basis. While you can have many different bases for the same space, every basis will always have exactly the same number of vectors.

For example, any basis for a plane will always contain exactly two vectors, so a plane has a dimension of 2. The entire 3D space requires three vectors in a basis, so it has a dimension of 3.

A.4 Rank and Null Space

Earlier in this booklet, we learned how to represent a system of linear equations using an augmented matrix and solve it using inverses or Cramer's Rule (for 2×2 and 3×3 systems). In university linear algebra, we formalise this by writing any system of equations as a single matrix equation:

$$A\mathbf{x} = \mathbf{b}$$

where:

- A is the **coefficient matrix** (size $m \times n$).
- $\mathbf{x}$ is the **variable vector** (a column of n unknowns).
- $\mathbf{b}$ is the **constant vector** (a column of m constants).

Existence and Uniqueness of Solutions

When dealing with $A\mathbf{x} = \mathbf{b}$, a fundamental question is: *Does a solution exist, and if so, is it unique?*

We can answer this by looking at the **rank** of the matrix (roughly, the number of non-zero rows after performing Gaussian elimination, which is also the number of linearly independent rows).

1. **Unique Solution:** A unique solution exists if and only if $\text{rank}(A) = \text{rank}([A \mid \mathbf{b}]) = n$. As a special case, if A is a square $n \times n$ matrix, a unique solution exists if and only if $\det(A) \neq 0$.
2. **No Solution (Inconsistent System):** A system has no solution if and only if $\text{rank}(A) \neq \text{rank}([A \mid \mathbf{b}])$. In Gaussian elimination, this results in a row like $\begin{bmatrix} 0 & 0 & \dots & 0 & \mid & c \end{bmatrix}$ where $c \neq 0$, translating to $0 = c$. This contradiction means the lines/planes never intersect.
3. **Infinitely Many Solutions:** A system has infinitely many solutions if and only if $\text{rank}(A) = \text{rank}([A \mid \mathbf{b}]) < n$. The system will have free variables, and geometrically, the intersection might be a line or a plane rather than a single point.

Homogeneous Systems

A special case occurs when $\mathbf{b} = \mathbf{0}$, giving the equation:

$$A\mathbf{x} = \mathbf{0}$$

This is called a **homogeneous system**. It is special because it *always* has at least one solution: the **trivial solution** where $\mathbf{x} = \mathbf{0}$ (i.e., all variables are zero).

Finding the non-trivial solutions (when there are infinitely many) is a core task in linear algebra. The set of all solutions, including the trivial solution, forms a subspace called the **null space** (or kernel) of the matrix A. Its non-zero elements are the non-trivial solutions.

Why it matters

Viewing equations as $A\mathbf{x} = \mathbf{b}$ shifts our perspective from algebra to geometry.

- It allows us to view matrix multiplication as a **linear transformation** mapping a vector $\mathbf{x}$ to a new vector $\mathbf{b}$.
- This perspective is crucial in programming, where arrays and matrices are the standard data structures. Solving $A\mathbf{x} = \mathbf{b}$ is so common that numerical computing libraries (like NumPy in Python or MATLAB) have highly optimized solvers specifically designed for this single equation.

A.5 Eigenvalues and Eigenvectors

When a matrix multiplies a vector, it typically changes both the vector's length and its direction. However, for a given square matrix A, there are often special vectors whose *direction* remains completely unchanged (or points in the exact opposite direction) after the multiplication. These are called **eigenvectors**.

The amount by which the eigenvector is stretched or squashed is called the **eigenvalue**.

The Core Definition

Let A be an $n \times n$ square matrix. A non-zero vector $\mathbf{v}$ is an eigenvector of A if there exists a scalar (number) λ such that:

$$A\mathbf{v} = \lambda\mathbf{v}$$

Here, λ is the eigenvalue associated with the eigenvector $\mathbf{v}$.

To find the eigenvalues, we rearrange the equation:

$$A\mathbf{v} - \lambda\mathbf{v} = \mathbf{0}$$

$$(A - \lambda I)\mathbf{v} = \mathbf{0}$$

where I is the identity matrix. For a non-zero vector $\mathbf{v}$ to exist, the matrix $(A - \lambda I)$ must not be invertible, which means its determinant must be zero:

$$\det(A - \lambda I) = 0$$

This equation is called the **characteristic equation**. Solving it gives the eigenvalues (λ). We can then substitute each λ back into $(A - \lambda I)\mathbf{v} = \mathbf{0}$ to find the corresponding eigenvectors.

Example

Let's find the eigenvalues of $A = \begin{bmatrix} 3 & 1 \\ 0 & 2 \end{bmatrix}$.

Step 1: Set up the characteristic equation $\det(A - \lambda I) = 0$.

$$A - \lambda I = \begin{bmatrix} 3 & 1 \\ 0 & 2 \end{bmatrix} - \begin{bmatrix} \lambda & 0 \\ 0 & \lambda \end{bmatrix} = \begin{bmatrix} 3-\lambda & 1 \\ 0 & 2-\lambda \end{bmatrix}$$

$$\det \begin{bmatrix} 3-\lambda & 1 \\ 0 & 2-\lambda \end{bmatrix} = (3-\lambda)(2-\lambda) - (1)(0) = (3-\lambda)(2-\lambda)$$

Step 2: Solve for λ.

$$(3-\lambda)(2-\lambda) = 0$$

So, the eigenvalues are $\lambda_1 = 3$ and $\lambda_2 = 2$.

Why it matters

Eigenvalues and eigenvectors are arguably the most important concept in applied linear algebra. They reveal the fundamental internal structure and behavior of a matrix.

- **Computer Science (PageRank):** Google's original PageRank algorithm models the internet as a giant matrix. The PageRank scores of all websites are the entries of the principal eigenvector of this matrix.
- **Data Science (PCA):** In Principal Component Analysis, eigenvectors of a covariance matrix point to the directions of maximum variance in a dataset, allowing us to compress images and data efficiently.
- **Physics & Engineering:** They are used to find the natural frequencies (harmonics) of buildings and bridges (to avoid resonance and collapse) and are central to the mathematics of Quantum Mechanics (where observables are eigenvalues of operators).

A.6 The Characteristic Polynomial

Definition

For a square matrix A of size $n \times n$, the characteristic polynomial $f(\lambda)$ is defined as the determinant:

$$f(\lambda) = \det(A - \lambda I_n)$$

where I_n represents the $n \times n$ identity matrix, and λ is a scalar variable.

The roots of this polynomial ($f(\lambda) = 0$) are the eigenvalues of A, which encode fundamental geometric and algebraic properties of the linear transformation represented by A.

Example

Consider the 2×2 matrix A:

$$A = \begin{pmatrix} 4 & 1 \\ 2 & 3 \end{pmatrix}$$

Construct $A - \lambda I_2$:

$$A - \lambda I_2 = \begin{pmatrix} 4 & 1 \\ 2 & 3 \end{pmatrix} - \begin{pmatrix} \lambda & 0 \\ 0 & \lambda \end{pmatrix} = \begin{pmatrix} 4-\lambda & 1 \\ 2 & 3-\lambda \end{pmatrix}$$

Compute the Determinant:

$$f(\lambda) = \det\begin{pmatrix} 4-\lambda & 1 \\ 2 & 3-\lambda \end{pmatrix}$$

$$f(\lambda) = (4-\lambda)(3-\lambda) - (1 \cdot 2)$$

$$f(\lambda) = \lambda^2 - 7\lambda + 12 - 2$$

$$f(\lambda) = \lambda^2 - 7\lambda + 10$$

Setting $f(\lambda) = 0$ yields $(\lambda - 2)(\lambda - 5) = 0$, giving the eigenvalues $\lambda_1 = 2$ and $\lambda_2 = 5$.

The Cayley–Hamilton Theorem

A cornerstone result of matrix algebra is the **Cayley–Hamilton Theorem**: *Every square matrix satisfies its own characteristic equation.* That is, if $f(\lambda) = \det(A - \lambda I) = 0$, then substituting the matrix A in place of λ yields the zero matrix:

$$f(A) = O.$$

For the matrix $A = \begin{pmatrix} 4 & 1 \\ 2 & 3 \end{pmatrix}$ above with $f(\lambda) = \lambda^2 - 7\lambda + 10$, this means:

$$A^2 - 7A + 10I = O.$$

We can directly verify this:

$$A^2 = \begin{pmatrix} 4 & 1 \\ 2 & 3 \end{pmatrix}\begin{pmatrix} 4 & 1 \\ 2 & 3 \end{pmatrix} = \begin{pmatrix} 18 & 7 \\ 14 & 11 \end{pmatrix},$$

and

$$A^2 - 7A + 10I = \begin{pmatrix} 18 & 7 \\ 14 & 11 \end{pmatrix} - \begin{pmatrix} 28 & 7 \\ 14 & 21 \end{pmatrix} + \begin{pmatrix} 10 & 0 \\ 0 & 10 \end{pmatrix} = \begin{pmatrix} 0 & 0 \\ 0 & 0 \end{pmatrix} = O.$$

NSW HSC Heritage (1977 Q9): Computing Inverses via Cayley–Hamilton

In the 1977 NSW HSC 4 Unit examination, students were asked:

> *The matrix A is given by $A = \begin{pmatrix} 5 & -4 \\ -4 & 5 \end{pmatrix}$.*
> *(i) Write down the characteristic equation of A.*
> *(ii) Prove that $A^2 - 10A + 9I = O$.*
> *(iii) Find A^{-1}.*

The Cayley–Hamilton relation provides a swift, cofactor-free method to compute the matrix inverse. Multiplying $A^2 - 10A + 9I = O$ through by A^{-1}:

$$A - 10I + 9A^{-1} = O \implies 9A^{-1} = 10I - A \implies A^{-1} = \frac{1}{9}(10I - A).$$

Substituting A gives:

$$A^{-1} = \frac{1}{9}\left[\begin{pmatrix} 10 & 0 \\ 0 & 10 \end{pmatrix} - \begin{pmatrix} 5 & -4 \\ -4 & 5 \end{pmatrix}\right] = \frac{1}{9}\begin{pmatrix} 5 & 4 \\ 4 & 5 \end{pmatrix}.$$

This technique bypasses minors, cofactors, and adjugate matrices entirely!

NSW HSC Heritage (1970 Q6, 1975 Q7): Classifying Conics via Eigenvalues

Symmetric 2×2 matrices $M = \begin{pmatrix} a & h \\ h & b \end{pmatrix}$ possess real eigenvalues λ_1, λ_2 and orthogonal eigenvectors. In the 1970 and 1975 HSC examinations, students used these eigenvalues to rotate Cartesian axes and eliminate the cross-term $2hxy$ from quadratic equations:

$$ax^2 + 2hxy + by^2 = \begin{pmatrix} x & y \end{pmatrix} \begin{pmatrix} a & h \\ h & b \end{pmatrix} \begin{pmatrix} x \\ y \end{pmatrix} = \lambda_1 x'^2 + \lambda_2 y'^2 = 1.$$

If $\lambda_1, \lambda_2 > 0$, the locus is an ellipse with principal semi-axes $1/\sqrt{\lambda_1}$ and $1/\sqrt{\lambda_2}$ aligned precisely along the eigenvectors. This was how 4 Unit originally unified linear algebra with analytic geometry.

A.7 Where Linear Algebra Goes Next: Least Squares, SVD, etc.

Matrix algebra is just the gateway to a vast landscape of advanced mathematics. If you continue into STEM at university, you will encounter these concepts again, applied to solve incredibly complex problems. Here is a brief look ahead.

Numerical Linear Algebra and SVD

When dealing with massive matrices (like millions of rows and columns), finding exact inverses or determinants is computationally impossible and unstable. **Numerical Linear Algebra** focuses on creating fast, stable algorithms to approximate solutions.

A crowning achievement of this field is the **Singular Value Decomposition (SVD)**. SVD factorises *any* matrix (even non-square ones) into three simpler matrices. It's the mathematical magic behind data compression, noise reduction in images, and recommender systems (like how Netflix suggests movies).

Linear Regression and Least Squares

In statistics and data science, you rarely have a perfect system of equations $A\mathbf{x} = \mathbf{b}$. Instead, you have messy, noisy real-world data, and you want to draw a "line of best fit." This is **Linear Regression**.

When there is no exact solution because there are more data points than variables (an overdetermined system), we use the **Method of Least Squares**. It uses matrix calculus and transposes to find the vector $\mathbf{x}$ that gets $A\mathbf{x}$ as close to $\mathbf{b}$ as mathematically possible.

Linear Transformations

Instead of just viewing matrices as grids of numbers, pure mathematics views them as **Linear Transformations**—functions that move and stretch vectors through space while keeping grid lines parallel and evenly spaced. From this perspective, multiplying two matrices isn't just an arithmetic algorithm; it is the *composition* of two geometric transformations (like rotating, then scaling).

Systems of Differential Equations

In physics and engineering, variables change continuously over time. A *differential equation* describes this change. When multiple variables affect each other (like predator and prey populations, or currents in an electrical circuit), you get a **System of Differential Equations**. Using matrices, eigenvectors, and the matrix exponential (e^A), you can solve these intertwined systems simultaneously.

Exterior Algebra and the Vector Cross Product

In multivariable calculus and modern geometry, determinants are understood through **exterior algebra** and differential forms. The determinant of three vectors in $\mathbb{R}^3$ computes the signed volume of the parallelepiped they span, which is identical to the scalar triple product:

$$\det(\mathbf{u}, \mathbf{v}, \mathbf{w}) = \mathbf{u} \cdot (\mathbf{v} \times \mathbf{w}).$$

When we set $\mathbf{w} = \mathbf{k} = (0, 0, 1)$, the 3×3 determinant projects directly onto the z-component of the cross product of two-dimensional vectors lying in the plane. This provides the mathematical foundation connecting 2×2 determinants, 3×3 coordinate area formulas, and 3D vector cross products (as explored in detail in our companion booklet *Vectors Mastery: HSC Math Extension 2*).

B Appendix B: Matrices in Computing and Applied Mathematics

B.1 NumPy and SymPy

For small matrices, calculating determinants, inverses, and solving equations by hand is good practice. But in the real world, matrices can have thousands or millions of entries. For these numerical calculations, we always use computers as assistants.

While languages like C++ or Julia might be faster for extreme performance, **Python** is widely considered the best starting point because its code is highly readable and it has powerful mathematical libraries.

Numerical Calculations with NumPy

`NumPy` is Python's standard library for numerical computing. It handles large arrays and matrices with high speed. It is ideal for calculating numerical answers quickly.

Here is a simple example of how to define two matrices and multiply them using NumPy:

```
import numpy as np

# Define two 2x2 matrices
A = np.array([[1, 2],
              [3, 4]])
B = np.array([[5, 6],
              [7, 8]])

# Matrix multiplication (using the @ operator)
C = A @ B

print("Matrix C is:")
print(C)
```

Symbolic Algebra with SymPy

Sometimes we don't just want numerical answers; we want exact algebraic expressions (like keeping π or $\sqrt{2}$ exact, or working with variables like x). For this, Python has a library called `SymPy` (Symbolic Python).

SymPy can do algebra exactly like you would on paper. Here is an example of finding the inverse of a matrix with exact fractions:

```python
import sympy as sp

# Define a matrix
M = sp.Matrix([[4, 7],
               [2, 6]])

# Find the exact inverse
M_inv = M.inv()

print("The exact inverse of M is:")
sp.pprint(M_inv)
```

Connecting Maths and Software

If you are interested in Software Engineering, writing code to automate your maths homework is one of the best ways to learn. By representing matrices in Python, you open the door to building physics engines for video games, writing data analysis scripts, and experimenting with machine learning algorithms.

B.2 Matrices and Graph Theory

This section connects two areas of mathematics: **matrices** and **graph theory**. A graph represents relationships between objects, while a matrix provides a compact numerical way to store and analyse those relationships.

This connection is widely used in computer science, transport networks, social networks and web search.

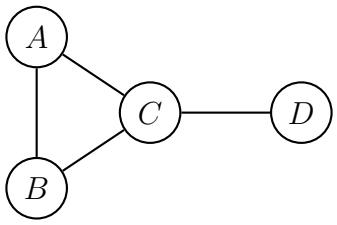
Graphs

A **graph** consists of:

- **vertices** (or nodes): the objects;
- **edges**: connections between pairs of vertices.

For example, suppose four railway stations A, B, C, D are connected as follows:

$$A - B, \qquad A - C, \qquad B - C, \qquad C - D.$$



We can think of the vertices as stations and the edges as direct railway connections.

The Adjacency Matrix

A graph can be represented by an **adjacency matrix**.

Let

$$M_{ij} = \begin{cases} 1, & \text{if vertices } i \text{ and } j \text{ are connected,} \\ 0, & \text{otherwise.} \end{cases}$$

For our railway network,

$$M = \begin{bmatrix} 0 & 1 & 1 & 0 \\ 1 & 0 & 1 & 0 \\ 1 & 1 & 0 & 1 \\ 0 & 0 & 1 & 0 \end{bmatrix}.$$

The rows and columns correspond to A, B, C, D.

For example,

$$M_{AC} = 1$$

because A and C are directly connected, while

$$M_{AD} = 0$$

because there is no direct connection between A and D.

Because this graph is **undirected**, its adjacency matrix is symmetric:

$$M = M^T.$$

Matrix Powers Reveal Routes

Here is the surprising connection.

If M is an adjacency matrix, then the entries of

$$M^2 = M \times M$$

count the number of **two-edge walks** between vertices.

For example, looking at the entry for A to D:

$$(M^2)_{AD} = M_{AA}M_{AD} + M_{AB}M_{BD} + M_{AC}M_{CD} + M_{AD}M_{DD}$$

Therefore,

$$(M^2)_{AD} = (0)(0) + (1)(0) + (1)(1) + (0)(0) = 1.$$

So there is exactly one two-edge route from A to D:

$$A \to C \to D.$$

More generally,

$$\boxed{(M^k)_{ij} = \text{number of walks of length } k \text{ from } i \text{ to } j.}$$

This gives matrix multiplication a concrete interpretation: multiplying matrices can reveal connectivity in a network. In fact, this example directly reinforces why matrix multiplication is defined as a row-by-column dot product!

Remark B.1 (Note)

A *walk* may revisit vertices or edges, so it is not necessarily a simple path.

Why It Matters

This matrix–graph connection appears in many real systems:

- **Transport:** stations are vertices and routes are edges.
- **Social networks:** people are vertices and friendships or follows are edges.
- **Computer networks:** devices are vertices and communication links are edges.
- **The Web:** webpages are vertices and hyperlinks form directed edges.

Large graphs can contain millions or billions of connections. Representing them using matrices allows computers to apply linear algebra to analyse these networks.

$$\boxed{\textbf{Graph} \longleftrightarrow \textbf{Adjacency Matrix} \longleftrightarrow \textbf{Matrix Algebra}}$$

Takeaway: Graph theory describes **who is connected to whom**; matrices turn those connections into numbers that we can calculate with.

B.3 Computer Graphics

Every video game and 3D animation relies entirely on matrices. A 3D object is just a list of coordinate vectors. To rotate, scale, or move (translate) a character on screen, the computer multiplies millions of coordinate vectors by 4×4 transformation matrices, 60 times a second. This intense matrix multiplication is exactly what GPUs (Graphics Processing Units) are physically designed to do.

B.4 Data Science and Machine Learning

Modern Artificial Intelligence, specifically deep neural networks, is fundamentally built on matrix operations. A single "layer" of a neural network mathematically takes an input vector $\mathbf{x}$, multiplies it by a "weight" matrix W, adds a "bias" vector $\mathbf{b}$, and then applies a non-linear activation function (like σ). The equation looks like this:

$$\mathbf{y} = \sigma(W\mathbf{x} + \mathbf{b})$$

Training an AI like ChatGPT involves using calculus (gradient descent) to slightly adjust billions of numbers inside the weight matrices W over and over until the network makes accurate predictions. This is why AI research relies so heavily on GPUs—because at its core, deep learning is just billions of matrix multiplications.